

# Pearson Edexcel Level 3 GCE

**Friday 20 June 2025**

Afternoon (Time: 1 hour 30 minutes)

Paper  
reference

**9FM0/3D**

## **Further Mathematics**

**Advanced**

**PAPER 3D: Decision Mathematics 1**

### **You must have:**

Mathematical Formulae and Statistical Tables (Green), calculator,  
Decision Mathematics Answer Book (enclosed)

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### **Instructions**

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
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- Do not return the question paper with the answer book
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– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

### **Information**

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### **Advice**

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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1.

4.7    2.9    5.5    1.4    5.8    2.8    3.8    6.5    5.1    6.3    4.1

- (a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 13 (3)
- (b) Use the first-fit decreasing bin packing algorithm to determine how the numbers listed above can be packed into bins of size 13 (2)

**(Total for Question 1 is 5 marks)**

---



2.

271    828    182    845    904    523    536    028    747    135

The list of ten numbers above is to be sorted into descending order.

- (a) Perform a quick sort on the list to obtain the sorted list. You should show the result of each pass and identify the pivots clearly. (4)

A list of  $n$  numbers is to be sorted into descending order using bubble sort.

- (b) Determine, in the worst case, the total number of comparisons required to sort the list. Give your answer as a simplified expression in terms of  $n$ . (2)

The following algorithm determines an approximation to the value of  $e$

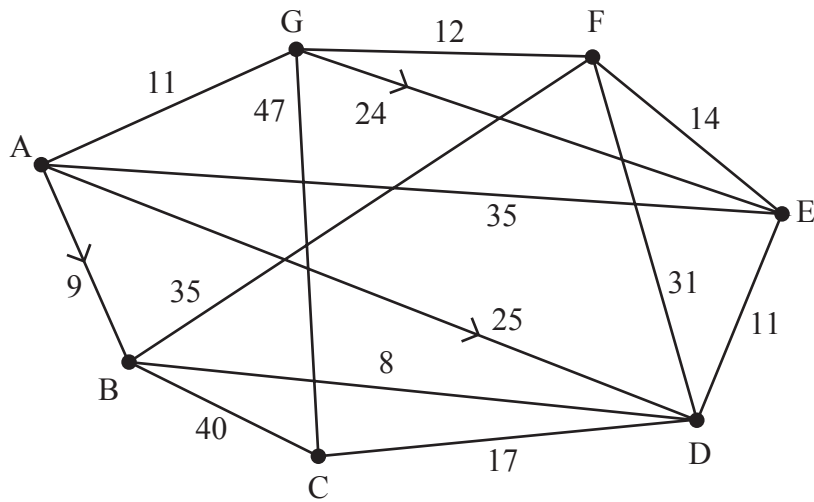
|         |                           |
|---------|---------------------------|
| Step 1  | Start                     |
| Step 2  | Let $a = 1$               |
| Step 3  | Let $b = 1$               |
| Step 4  | Let $c = 1$               |
| Step 5  | Let $d = a$               |
| Step 6  | Let $c = c \times b$      |
| Step 7  | Let $d = d + \frac{1}{c}$ |
| Step 8  | If $b = 6$ go to Step 11  |
| Step 9  | Let $b = b + 1$           |
| Step 10 | Go to Step 6              |
| Step 11 | Output $d$                |
| Step 12 | Stop                      |

- (c) Complete the table in the answer book to show the results obtained at each step of the algorithm. (3)

- (d) Calculate, to 3 significant figures, the percentage error in using the value found in (c) to approximate the value of  $e$  (1)

(Total for Question 2 is 10 marks)

3.



**Figure 1**

Direct roads between seven villages, A, B, C, D, E, F and G, are represented in Figure 1. The weight on each arc is the time, in minutes, taken to travel along the corresponding road. Three roads, AB, AD and GE, are one-way, as indicated by the arrow on the corresponding arc.

Floyd's algorithm is to be used to find the complete network of shortest times between the seven villages.

(a) Set up an initial time matrix for this network.

(2)

The time matrix after three iterations of Floyd's algorithm is shown below.

|   | A        | B  | C  | D  | E        | F  | G  |
|---|----------|----|----|----|----------|----|----|
| A | –        | 9  | 49 | 17 | 35       | 44 | 11 |
| B | $\infty$ | –  | 40 | 8  | $\infty$ | 35 | 87 |
| C | $\infty$ | 40 | –  | 17 | $\infty$ | 75 | 47 |
| D | $\infty$ | 8  | 17 | –  | 11       | 31 | 64 |
| E | 35       | 44 | 84 | 11 | –        | 14 | 46 |
| F | $\infty$ | 35 | 75 | 31 | 14       | –  | 12 |
| G | 11       | 20 | 47 | 28 | 24       | 12 | –  |

(b) Perform the next two iterations of Floyd's algorithm that follow from the table above.

You should show only the time matrix after each iteration.

(5)

The final time matrix after completion of Floyd's algorithm is shown below.

|   | A  | B  | C  | D  | E  | F  | G  |
|---|----|----|----|----|----|----|----|
| A | –  | 9  | 34 | 17 | 28 | 23 | 11 |
| B | 54 | –  | 25 | 8  | 19 | 33 | 45 |
| C | 58 | 25 | –  | 17 | 28 | 42 | 47 |
| D | 46 | 8  | 17 | –  | 11 | 25 | 37 |
| E | 35 | 19 | 28 | 11 | –  | 14 | 26 |
| F | 23 | 32 | 42 | 25 | 14 | –  | 12 |
| G | 11 | 20 | 45 | 28 | 24 | 12 | –  |

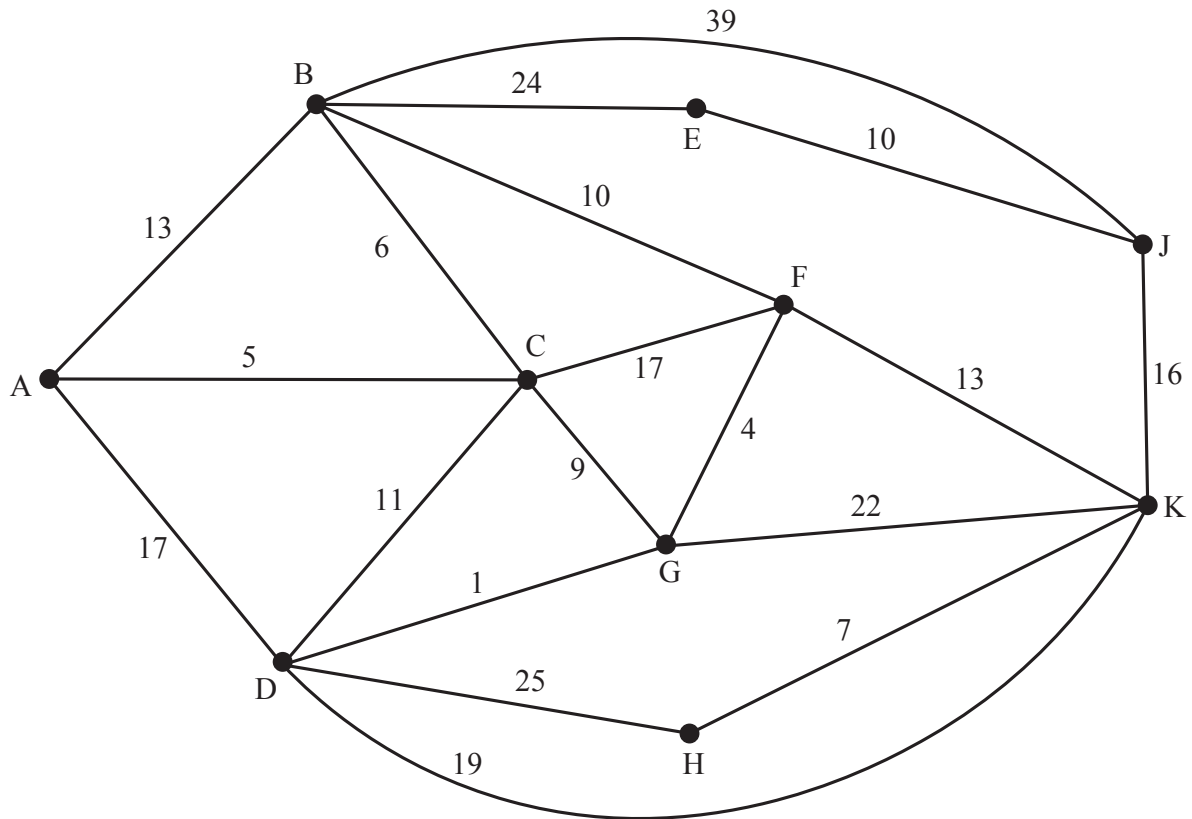
Albert must visit each village. He will start and finish at B and wishes to minimise the total time taken to visit each village.

- (c) (i) Use the nearest neighbour algorithm, starting at B, to find a Hamiltonian cycle in the complete network of shortest times shown above.
- (ii) Find the total travel time for this cycle.
- (iii) Given that he travels the Hamiltonian cycle found in (c)(i), interpret this cycle in terms of the actual villages visited by Albert.

(3)

(Total for Question 3 is 10 marks)

4.



**Figure 2**

[The total weight of the network is 268]

Figure 2 represents a network of corridors between 10 offices, A, B, C, D, E, F, G, H, J and K, in a building. The number on each edge represents the length, in metres, of the corresponding corridor.

On Monday morning, Turvi needs to walk from J to K via A. She wishes to minimise the distance she travels along the corridors.

- (a) (i) Use Dijkstra's algorithm, starting at A, to find the shortest route from J to K via A.
- (ii) State the length of this route.

(6)

On Monday afternoon, Turvi needs a route that traverses each corridor at least once. She plans to start at J and finish at A and wants to minimise the distance travelled.

- (b) (i) By considering the pairings of all relevant nodes, find the corridors that would need to be traversed twice. You must make your method and working clear.
- (ii) State the total length of this route.

(5)

Turvi discovers that she cannot walk along corridors AB, CD and JK because they are being decorated. Turvi walks from J to A travelling along all the other corridors at least once. She does this in the minimum distance possible.

- (c) Determine the difference between the length of the route Turvi travels and the length of the route found in (b). You must make the values used in your calculation clear.

(2)

(Total for Question 4 is 13 marks)

5. Consider the following linear programming problem in  $x$ ,  $y$  and  $z$ .

$$\text{Maximise } P = 3x + 4y + 2z$$

$$\text{subject to } 2x + 2y + 2z \leq 21$$

$$2x - y - z \leq 18$$

$$-3x + y - 2z \leq 1$$

$$x \geq 4 \quad y \geq 1 \quad z \leq -3$$

(a) Explain why the Simplex algorithm cannot be used to solve this linear programming problem in its current form.

(1)

(b) Use the substitutions  $x = X + 4$ ,  $y = Y + 1$ ,  $z = -Z - 3$  and  $P = Q + 10$  to reformulate the linear programming problem into a form in which the Simplex algorithm can be used to solve the problem. You should write the constraints as simplified inequalities with integer coefficients.

(3)

After a second iteration of the Simplex algorithm, a possible tableau  $T$  is

| b.v.  | $X$ | $Y$ | $Z$            | $s_1$          | $s_2$ | $s_3$          | Value           |
|-------|-----|-----|----------------|----------------|-------|----------------|-----------------|
| $X$   | 1   | 0   | $-\frac{3}{4}$ | $\frac{1}{8}$  | 0     | $-\frac{1}{4}$ | $\frac{5}{8}$   |
| $s_2$ | 0   | 0   | $\frac{9}{4}$  | $\frac{1}{8}$  | 1     | $\frac{3}{4}$  | $\frac{117}{8}$ |
| $Y$   | 0   | 1   | $-\frac{1}{4}$ | $\frac{3}{8}$  | 0     | $\frac{1}{4}$  | $\frac{63}{8}$  |
| $Q$   | 0   | 0   | $-\frac{5}{4}$ | $\frac{15}{8}$ | 0     | $\frac{1}{4}$  | $\frac{267}{8}$ |

(c) Write down the profit equation given by  $T$ .

(1)

(d) Using your answer to (c), explain why  $T$  is not optimal.

(1)

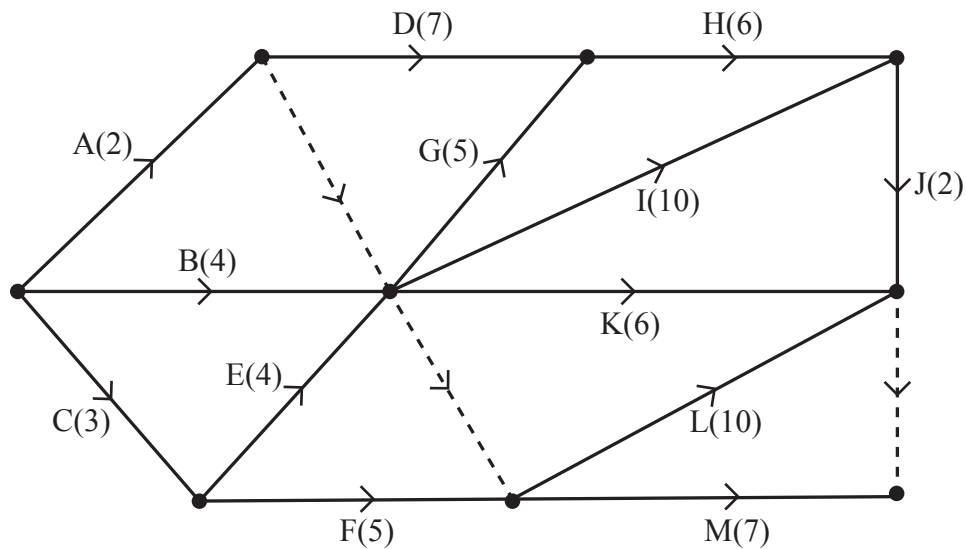
(e) (i) Perform the third iteration of the Simplex algorithm. You should state the row operations you use.

(ii) Hence determine the maximum value of  $P$ , and the corresponding values of  $x$ ,  $y$  and  $z$ .

(6)

(Total for Question 5 is 12 marks)

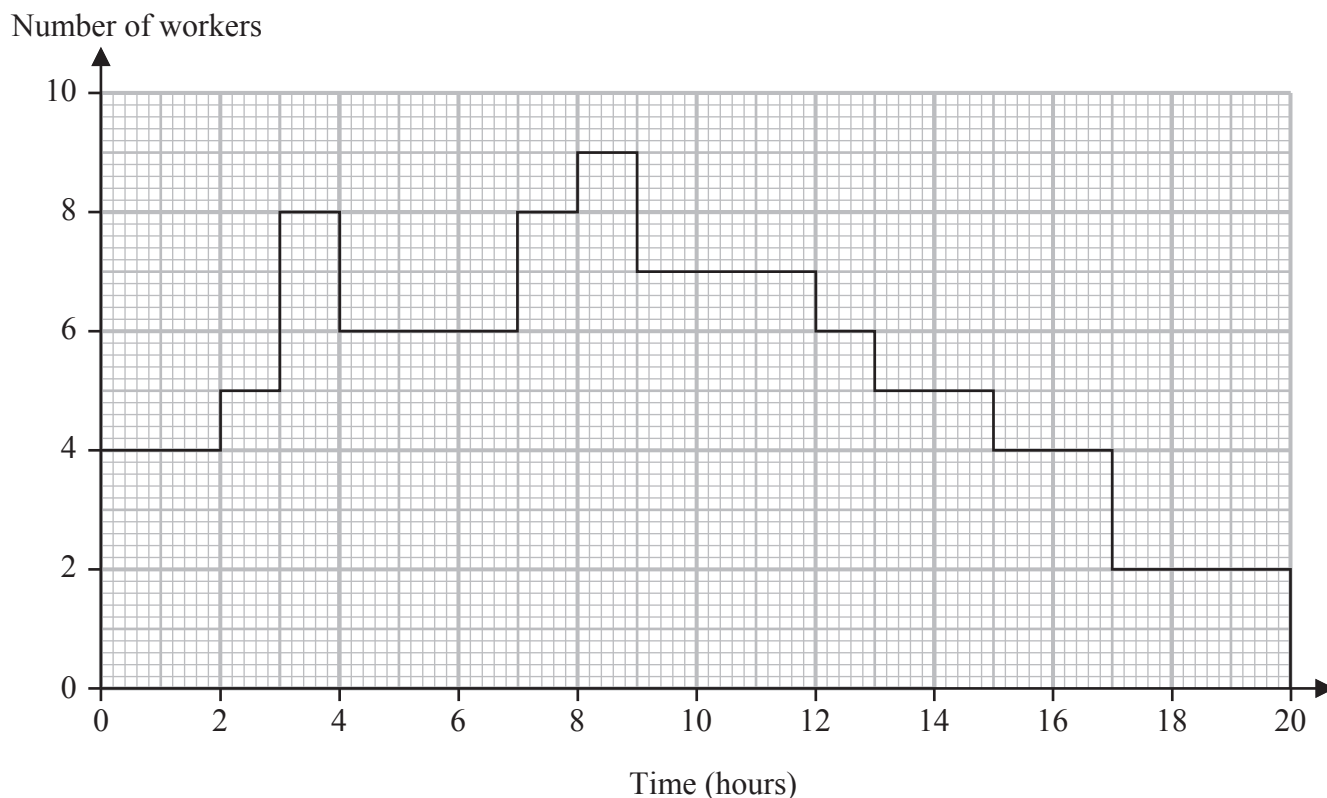
6.



**Figure 3**

A project is modelled by the activity network shown in Figure 3. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the corresponding activity. The project is to be completed in the shortest possible time.

- (a) Complete Table 1 in the answer book to show the immediately preceding activities for each activity. (2)
- (b) Complete Diagram 1 in the answer book to show the early event times and the late event times. (4)
- (c) Draw a Gantt chart for this project on Grid 1 in the answer book. (4)



**Figure 4**

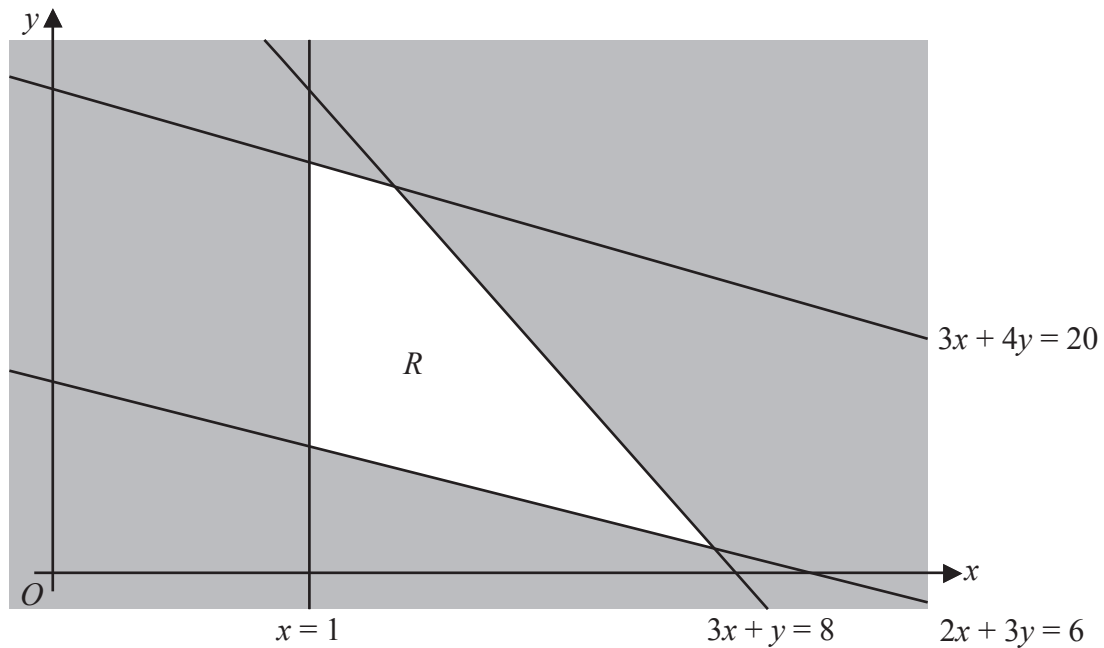
The resource histogram in Figure 4 shows the number of workers required when each activity starts at its earliest possible time. When an activity is started it must be completed without interruption. Each activity requires at least one worker.

- (d) By considering which activities are happening each hour, and by working backwards from the minimum project completion time, complete Table 2 in the answer book to show the number of workers needed for each activity.

**(3)**

**(Total for Question 6 is 13 marks)**

7.



**Figure 5**

Figure 5 shows the constraints of a linear programming problem in  $x$  and  $y$ , where  $R$  is the feasible region.

The objective is to maximise  $P = 11x + ky$ , where  $k$  is a positive constant.

The optimal value of  $P$  is to be found using the big-M method.

- (a) Set up an initial tableau for solving this linear programming problem using the big-M method.

You should use exactly 2 slack variables, 2 surplus variables and 2 artificial variables.

(7)

After a third iteration of the big-M method, a possible tableau is

| b.v.  | $x$ | $y$ | $s_1$ | $s_2$ | $s_3$                         | $s_4$                         | $a_1$ | $a_2$                             | Value                          |
|-------|-----|-----|-------|-------|-------------------------------|-------------------------------|-------|-----------------------------------|--------------------------------|
| $s_1$ | 0   | 0   | 1     | 0     | $\frac{9}{7}$                 | $-\frac{1}{7}$                | 0     | $-\frac{9}{7}$                    | $\frac{78}{7}$                 |
| $x$   | 1   | 0   | 0     | 0     | $\frac{1}{7}$                 | $\frac{3}{7}$                 | 0     | $-\frac{1}{7}$                    | $\frac{18}{7}$                 |
| $y$   | 0   | 1   | 0     | 0     | $-\frac{3}{7}$                | $-\frac{2}{7}$                | 0     | $\frac{3}{7}$                     | $\frac{2}{7}$                  |
| $s_2$ | 0   | 0   | 0     | 1     | $\frac{1}{7}$                 | $\frac{3}{7}$                 | -1    | $-\frac{1}{7}$                    | $\frac{11}{7}$                 |
| $P$   | 0   | 0   | 0     | 0     | $\frac{11}{7} - \frac{3}{7}k$ | $\frac{33}{7} - \frac{2}{7}k$ | $M$   | $M - \frac{11}{7} + \frac{3}{7}k$ | $\frac{198}{7} + \frac{2}{7}k$ |

(b) Given that the third iteration gives the optimal value for  $P$ , determine this value.

(5)

(Total for Question 7 is 12 marks)

**TOTAL FOR PAPER IS 75 MARKS**



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1.

17   8   16   12   24   19   23   11   20   13   4

The eleven numbers listed above are to be packed into bins of size  $n$  where  $n$  is a positive integer. When the first-fit bin packing algorithm is applied to the eleven numbers, the bins are packed as shown below.

Bin 1:   17       8       12

Bin 2:   16       24

Bin 3:   19       11       4

Bin 4:   23       13

Bin 5:   20

(a) Explain why this packing means that the value of  $n$  must be 40

(1)

The original list of eleven numbers is to be sorted into descending order.

(b) Use a quick sort to obtain the fully sorted list. You must make your pivots clear.

(4)

(c) Apply the first-fit decreasing bin packing algorithm to the fully sorted list to pack the numbers into bins of size 40

(2)

**(Total for Question 1 is 7 marks)**

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2. The table below represents a network of shortest distances, in miles, to travel between nine castles, A, B, C, D, E, F, G, H and J.

|   | A  | B  | C  | D  | E  | F  | G  | H  | J  |
|---|----|----|----|----|----|----|----|----|----|
| A | –  | 50 | 59 | 26 | 50 | 40 | 87 | 63 | 59 |
| B | 50 | –  | 28 | 61 | 79 | 63 | 45 | 64 | 48 |
| C | 59 | 28 | –  | 33 | 57 | 35 | 70 | 36 | 45 |
| D | 26 | 61 | 33 | –  | 24 | 64 | 71 | 37 | 33 |
| E | 50 | 79 | 57 | 24 | –  | 40 | 64 | 30 | 31 |
| F | 40 | 63 | 35 | 64 | 40 | –  | 47 | 70 | 71 |
| G | 87 | 45 | 70 | 71 | 64 | 47 | –  | 34 | 67 |
| H | 63 | 64 | 36 | 37 | 30 | 70 | 34 | –  | 33 |
| J | 59 | 48 | 45 | 33 | 31 | 71 | 67 | 33 | –  |

- (a) Use Prim's algorithm, starting at D, to find the minimum spanning tree for this network. You must clearly state the order in which you select the arcs of your tree. (3)

- (b) State the weight of the minimum spanning tree found in part (a). (1)

- (c) Draw the minimum spanning tree using the vertices given in Diagram 1 in the answer book. (1)

A historian needs to visit all of the castles, starting and finishing at the same castle, and wishes to minimise the total distance travelled.

- (d) Use your answer to part (b) to calculate an initial upper bound for the length of the historian's route. (1)

- (e) (i) Use the nearest neighbour algorithm, starting at D, to find an upper bound for the length of the historian's route.  
(ii) Write down the route which gives this upper bound. (3)

Using the nearest neighbour algorithm, starting at F, an upper bound of length 352 miles was found.

- (f) State the best upper bound that can be obtained by using this information and your answers from parts (d) and (e). Give the reason for your answer. (1)



(g) By deleting A and all of its arcs, find a lower bound for the length of the historian's route.

(2)

By deleting J and all of its arcs, a lower bound of length 274 miles was found.

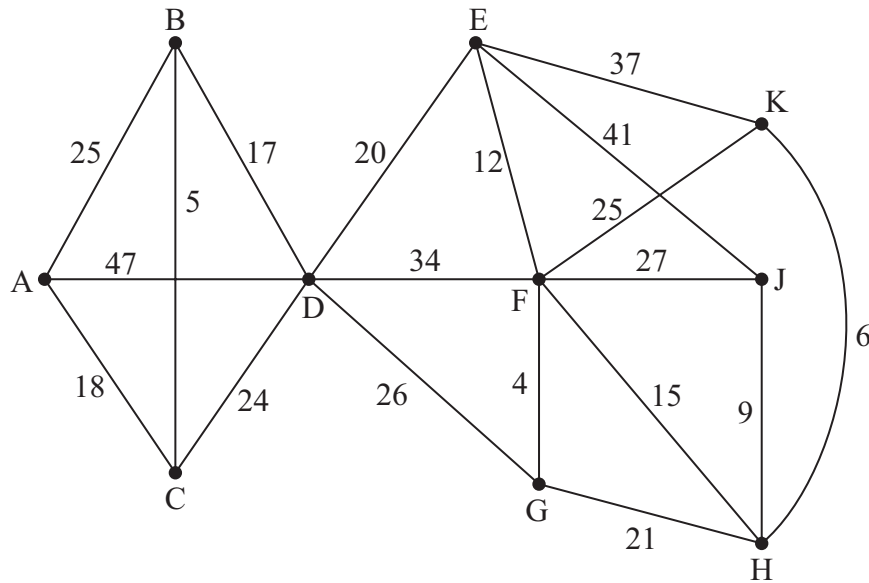
(h) State the best lower bound that can be obtained by using this information and your answer to part (g). Give the reason for your answer.

(1)

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**(Total for Question 2 is 13 marks)**

3.



**Figure 1**

*[The total weight of the network is 413]*

Figure 1 represents a network of cycle tracks between ten towns, A, B, C, D, E, F, G, H, J and K. The number on each arc represents the length, in kilometres, of the corresponding track.

(a) Use Dijkstra's algorithm to find the shortest path from A to J.

(5)

Abi needs to travel along every track shown in Figure 1 to check that they are all in good repair. She needs to start her inspection route at town G and finish her route at either town J or town K.

Abi wishes to minimise the total distance required to traverse every track.

(b) By considering all relevant pairings of vertices, determine whether Abi should finish her inspection route at town J or town K. You must

- state which tracks she will repeat in her route
- state the total length of her route

(6)

The direct track between town B and town C **and** the direct track between town H and town K are **now** closed to all users. A second person, Tarig, is asked to check all the remaining tracks starting at G and finishing at H.

Tarig wishes to minimise the total length of his inspection route.

(c) Determine which route, Abi's or Tarig's, is shorter. You must make your working clear.

(2)

**(Total for Question 3 is 13 marks)**

4. (a) Explain why it is **not** possible to draw a graph with exactly six nodes with degrees 1, 2, 3, 4, 5 and 6

(1)

A tree,  $T$ , has exactly six nodes. The degrees of the six nodes of  $T$  are

$$1 \quad 2 \quad (4 - x) \quad (2x - 5) \quad (4x - 11) \quad (3x - 5)$$

where  $x$  is an integer.

- (b) Explain how you know that  $T$  cannot be Eulerian.

(1)

- (c) (i) Determine the value of  $x$

- (ii) Hence state whether  $T$  is semi-Eulerian or not. You must justify your answer.

(5)

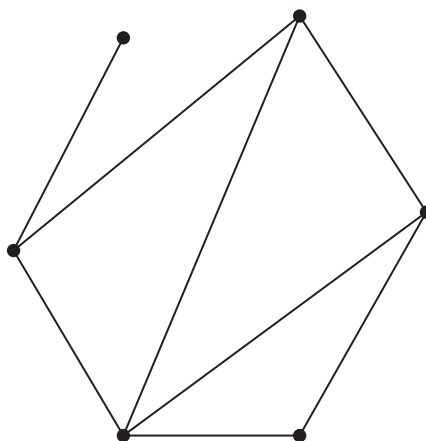


Figure 2

Figure 2 shows a graph,  $G$ , with six nodes with degrees 1, 2, 3, 3, 3 and 4

- (d) Using the vertices in Diagram 1 in the answer book, draw a graph with exactly six nodes with degrees 1, 2, 3, 3, 3 and 4 that is **not** isomorphic to  $G$ .

(1)

(Total for Question 4 is 8 marks)

5. Two friends, Anaira and Tommi, play a game involving two positive numbers  $x$  and  $y$

Anaira gives Tommi the following clues to see if he can correctly determine the value of  $x$  and the value of  $y$

- $x$  is greater than  $y$  and the difference between the two is at least 100
- $x$  is at most 5 times as large as  $y$
- the sum of  $2x$  and  $3y$  is at least 350
- the sum of  $x$  and  $y$  is as small as possible

Tommi decides to solve this problem by using the big-M method.

(a) Set up an initial tableau for solving this problem using the big-M method.

As part of your solution, you must show

- how the constraints were made into equations using one slack variable, exactly two surplus variables and exactly two artificial variables
- how the objective function was formed

(6)

The big-M method is applied until the tableau containing the optimal solution to the problem is found. One row of this final tableau is as follows.

| b.v. | $x$ | $y$ | $s_1$          | $s_2$ | $s_3$          | $a_1$         | $a_2$         | Value |
|------|-----|-----|----------------|-------|----------------|---------------|---------------|-------|
| $x$  | 1   | 0   | $-\frac{3}{5}$ | 0     | $-\frac{1}{5}$ | $\frac{3}{5}$ | $\frac{1}{5}$ | 130   |

(b) (i) State the value of  $x$

(ii) Hence deduce the value of  $y$ , making your reasoning clear.

(3)

(Total for Question 5 is 9 marks)



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6. The precedence table below shows the 12 activities required to complete a project.

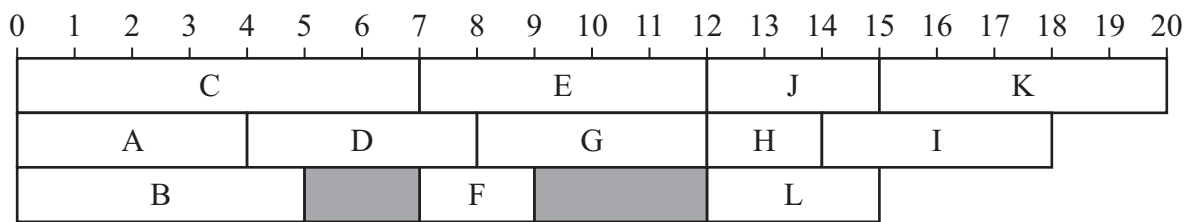
| Activity | Immediately preceding activities |
|----------|----------------------------------|
| A        | —                                |
| B        | —                                |
| C        | —                                |
| D        | A                                |
| E        | A, B, C                          |
| F        | A, B, C                          |
| G        | C                                |
| H        | D, E                             |
| I        | D, E                             |
| J        | D, E                             |
| K        | F, G, J                          |
| L        | F, G                             |

(a) Draw the activity network described in the precedence table, using activity on arc.

Your activity network must contain the minimum number of dummies only.

(5)

Each of the activities shown in the precedence table requires one worker. The project is to be completed in the minimum possible time.



**Figure 3**

Figure 3 shows a schedule for the project using three workers.

(b) (i) State the critical path for the network.

(ii) State the minimum completion time for the project.

(iii) Calculate the total float on activity B.

(iv) Calculate the total float on activity G.

**(4)**

Immediately after the start of the project, it is found that the duration of activity I, as shown in Figure 3, is incorrect. In fact, activity I will take 8 hours.

The durations of all the other activities remain as shown in Figure 3.

(c) Determine whether the project can still be completed in the minimum completion time using only three workers when the duration of activity I is 8 hours.

Your answer must make specific reference to workers, times and activities.

**(2)**

**(Total for Question 6 is 11 marks)**

7. A maximisation linear programming problem in  $x$ ,  $y$  and  $z$  is to be solved using the Simplex method.

The tableau after the 1st iteration is shown below.

| b.v.  | $x$ | $y$            | $z$             | $s_1$ | $s_2$          | $s_3$ | Value |
|-------|-----|----------------|-----------------|-------|----------------|-------|-------|
| $s_1$ | 0   | $-\frac{1}{2}$ | $\frac{3}{2}$   | 1     | $-\frac{1}{2}$ | 0     | 30    |
| $x$   | 1   | $\frac{1}{4}$  | $-\frac{1}{4}$  | 0     | $\frac{1}{4}$  | 0     | 10    |
| $s_3$ | 0   | 1              | 1               | 0     | 0              | 1     | 26    |
| $P$   | 0   | $-\frac{1}{4}$ | $-\frac{11}{4}$ | 0     | $\frac{3}{4}$  | 0     | 30    |

- (a) State the column that contains the pivot value for the 1st iteration. You must give a reason for your answer. (2)
- (b) By considering the equations represented in the above tableau, formulate the linear programming problem in  $x$ ,  $y$  and  $z$  only. State the objective and list the constraints as inequalities with integer coefficients. (5)
- (c) Taking the most negative number in the profit row to indicate the pivot column, perform the 2nd iteration of the Simplex algorithm, to obtain a new tableau, T. Make your method clear by stating the row operations you use. (4)
- (d) (i) Explain, using T, how you know that an optimal solution to the original linear programming problem has **not** been found after the 2nd iteration. (2)
- (ii) State the values of the basic variables after the 2nd iteration. (2)



A student attempts the 3rd iteration of the Simplex algorithm and obtains the tableau below.

| b.v. | $x$ | $y$ | $z$ | $s_1$          | $s_2$          | $s_3$          | Value           |
|------|-----|-----|-----|----------------|----------------|----------------|-----------------|
| $z$  | 0   | 0   | 1   | $\frac{1}{2}$  | $-\frac{1}{4}$ | $\frac{1}{4}$  | $\frac{43}{2}$  |
| $x$  | 1   | 0   | 0   | $\frac{1}{4}$  | $\frac{1}{8}$  | $-\frac{1}{8}$ | $\frac{57}{4}$  |
| $y$  | 0   | 1   | 0   | $-\frac{1}{2}$ | $\frac{1}{4}$  | $\frac{3}{4}$  | $\frac{9}{2}$   |
| $P$  | 0   | 1   | 0   | $\frac{5}{4}$  | $\frac{1}{8}$  | $\frac{7}{8}$  | $\frac{361}{4}$ |

(e) Explain how you know that the student's attempt at the 3rd iteration is **not** correct.

(1)

(Total for Question 7 is 14 marks)

**TOTAL FOR PAPER IS 75 MARKS**



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# Pearson Edexcel Level 3 GCE

**Friday 23 June 2023**

Afternoon (Time: 1 hour 30 minutes)

Paper  
reference

**9FM0/3D**

## Further Mathematics

**Advanced**

**PAPER 3D: Decision Mathematics 1**

### You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,  
Decision Mathematics Answer Book (enclosed)

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.  
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

### Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

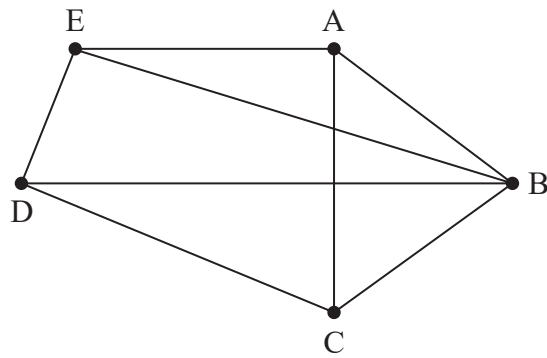
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**Pearson**

1.

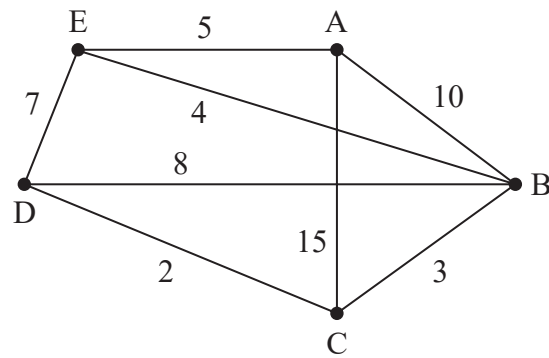


**Figure 1**

Figure 1 shows the graph  $G$ .

- (a) State whether  $G$  is Eulerian, semi-Eulerian, or neither, giving a reason for your answer. (1)
- (b) Write down an example of a Hamiltonian cycle on  $G$ . (1)
- (c) State whether or not  $G$  is planar, justifying your answer. (1)
- (d) State the number of arcs that would need to be added to  $G$  to make the graph  $K_5$ . (1)





**Figure 2**

Direct roads between five villages, A, B, C, D and E, are represented in Figure 2. The weight on each arc is the time, in minutes, required to travel along the corresponding road. Floyd's algorithm is to be used to find the complete network of shortest times between the five villages.

- (e) For the network represented in Figure 2, complete the initial time matrix in the answer book.

(1)

The time matrix after four iterations of Floyd's algorithm is shown in Table 1.

|   | A  | B  | C  | D  | E |
|---|----|----|----|----|---|
| A | —  | 10 | 13 | 15 | 5 |
| B | 10 | —  | 3  | 5  | 4 |
| C | 13 | 3  | —  | 2  | 7 |
| D | 15 | 5  | 2  | —  | 7 |
| E | 5  | 4  | 7  | 7  | — |

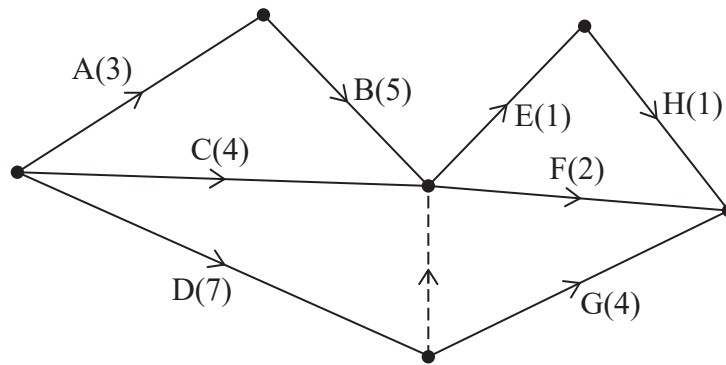
**Table 1**

- (f) Perform the final iteration of Floyd's algorithm that follows from Table 1, showing the time matrix for this iteration.

(2)

**(Total for Question 1 is 7 marks)**

2.



**Figure 3**

The network in Figure 3 shows the activities that need to be undertaken to complete a project. Each activity is represented by an arc and the duration, in hours, of the corresponding activity is shown in brackets.

- (a) (i) Complete Diagram 1 in the answer book to show the early event times and the late event times.
- (ii) State the minimum completion time of the project.

**(3)**

The table below lists the number of workers required for each activity in the project.

| Activity | Number of workers |
|----------|-------------------|
| A        | 2                 |
| B        | 1                 |
| C        | 2                 |
| D        | 2                 |
| E        | 3                 |
| F        | 2                 |
| G        | 1                 |
| H        | 3                 |

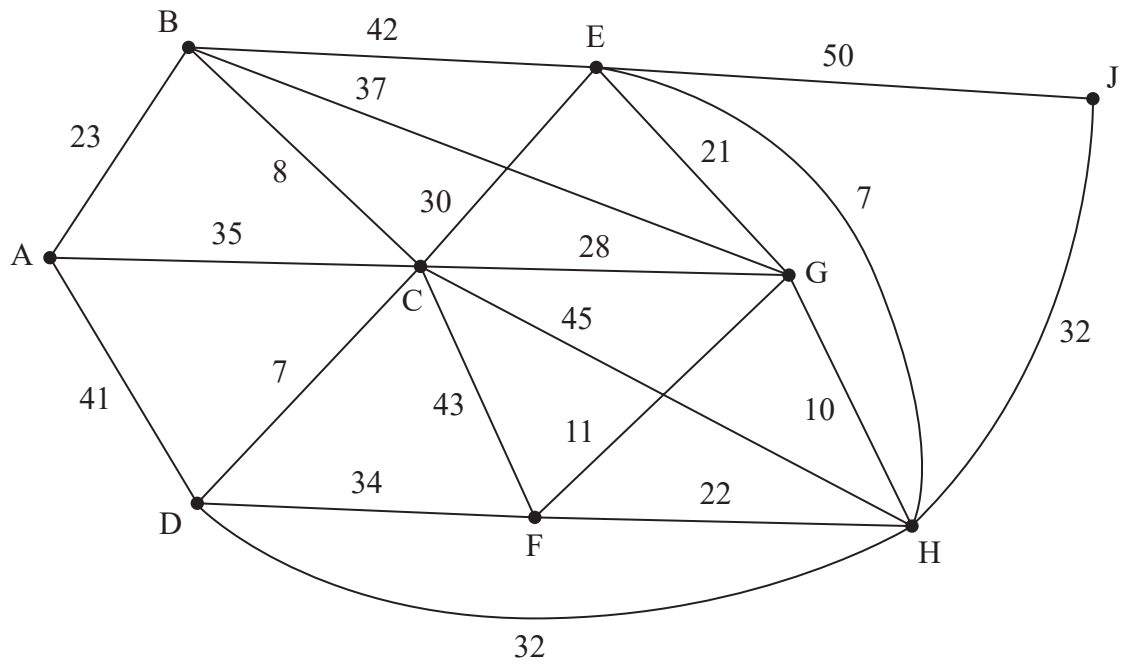
Each worker is able to do any of the activities. Once an activity is started it must be completed without interruption. It is given that each activity begins at its earliest possible start time.

- (b) (i) On Grid 1 in the answer book, draw a resource histogram to show the number of workers required at each time.
- (ii) Hence state the time interval(s) when six workers are required.

**(4)**

**(Total for Question 2 is 7 marks)**

3.



**Figure 4**

Figure 4 represents a network with nodes, A, B, C, D, E, F, G, H and J. The number on each edge gives the length of the corresponding edge.

- (a) (i) Use Dijkstra's algorithm to find the shortest path from A to J.  
(ii) State the length of the shortest path from A to J.

(6)

One application of Dijkstra's algorithm has order  $n^2$ , where  $n$  is the number of nodes in the network.

It takes a computer 0.0312 seconds to find the shortest path from a given start node to a given end node in a network of 9 nodes.

- (b) Calculate approximately how long it would take, in minutes, for the computer to find the shortest path from a given start node to a given end node for a network of 9000 nodes.

(2)

**(Total for Question 3 is 8 marks)**

4. The eleven distinct numbers listed below are to be packed into bins of size 40

15      22      3      9      23       $x$       5      4      18      20      13

It is known that  $x$

- is an integer less than 40
- is the largest number in the list

(a) Explain why it **is not** possible to pack the numbers into 3 bins of size 40 (1)

Given that it **is** possible to pack the numbers into 4 bins of size 40

(b) determine the range of values for  $x$  (2)

(c) Use the first-fit bin packing algorithm to determine how the numbers can be packed into bins of size 40 (3)

(d) Carry out a quick sort to produce a list of the numbers in descending order. You should show the result of each pass and identify your pivots clearly. (4)

When the first-fit decreasing bin packing algorithm is applied to the list, neither the 15 nor the 13 is placed in the first bin.

(e) Determine the value of  $x$ . You must give reasons for your answer. (2)

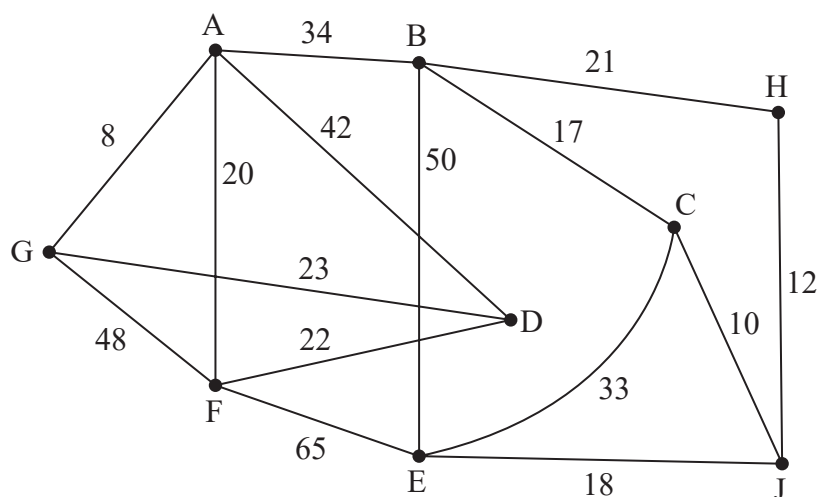
(Total for Question 4 is 12 marks)



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5.



**Figure 5**

[The total weight of the network is 423]

Direct roads between nine towns, A, B, C, D, E, F, G, H and J, are represented in Figure 5. The number on each arc represents the length, in miles, of the corresponding road.

The table below shows the shortest distances, in miles, between the nine towns.

|   | A  | B  | C  | D  | E  | F  | G  | H  | J  |
|---|----|----|----|----|----|----|----|----|----|
| A | –  | 34 | 51 | 31 | 79 | 20 | 8  | 55 | 61 |
| B | 34 | –  | 17 | 65 | 45 | 54 | 42 | 21 | 27 |
| C | 51 | 17 | –  | 82 | 28 | 71 | 59 | 22 | 10 |
| D | 31 | 65 | 82 | –  | 87 | 22 | 23 | 86 | 92 |
| E | 79 | 45 | 28 | 87 | –  | 65 | 87 | 30 | 18 |
| F | 20 | 54 | 71 | 22 | 65 | –  | 28 | 75 | 81 |
| G | 8  | 42 | 59 | 23 | 87 | 28 | –  | 63 | 69 |
| H | 55 | 21 | 22 | 86 | 30 | 75 | 63 | –  | 12 |
| J | 61 | 27 | 10 | 92 | 18 | 81 | 69 | 12 | –  |

**Table of shortest distances**

A route is needed that minimises the total distance required to traverse each road at least once.

The route must start at F and finish at J.

- (a) (i) By considering the pairings of all relevant nodes, find the roads that would need to be traversed twice.  
(ii) State the total length of this route. (5)
- (b) Starting at A, use Prim's algorithm to find the minimum spanning tree for the table of shortest distances. You must state the order in which you select the arcs of your tree. (3)

Pete needs to visit all nine towns, starting and finishing in the same town, and wishes to minimise the total distance he travels.

- (c) Starting at G, use the nearest neighbour algorithm on the table of shortest distances to find an upper bound for the length of Pete's route. Write down the route that gives this upper bound. (2)
- (d) By deleting G and all of its arcs, find a lower bound for the length of Pete's route. (2)

Pete decides to take the route he found in (c).

- (e) Interpret the route in terms of the actual towns visited. (1)

---

(Total for Question 5 is 13 marks)

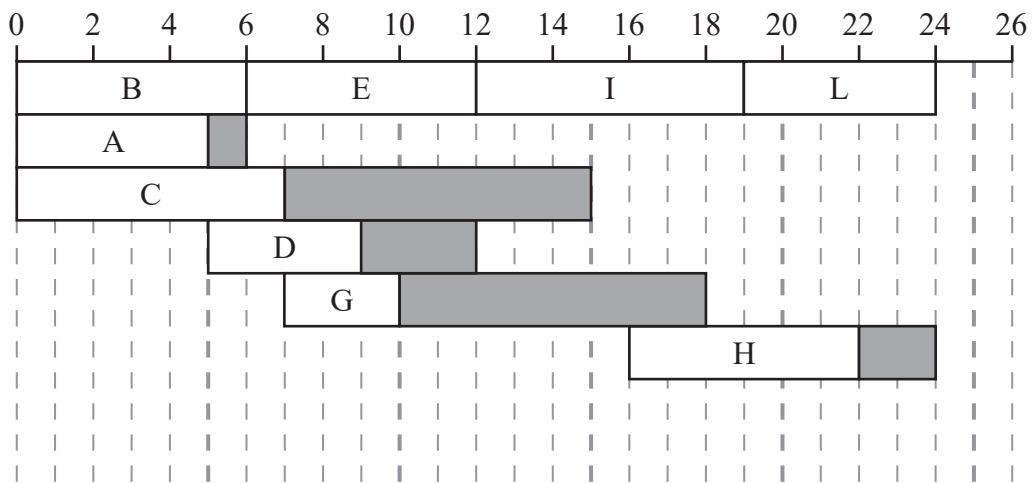
6. The precedence table below shows the twelve activities required to complete a project.

| Activity | Immediately preceding activities |
|----------|----------------------------------|
| A        | —                                |
| B        | —                                |
| C        | —                                |
| D        | A                                |
| E        | A, B                             |
| F        | D, E                             |
| G        | A, B, C                          |
| H        | F, G                             |
| I        | D, E                             |
| J        | D, E                             |
| K        | F, G, I, J                       |
| L        | I                                |

- (a) Draw the activity network described in the precedence table, using activity on arc. Your activity network must contain the minimum number of dummies only.

(5)





**Figure 6**

Figure 6 shows a partially completed cascade chart for the project. The non-critical activities F, J and K are **not** shown in Figure 6.

The time taken to complete each activity is given in hours and the project is to be completed in the minimum possible time.

(b) State the critical activities.

(1)

Given that the total float of activity F is 2 hours,

(c) state the duration of activity F.

(1)

The duration of activity J is  $x$  hours, and the duration of activity K is  $y$  hours, where  $x > 0$  and  $y > 0$

(d) (i) State, in terms of  $y$ , the maximum possible total float for activity K.

(ii) State, in terms of  $x$  and  $y$ , the total float for activity J.

(2)

**(Total for Question 6 is 9 marks)**

7. A publisher plans to produce three versions of the same book: a paperback, a hardcover, and a deluxe edition.

- Each paperback takes 4 minutes to print and 1 minute to bind
- Each hardcover takes 8 minutes to print and 5 minutes to bind
- Each deluxe edition takes 15 minutes to print and 12 minutes to bind

The printing machine is available for at most 150 hours and the binding machine must be used for at least 60 hours.

The publisher decides to produce

- at least 1600 books in total
- at least three times as many paperbacks as hardcovers

The profit on each paperback sold is £8, the profit on each hardcover sold is £20 and the profit on each deluxe edition sold is £40

Let  $x$ ,  $y$  and  $z$  represent the number of paperbacks, hardcovers and deluxe editions produced.

- (a) Formulate this as a linear programming problem, stating the objective and listing the constraints as simplified inequalities with integer coefficients.

(5)

The publisher decides to solve this linear programming problem by using the two-stage simplex method.

- (b) Set up an initial tableau for solving this problem using the two-stage simplex method.

As part of your solution, you must show how

- the constraints have been made into equations by using slack variables, exactly two surplus variables and exactly two artificial variables
- the rows for the two objective functions are formed

(6)



The following tableau is obtained after two iterations of the first stage of the two-stage simplex method.

| b.v.  | $x$ | $y$             | $z$ | $s_1$ | $s_2$            | $s_3$            | $s_4$ | $a_1$           | $a_2$           | Value               |
|-------|-----|-----------------|-----|-------|------------------|------------------|-------|-----------------|-----------------|---------------------|
| $s_1$ | 0   | 0               | 0   | 1     | 1                | 3                | 0     | -1              | -3              | 600                 |
| $z$   | 0   | $\frac{4}{11}$  | 1   | 0     | $-\frac{1}{11}$  | $\frac{1}{11}$   | 0     | $\frac{1}{11}$  | $-\frac{1}{11}$ | $\frac{2000}{11}$   |
| $x$   | 1   | $\frac{7}{11}$  | 0   | 0     | $\frac{1}{11}$   | $-\frac{12}{11}$ | 0     | $-\frac{1}{11}$ | $\frac{12}{11}$ | $\frac{15600}{11}$  |
| $s_4$ | 0   | $\frac{40}{11}$ | 0   | 0     | $\frac{1}{11}$   | $-\frac{12}{11}$ | 1     | $-\frac{1}{11}$ | $\frac{12}{11}$ | $\frac{15600}{11}$  |
| $P$   | 0   | $-\frac{4}{11}$ | 0   | 0     | $-\frac{32}{11}$ | $-\frac{56}{11}$ | 0     | $\frac{32}{11}$ | $\frac{56}{11}$ | $\frac{204800}{11}$ |
| $I$   | 0   | 0               | 0   | 0     | 0                | 0                | 0     | 1               | 1               | 0                   |

- (c) Taking the most negative number in the profit row to indicate the pivot column, perform one complete iteration of the second stage of the two-stage simplex method to obtain a new tableau. Make your method clear by stating the row operations you use.

(5)

After three iterations of the second stage of the two-stage simplex method, the following tableau is obtained.

| b.v.  | $x$ | $y$ | $z$ | $s_1$           | $s_2$ | $s_3$          | $s_4$           | Value |
|-------|-----|-----|-----|-----------------|-------|----------------|-----------------|-------|
| $s_2$ | 0   | 0   | 0   | 1               | 1     | 3              | 0               | 600   |
| $z$   | 0   | 0   | 1   | $\frac{1}{10}$  | 0     | $\frac{1}{2}$  | $-\frac{1}{10}$ | 100   |
| $x$   | 1   | 0   | 0   | $-\frac{3}{40}$ | 0     | $-\frac{9}{8}$ | $-\frac{7}{40}$ | 1125  |
| $y$   | 0   | 1   | 0   | $-\frac{1}{40}$ | 0     | $-\frac{3}{8}$ | $\frac{11}{40}$ | 375   |
| $P$   | 0   | 0   | 0   | $\frac{29}{10}$ | 0     | $\frac{7}{2}$  | $\frac{1}{10}$  | 20500 |

Given that the publisher produces the optimal number of each version of the book,

(d) (i) state the maximum profit the publisher can earn,

(ii) find the number of hours the binding machine will be used.

(2)

(e) Give a reason why the publisher may not earn the profit stated in (d)(i).

(1)

(Total for Question 7 is 19 marks)

**TOTAL FOR PAPER IS 75 MARKS**



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# Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper  
reference

**9FM0/3D**

## Further Mathematics

Advanced

**PAPER 3D: Decision Mathematics 1**

### You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,  
Decision Mathematics Answer Book (enclosed)

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- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
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### Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. A gardener needs the following lengths of string. All lengths are in metres.

4.3    6.1    5.1    4.7    2.5    5.9    3.4    1.7    2.1    0.4    1.3

She cuts the lengths from balls of string. Each ball contains 10 m of string.

- (a) Calculate a lower bound for the number of balls of string the gardener needs.  
You must make your method clear.

(2)

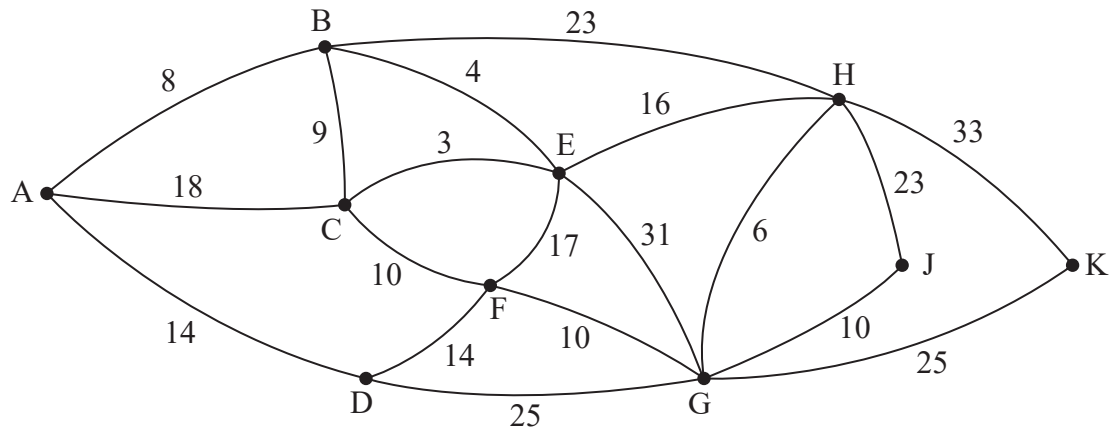
- (b) Use the first-fit bin packing algorithm to determine how the lengths could be cut  
from the balls of string.

(3)

**(Total for Question 1 is 5 marks)**

---

2.



**Figure 1**

[The total weight of the network is 299]

Figure 1 represents a network of cycle tracks between 10 landmarks, A, B, C, D, E, F, G, H, J and K. The number on each edge represents the length, in kilometres, of the corresponding track.

One day, Blanche wishes to cycle from A to K. She wishes to minimise the distance she travels.

(a) (i) Use Dijkstra's algorithm to find the shortest path from A to K.

(ii) State the length of the shortest path from A to K.

(6)

The cycle tracks between the landmarks now need to be inspected. Blanche must travel along each track at least once. She wishes to minimise the length of her inspection route. Blanche will start her inspection route at D and finish at E.

(b) (i) State the edges that will need to be traversed twice.

(ii) Find the length of Blanche's route.

(2)

It is now decided to start the inspection route at A and finish at K. Blanche must minimise the length of her route and travel along each track at least once.

(c) By considering the pairings of all relevant nodes, find the length of Blanche's new route. You must make your method and working clear.

(5)

(Total for Question 2 is 13 marks)

3. The initial distance matrix (Table 1) shows the lengths, in metres, of the corridors connecting six classrooms, A, B, C, D, E and F, in a school. For safety reasons, some of the corridors are one-way only.

|   | A        | B        | C        | D  | E        | F        |
|---|----------|----------|----------|----|----------|----------|
| A | –        | 12       | 32       | 24 | 29       | 11       |
| B | 12       | –        | 17       | 8  | $\infty$ | $\infty$ |
| C | 32       | 17       | –        | 4  | 12       | $\infty$ |
| D | 24       | $\infty$ | 4        | –  | $\infty$ | 13       |
| E | $\infty$ | $\infty$ | 12       | 18 | –        | 12       |
| F | 11       | $\infty$ | $\infty$ | 13 | 12       | –        |

**Table 1**

- (a) By adding the arcs from vertex A, along with their weights, complete the drawing of this network on Diagram 1 in the answer book.

(2)

Floyd's algorithm is to be used to find the complete network of shortest distances between the six classrooms.

The distance matrix after **two** iterations of Floyd's algorithm is shown in Table 2.

|   | A        | B        | C  | D  | E  | F  |
|---|----------|----------|----|----|----|----|
| A | –        | 12       | 29 | 20 | 29 | 11 |
| B | 12       | –        | 17 | 8  | 41 | 23 |
| C | 29       | 17       | –  | 4  | 12 | 40 |
| D | 24       | 36       | 4  | –  | 53 | 13 |
| E | $\infty$ | $\infty$ | 12 | 18 | –  | 12 |
| F | 11       | 23       | 40 | 13 | 12 | –  |

**Table 2**

- (b) Perform the next two iterations of Floyd's algorithm that follow from Table 2. You should show the distance matrix after each iteration.

(4)

The final distance matrix after completion of Floyd's algorithm is shown in Table 3.

|   | A  | B  | C  | D  | E  | F  |
|---|----|----|----|----|----|----|
| A | –  | 12 | 24 | 20 | 23 | 11 |
| B | 12 | –  | 12 | 8  | 24 | 21 |
| C | 28 | 17 | –  | 4  | 12 | 17 |
| D | 24 | 21 | 4  | –  | 16 | 13 |
| E | 23 | 29 | 12 | 16 | –  | 12 |
| F | 11 | 23 | 17 | 13 | 12 | –  |

**Table 3**

Yinka must visit each classroom. He will start and finish at E and wishes to minimise the total distance travelled.

- (c) (i) Use the nearest neighbour algorithm, starting at E, to find two Hamiltonian cycles in the completed network of shortest distances.
- (ii) Find the length of each of the two cycles.
- (iii) State, with a reason, which of the two cycles provides the better upper bound for the length of Yinka's route.

**(4)**

**(Total for Question 3 is 10 marks)**

4. A linear programming problem in  $x$ ,  $y$  and  $z$  is to be solved using the big-M method. The initial tableau is shown below.

| b.v.  | $x$      | $y$       | $z$       | $s_1$ | $s_2$ | $s_3$ | $a_1$ | $a_2$ | Value  |
|-------|----------|-----------|-----------|-------|-------|-------|-------|-------|--------|
| $s_1$ | 2        | 3         | 4         | 1     | 0     | 0     | 0     | 0     | 13     |
| $a_1$ | 1        | -2        | 2         | 0     | -1    | 0     | 1     | 0     | 8      |
| $a_2$ | 3        | 0         | -4        | 0     | 0     | -1    | 0     | 1     | 12     |
| $P$   | $2 - 4M$ | $-3 + 2M$ | $-1 + 2M$ | 0     | $M$   | $M$   | 0     | 0     | $-20M$ |

- (a) Using the information in the above tableau, formulate the linear programming problem. You should

- list each of the constraints as an inequality
- state the two possible objectives

(4)

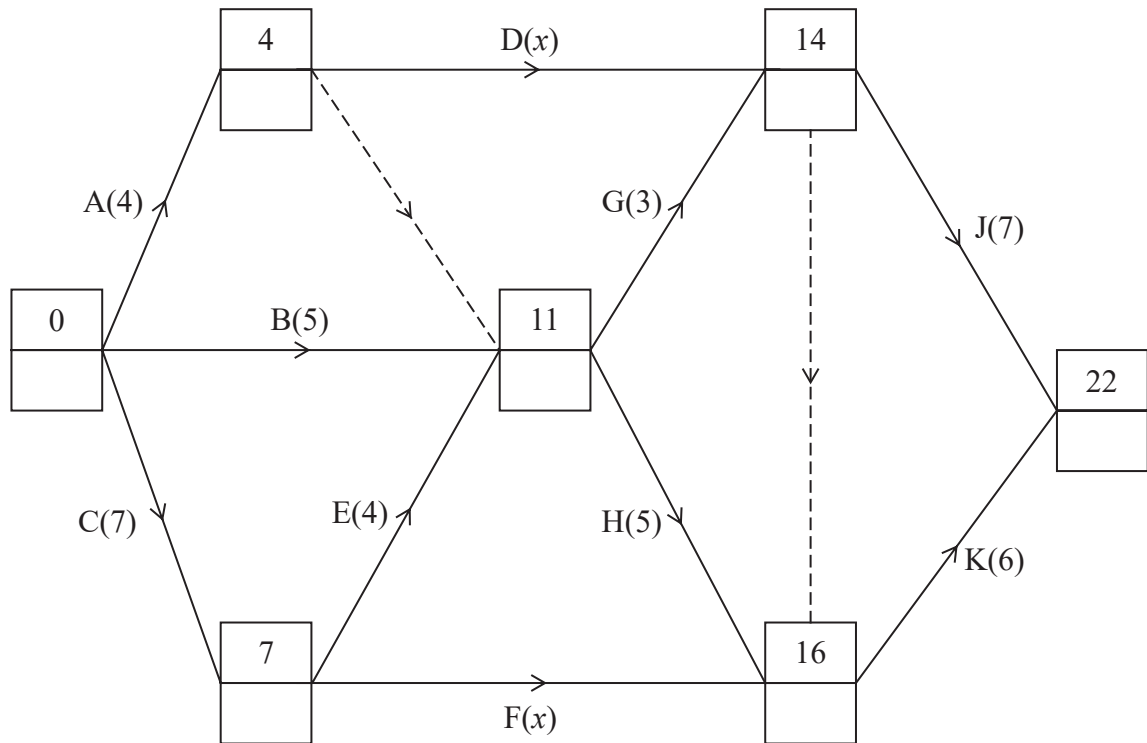
- (b) Obtain the most efficient pivot for a first iteration of the big-M method. You must give reasons for your answer.

(2)

(Total for Question 4 is 6 marks)

**Please turn over for Question 5**

5.



**Key:**

|                  |
|------------------|
| Early event time |
| Late event time  |

**Figure 2**

The network in Figure 2 shows the activities that need to be completed for a project. Each activity is represented by an arc and the duration of the activity, in days, is shown in brackets. The early event times are shown in Figure 2.

- (a) Complete Table 1 in the answer book to show the immediately preceding activities for each activity. (2)

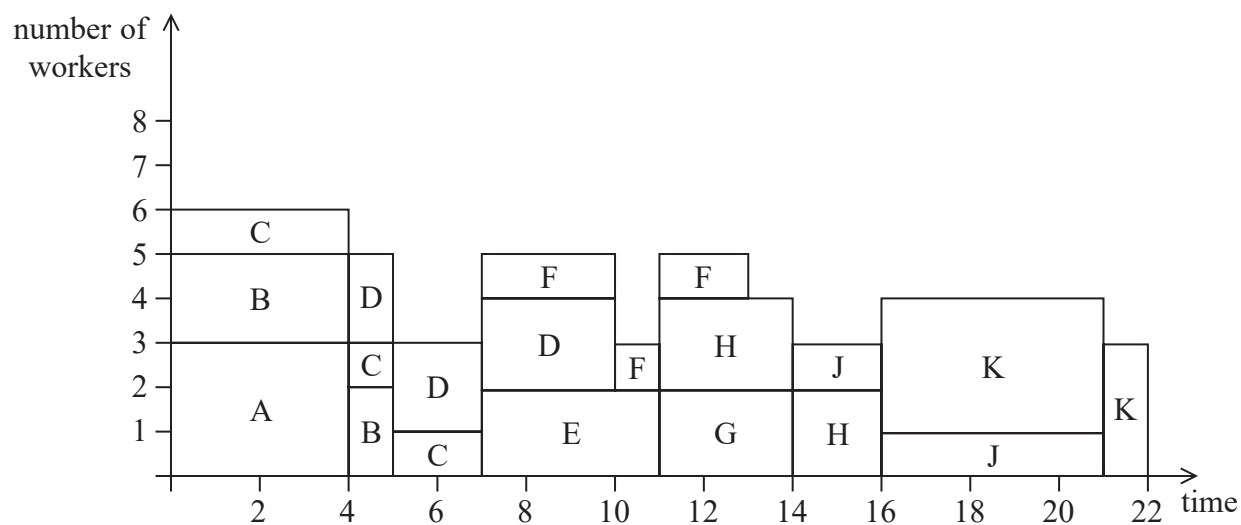
It is given that  $4 < x \leq m$

- (b) State the largest possible integer value of  $m$ . (1)

- (c) (i) Complete Diagram 1 in the answer book to show the late event times.  
 (ii) State the activities that must be critical. (3)

- (d) Calculate the total float for activity G. (1)

The resource histogram in Figure 3 shows the number of workers required when each activity starts at its earliest possible time. The histogram also shows which activities happen at each time.



**Figure 3**

- (e) Complete Table 2 in the answer book to show the number of workers required for each activity of the project. (2)
- (f) Draw a Gantt chart on Grid 1 in the answer book to represent the activity network. (5)

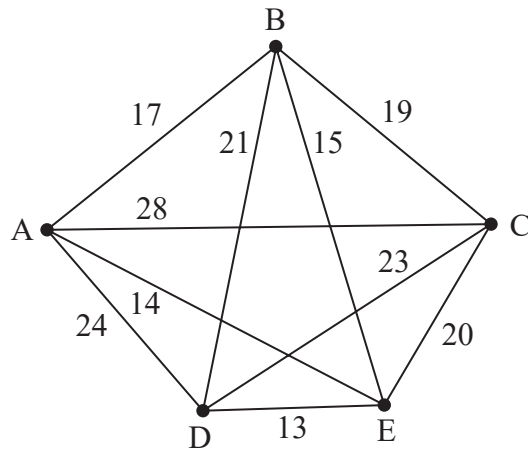
**(Total for Question 5 is 14 marks)**

6. The following algorithm determines the number of comparisons made when Prim's algorithm is applied to  $K_n$

|         |                                      |
|---------|--------------------------------------|
| Step 1  | Start                                |
| Step 2  | Input the value of $n$               |
| Step 3  | Let $a = 1$                          |
| Step 4  | Let $b = n - 2$                      |
| Step 5  | Let $c = b$                          |
| Step 6  | Let $a = a + 1$                      |
| Step 7  | Let $b = b - 1$                      |
| Step 8  | Let $c = c + (a \times b) + (a - 1)$ |
| Step 9  | If $b > 0$ go to Step 6              |
| Step 10 | Output $c$                           |
| Step 11 | Stop                                 |

- (a) For  $K_5$ , complete the table in the answer book to show the results obtained at each step of the algorithm.

(3)



**Figure 4**

The weights of the ten arcs in Figure 4 are

17    21    24    14    23    13    15    19    28    20

- (b) (i) Starting at the left-hand end of the above list, sort the list into ascending order using bubble sort. You need only write down the state of the list at the end of each pass.
- (ii) Find the total number of comparisons performed during the sort. (5)
- (c) Find the maximum total number of comparisons required to sort the weights of the 10 arcs of  $K_5$  into ascending order using bubble sort. (1)

It is given that the maximum total number of comparisons required to sort the weights of the arcs of  $K_n$  into ascending order using bubble sort is

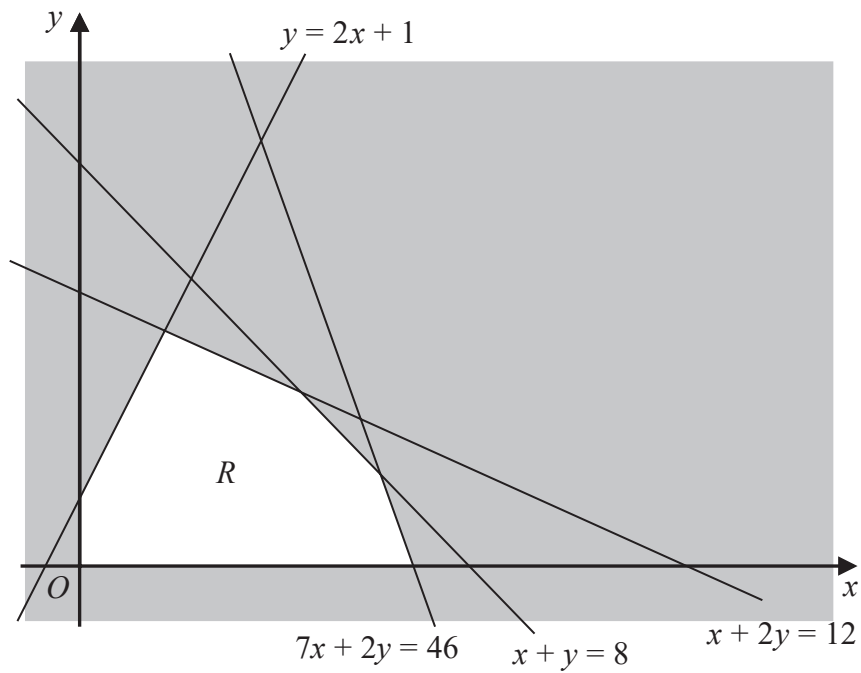
$$\lambda n(n-1)(n+1)(n-2)$$

where  $\lambda$  is a constant.

- (d) Determine the maximum total number of comparisons required to sort the weights of the arcs of  $K_{50}$  into ascending order using bubble sort. You must make your method and working clear. (3)

(Total for Question 6 is 12 marks)

7.



**Figure 5**

Figure 5 shows the constraints of a linear programming problem in  $x$  and  $y$  where  $R$  is the feasible region.

The objective is to maximise  $P = x + ky$ , where  $k$  is a positive constant.

The optimal vertex of  $R$  is to be found using the Simplex algorithm.

- (a) Set up an initial tableau for solving this linear programming problem using the Simplex algorithm.

(5)

After two iterations of the Simplex algorithm a possible tableau  $T$  is

| b.v.  | $x$ | $y$ | $s_1$ | $s_2$                        | $s_3$ | $s_4$                         | Value    |
|-------|-----|-----|-------|------------------------------|-------|-------------------------------|----------|
| $s_1$ | 0   | 0   | 1     | $-\frac{3}{5}$               | 0     | $\frac{1}{5}$                 | 1        |
| $x$   | 1   | 0   | 0     | $\frac{1}{5}$                | 0     | $-\frac{2}{5}$                | 2        |
| $s_3$ | 0   | 0   | 0     | $-\frac{11}{5}$              | 1     | $\frac{12}{5}$                | 22       |
| $y$   | 0   | 1   | 0     | $\frac{2}{5}$                | 0     | $\frac{1}{5}$                 | 5        |
| $P$   | 0   | 0   | 0     | $\frac{1}{5} + \frac{2}{5}k$ | 0     | $-\frac{2}{5} + \frac{1}{5}k$ | $5k + 2$ |

(b) State the value of each variable after the second iteration.

(1)

It is given that  $T$  does not give an optimal solution to the linear programming problem.

After a third iteration of the Simplex algorithm the resulting tableau does give an optimal solution to the problem.

(c) Perform the third iteration of the Simplex algorithm and hence determine the range of possible values for  $P$ . You should state the row operations you use and make your method and working clear.

(9)

(Total for Question 7 is 15 marks)

**TOTAL FOR PAPER IS 75 MARKS**

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# Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper  
reference

9FM0/3D

## Further Mathematics

Advanced

PAPER 3D: Decision Mathematics 1

### You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,  
Decision Mathematics Answer Book (enclosed)

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

### Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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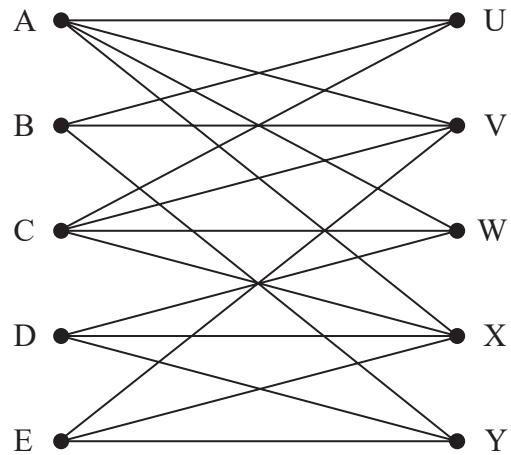
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1.



**Figure 1**

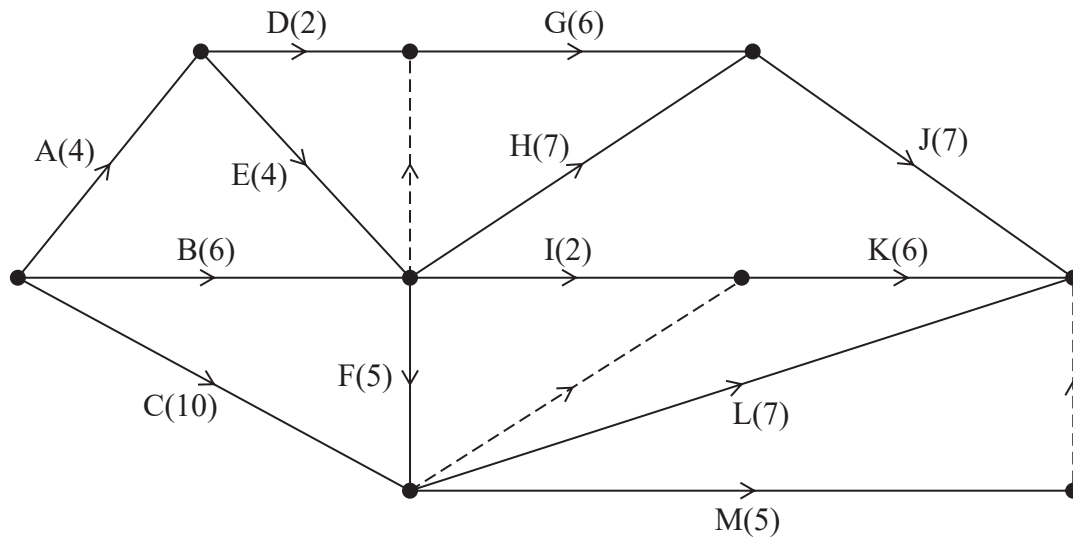
A Hamiltonian cycle for the graph in Figure 1 begins C, V, E, X, A, W, ....

- (a) Complete the Hamiltonian cycle. (1)
- (b) Hence use the planarity algorithm to determine whether the graph shown in Figure 1 is planar. You must make your working clear and justify your answer. (3)

(Total for Question 1 is 4 marks)

---

2.



**Figure 2**

A project is modelled by the activity network shown in Figure 2. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the corresponding activity.

- (a) Complete Diagram 1 in the answer book to show the early event times and the late event times. (4)

Each activity requires one worker and the project must be completed in the shortest possible time using as few workers as possible.

- (b) Calculate a lower bound for the number of workers needed to complete the project in the shortest possible time. You must show your working. (2)

- (c) Schedule the activities using Grid 1 in the answer book. (3)

**(Total for Question 2 is 9 marks)**

3.

|   | A  | B  | C  | D  | E  | F  | G  | H  |
|---|----|----|----|----|----|----|----|----|
| A | –  | 24 | 42 | 48 | 34 | 37 | 32 | 22 |
| B | 24 | –  | 40 | 35 | 30 | 41 | 39 | 44 |
| C | 42 | 40 | –  | 21 | 26 | 45 | 38 | 36 |
| D | 48 | 35 | 21 | –  | 32 | 37 | 29 | 27 |
| E | 34 | 30 | 26 | 32 | –  | 34 | 40 | 28 |
| F | 37 | 41 | 45 | 37 | 34 | –  | 43 | 41 |
| G | 32 | 39 | 38 | 29 | 40 | 43 | –  | 38 |
| H | 22 | 44 | 36 | 27 | 28 | 41 | 38 | –  |

**Table 1**

Table 1 shows the shortest distances, in miles, between eight towns, A, B, C, D, E, F, G and H.

- (a) Use Prim's algorithm, starting at A, to find the minimum spanning tree for this table of distances. You must clearly state the order in which you select the edges of your tree. (3)

- (b) State the weight of the minimum spanning tree. (1)

|   | A  | B  | C  | D  | E  | F  | G  | H  |
|---|----|----|----|----|----|----|----|----|
| J | 31 | 27 | 50 | 29 | 43 | 25 | 49 | 35 |

**Table 2**

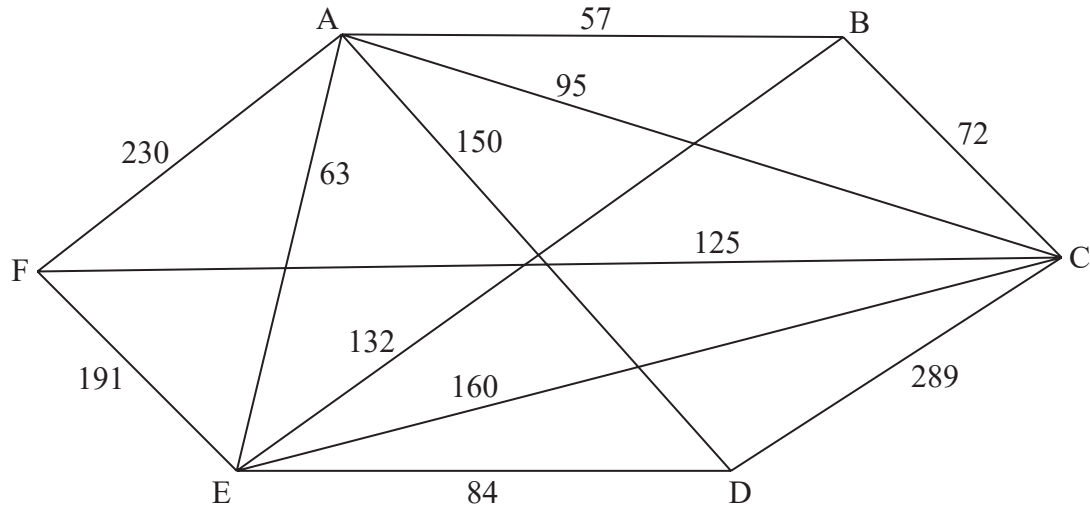
Table 2 shows the distances, in miles, between town J and towns A, B, C, D, E, F, G and H.

Pranav needs to visit all of the towns, starting and finishing at J, and wishes to minimise the total distance he travels.

- (c) Starting at J, use the nearest neighbour algorithm to obtain an upper bound for the length of Pranav's route. You must state your route and its length. (2)
- (d) Starting by deleting J, and all of its edges, find a lower bound for the length of Pranav's route. (2)

**(Total for Question 3 is 8 marks)**

4.



**Figure 3**

[The total weight of the network is 1648]

Direct roads between six cities, A, B, C, D, E and F, are represented in Figure 3. The weight on each arc is the time, in minutes, required to travel along the corresponding road.

Floyd's algorithm is to be used to find the complete network of shortest times between the six cities.

An initial route matrix is given in the answer book.

- (a) Set up the initial time matrix. (1)
- (b) Perform the first iteration of Floyd's algorithm. You should show the time and route matrices after this iteration. (2)

The final time matrix after completion of Floyd's algorithm is shown below.

|   | A   | B   | C   | D   | E   | F   |
|---|-----|-----|-----|-----|-----|-----|
| A | –   | 57  | 95  | 147 | 63  | 220 |
| B | 57  | –   | 72  | 204 | 120 | 197 |
| C | 95  | 72  | –   | 242 | 158 | 125 |
| D | 147 | 204 | 242 | –   | 84  | 275 |
| E | 63  | 120 | 158 | 84  | –   | 191 |
| F | 220 | 197 | 125 | 275 | 191 | –   |

A route is needed that minimises the total time taken to traverse each road at least once.

The route must start at B and finish at E.

- (c) Use an appropriate algorithm to find the roads that will need to be traversed twice. You should make your method and working clear. (4)
- (d) Write down the length of the route. (1)

(Total for Question 4 is 8 marks)

5.                    30      12      5      2      23      18      36      10      15      24

- (a) The list of ten numbers above is to be sorted into descending order. Use a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

The ten numbers are to be packed into bins of size  $n$ , where  $n$  is a positive integer.

When the first-fit bin packing algorithm is applied to the original list of ten numbers, the following allocation is obtained.

Bin 1:      30      12      2

Bin 2:      5      23      10

Bin 3:      18      15

Bin 4:      36

Bin 5:      24

- (b) Explain why the value of the integer  $n$  must be either 44 or 45

(3)

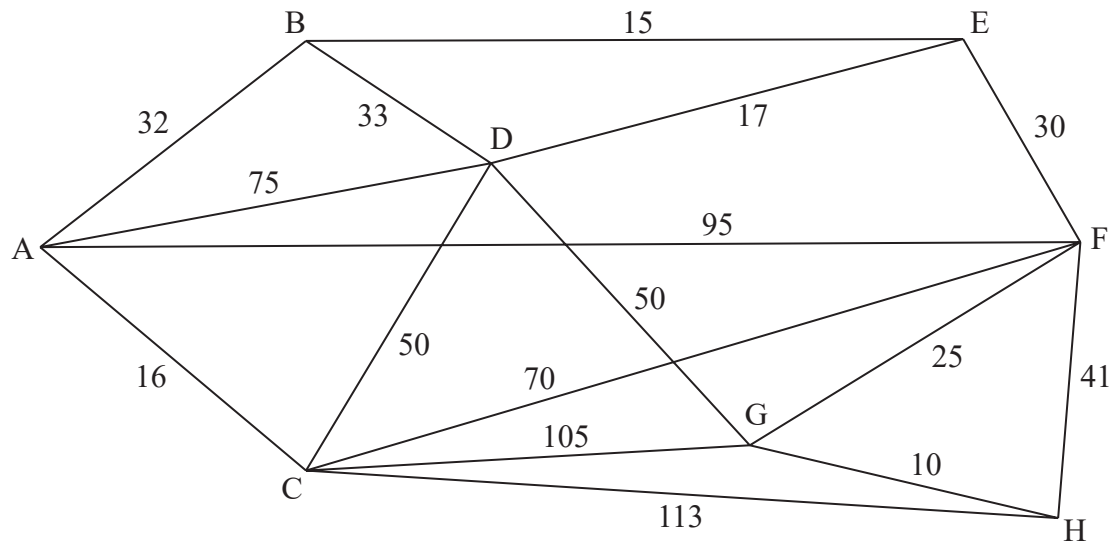
- (c) Use the first-fit decreasing bin packing algorithm to determine how the numbers can be packed into bins of size 45

(3)

---

(Total for Question 5 is 10 marks)

6.



**Figure 4**

In Figure 4 the weights on the arcs represent distances.

(a) (i) Use Dijkstra's algorithm to find the shortest path from A to H.

(ii) State the length of the shortest path from A to H.

**(6)**

One application of Dijkstra's algorithm has order  $n^2$ , where  $n$  is the number of nodes in the network. A computer produces a table of shortest distances between any two different nodes by repeatedly applying Dijkstra's algorithm from each node of the network.

It takes the computer 0.082 seconds to produce a table of shortest distances for a network of 10 nodes.

(b) Calculate approximately how long it will take, in seconds, for the computer to produce a table of shortest distances for a network with 200 nodes. You must give a reason for your answer.

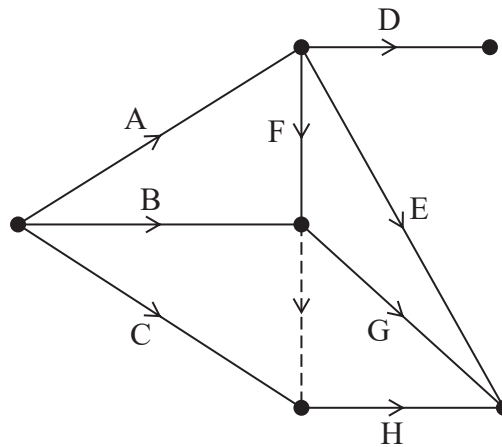
**(3)**

(c) Explain why your answer to part (b) can only be an approximation.

**(1)**

**(Total for Question 6 is 10 marks)**

7.



**Figure 5**

Figure 5 shows a partially completed activity network for a project that consists of 14 activities.

- (a) Complete the precedence table in the answer book for the 8 activities in Figure 5. (2)

The precedence table for the remaining 6 activities is given below.

| Activity | Immediately preceding activities |
|----------|----------------------------------|
| I        | D, E, G, H                       |
| J        | D, E, G, H                       |
| K        | E, G, H                          |
| L        | I, J, K                          |
| M        | J, K                             |
| N        | J, K                             |

- (b) Complete the activity network in the answer book for the project. Your completed activity network must contain only the minimum number of dummies. (4)

Given that all 14 activities have the same duration,

- (c) explain why activity D cannot be critical. (2)

(Total for Question 7 is 8 marks)

8. Susie is preparing for a triathlon event that is taking place next month. A triathlon involves three activities: swimming, cycling and running.

Susie decides that in her training next week she should

- maximise the total time spent cycling and running
- train for at most 39 hours
- spend at least 40% of her time swimming
- spend a total of at least 28 hours of her time swimming and running

Susie needs to determine how long she should spend next week training for each activity. Let

- $x$  represent the number of hours swimming
- $y$  represent the number of hours cycling
- $z$  represent the number of hours running

- (a) Formulate the information above as a linear programming problem. State the objective and list the constraints as simplified inequalities with integer coefficients.

(5)

Susie decides to solve this linear programming problem by using the two-stage Simplex method.

- (b) Set up an initial tableau for solving this problem using the two-stage Simplex method.

As part of your solution you must show how

- the constraints have been made into equations using slack variables, exactly one surplus variable and exactly one artificial variable
- the rows for the two objective functions are formed

(6)

The following tableau  $T$  is obtained after one iteration of the second stage of the two-stage Simplex method.

| b.v.  | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | Value |
|-------|-----|-----|-----|-------|-------|-------|-------|
| $y$   | 0   | 1   | 0   | 1     | 0     | 1     | 11    |
| $s_2$ | 0   | 0   | 5   | -2    | 1     | -5    | 62    |
| $x$   | 1   | 0   | 1   | 0     | 0     | -1    | 28    |
| $P$   | 0   | 0   | -1  | 1     | 0     | 1     | 11    |

- (c) Obtain a suitable pivot for a second iteration. You must give reasons for your answer.

(2)

- (d) Starting from tableau  $T$ , solve the linear programming problem by performing one further iteration of the second stage of the two-stage Simplex method. You should make your method clear by stating the row operations you use.

(5)

(Total for Question 8 is 18 marks)

**TOTAL FOR PAPER IS 75 MARKS**

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**Pearson Edexcel Level 3 GCE**

**Wednesday 21 October 2020**

Afternoon (Time: 1 hour 30 minutes)

Paper Reference **9FM0/3D**

## **Further Mathematics**

**Advanced**

**Paper 3D: Decision Mathematics 1**

### **You must have:**

Mathematical Formulae and Statistical Tables (Green), calculator,  
Decision Mathematics Answer Book (enclosed)

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### **Instructions**

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

### **Information**

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### **Advice**

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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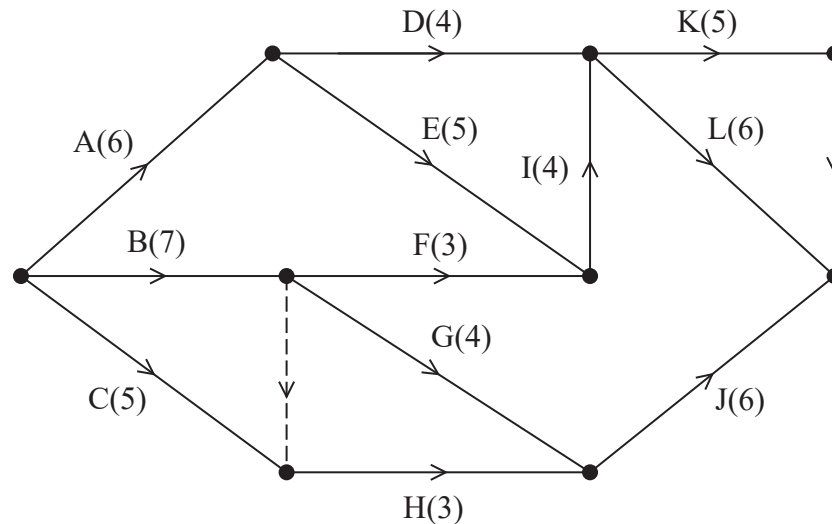
1. The table below shows the lengths, in km, of the roads in a network connecting seven towns, A, B, C, D, E, F and G.

|   | A  | B  | C  | D  | E  | F  | G  |
|---|----|----|----|----|----|----|----|
| A | –  | 24 | –  | 22 | 35 | –  | –  |
| B | 24 | –  | 25 | 27 | –  | –  | –  |
| C | –  | 25 | –  | 33 | 31 | 36 | 26 |
| D | 22 | 27 | 33 | –  | –  | 42 | –  |
| E | 35 | –  | 31 | –  | –  | 37 | 29 |
| F | –  | –  | 36 | 42 | 37 | –  | 40 |
| G | –  | –  | 26 | –  | 29 | 40 | –  |

- (a) By adding the arcs from vertex D along with their weights, complete the drawing of the network on Diagram 1 in the answer book. (2)
- (b) Use Kruskal's algorithm to find a minimum spanning tree for the network. You should list the arcs in the order that you consider them. In each case, state whether you are adding the arc to your minimum spanning tree. (3)
- (c) State the weight of the minimum spanning tree. (1)

**(Total for Question 1 is 6 marks)**

2.



**Figure 1**

The network in Figure 1 shows the activities that need to be undertaken to complete a project. Each activity is represented by an arc and the duration, in hours, of the corresponding activity is shown in brackets.

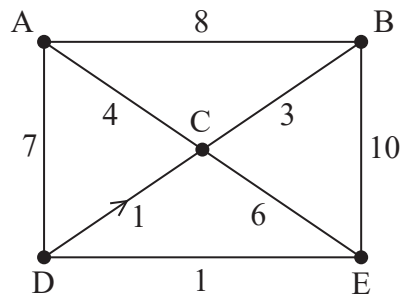
- Explain why each of the dummy activities is required. (2)
- Complete the table in the answer book to show the immediately preceding activities for each activity. (2)
- Complete Diagram 1 in the answer book to show the early event times and the late event times.
  - State the minimum completion time for the project.
  - State the critical activities. (6)

Each activity requires one worker. Each worker is able to do any of the activities. Once an activity is started it must be completed without interruption.

- On Grid 1 in the answer book, draw a resource histogram to show the number of workers required at each time when each activity begins at its earliest possible start time. (3)
- Determine whether or not the project can be completed in the minimum possible time using fewer workers than the number indicated by the resource histogram in (d). You must justify your answer with reference to the resource histogram and the completed Diagram 1. (2)

**(Total for Question 2 is 15 marks)**

3.



**Figure 2**

Direct roads between five villages, A, B, C, D and E, are shown in Figure 2. The weight on each arc is the time, in minutes, it takes to travel along the corresponding road. The road from D to C is one-way as indicated by the arrow on the corresponding arc.

Floyd's algorithm is to be used to find the complete network of shortest times between the five villages.

(a) Set up initial time and route matrices.

(2)

The matrices after two iterations of Floyd's algorithm are shown below.

Time matrix

|   | A  | B  | C | D  | E  |
|---|----|----|---|----|----|
| A | –  | 8  | 4 | 7  | 18 |
| B | 8  | –  | 3 | 15 | 10 |
| C | 4  | 3  | – | 11 | 6  |
| D | 7  | 15 | 1 | –  | 1  |
| E | 18 | 10 | 6 | 1  | –  |

Route matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A | A | B | C | D | B |
| B | A | B | C | A | E |
| C | A | B | C | A | E |
| D | A | A | C | D | E |
| E | B | B | C | D | E |

(b) Perform the next two iterations of Floyd's algorithm that follow from the tables above. You should show the time and route matrices after each iteration.

(4)

The final time matrix after completion of Floyd's algorithm is shown below.

Final time matrix

|   | A | B | C | D  | E |
|---|---|---|---|----|---|
| A | – | 7 | 4 | 7  | 8 |
| B | 7 | – | 3 | 10 | 9 |
| C | 4 | 3 | – | 7  | 6 |
| D | 5 | 4 | 1 | –  | 1 |
| E | 6 | 5 | 2 | 1  | – |

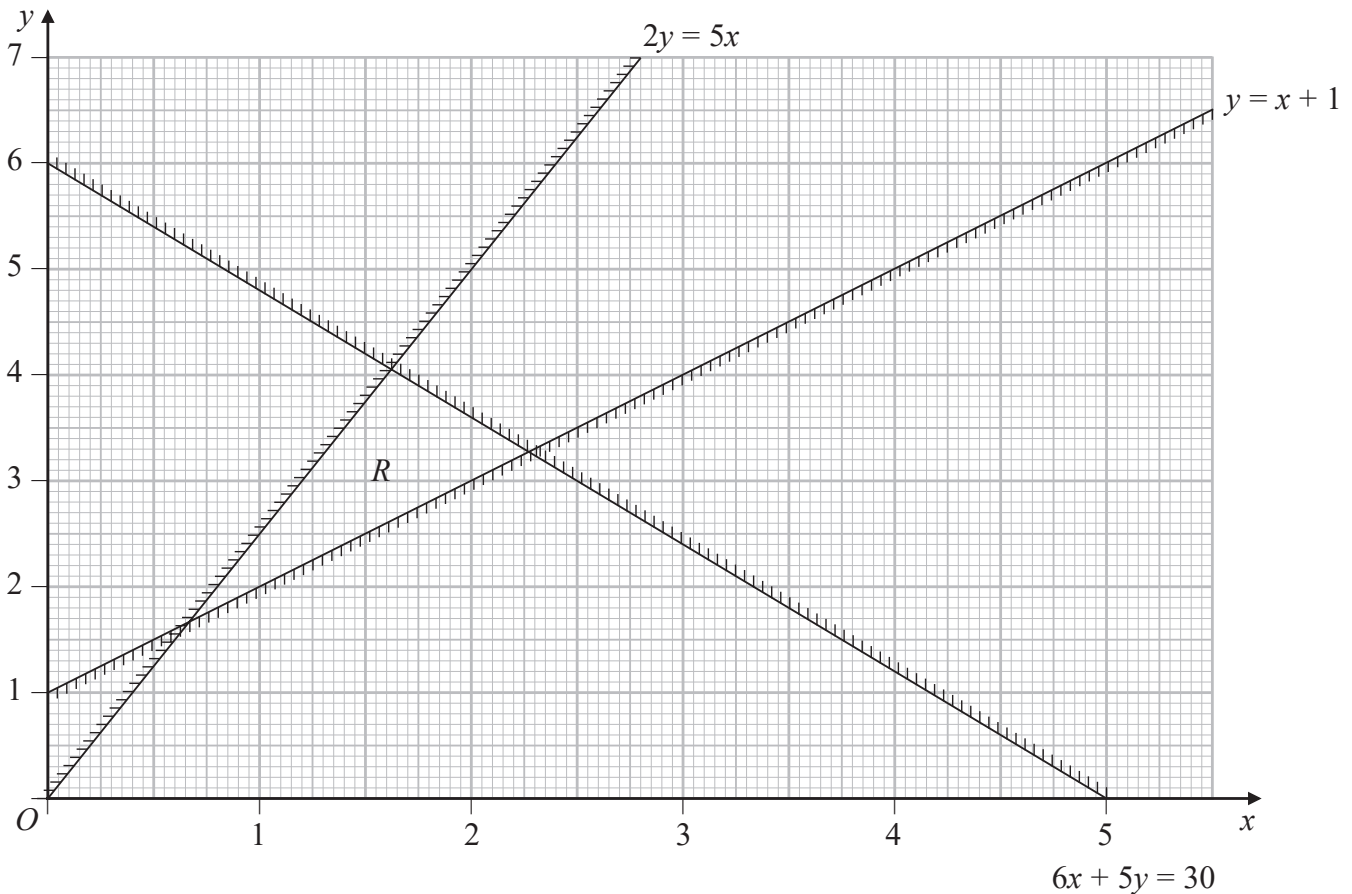
- (c) (i) Use the nearest neighbour algorithm, starting at A, to find a Hamiltonian cycle in the complete network of shortest times.
- (ii) Find the time taken for this cycle.
- (iii) Interpret the cycle in terms of the actual villages visited.

**(3)**

---

**(Total for Question 3 is 9 marks)**

4.



**Figure 3**

Figure 3 shows the constraints of a linear programming problem in  $x$  and  $y$ , where  $R$  is the feasible region.

- (a) Write down the inequalities that define  $R$ . (2)

The objective is to maximise  $P$ , where  $P = 3x + y$

- (b) Obtain the exact value of  $P$  at each of the three vertices of  $R$  and hence find the optimal vertex,  $V$ . (4)

The objective is changed to maximise  $Q$ , where  $Q = 3x + ay$ . Given that  $a$  is a constant and the optimal vertex is still  $V$ ,

- (c) find the range of possible values of  $a$ . (4)

(Total for Question 4 is 10 marks)

5. The nine distinct numbers in the following list are to be packed into bins of size 50

23    17    19     $x$     24    8    18    10    21

When the first-fit bin packing algorithm is applied to the numbers in the list it results in the following allocation.

Bin 1: 23    17    8

Bin 2: 19     $x$     10

Bin 3: 24    18

Bin 4: 21

- (a) Explain why  $13 < x < 21$

(3)

The same list of numbers is to be sorted into descending order. A bubble sort, starting at the left-hand end of the list, is to be used to obtain the sorted list. After the first complete pass the list is

23    19    17    24     $x$     18    10    21    8

- (b) Using this information, write down the smallest interval that must contain  $x$ , giving your answer as an inequality.

(2)

When the first-fit decreasing bin packing algorithm is applied to the nine distinct numbers it results in the following allocation.

Bin 1: 24    23

Bin 2: 21    19    10

Bin 3: 18    17     $x$

Bin 4: 8

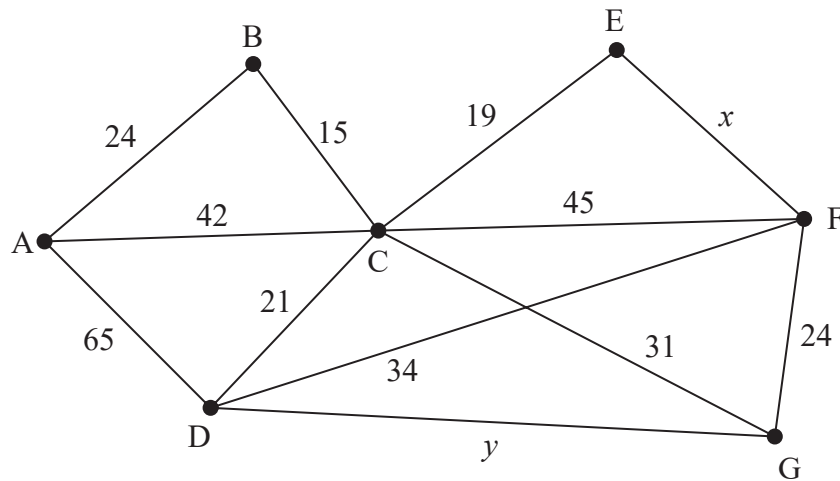
Given that only one of the bins is full and that  $x$  is an integer,

- (c) calculate the value of  $x$ . You must give reasons for your answer.

(2)

(Total for Question 5 is 7 marks)

6.



**Figure 4**

[The total weight of the network is  $320 + x + y$ ]

- (a) State, with justification, whether the graph in Figure 4 is Eulerian, semi-Eulerian or neither. (2)

The weights on the arcs in Figure 4 represent distances. The weight on arc EF is  $x$  where  $12 < x < 26$  and the weight on arc DG is  $y$  where  $0 < y < 10$

An inspection route of minimum length that traverses each arc at least once is found. The inspection route starts and finishes at A and has a length of 409

It is also given that the length of the shortest route from F to G via A is 140

- (b) Using appropriate algorithms, find the value of  $x$  and the value of  $y$ . (9)

(Total for Question 6 is 11 marks)

7. A maximisation linear programming problem in  $x$ ,  $y$  and  $z$  is to be solved using the two-stage simplex method.

The partially completed initial tableau is shown below.

| Basic variable | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $a_1$ | $a_2$ | Value |
|----------------|-----|-----|-----|-------|-------|-------|-------|-------|-------|
| $s_1$          | 1   | 2   | 3   | 1     | 0     | 0     | 0     | 0     | 45    |
| $a_1$          | 3   | 2   | 0   | 0     | -1    | 0     | 1     | 0     | 9     |
| $a_2$          | -1  | 0   | 4   | 0     | 0     | -1    | 0     | 1     | 4     |
| $P$            | -2  | -1  | -3  | 0     | 0     | 0     | 0     | 0     | 0     |
| $A$            |     |     |     |       |       |       |       |       |       |

- (a) Using the information in the above tableau, formulate the linear programming problem. State the objective and list the constraints as inequalities.

(4)

- (b) Complete the bottom row of Table 1 in the answer book. You should make your method and working clear.

(2)

The following tableau is obtained after two iterations of the first stage of the two-stage simplex method.

| Basic variable | $x$ | $y$           | $z$ | $s_1$ | $s_2$            | $s_3$          | $a_1$           | $a_2$          | Value           |
|----------------|-----|---------------|-----|-------|------------------|----------------|-----------------|----------------|-----------------|
| $s_1$          | 0   | $\frac{5}{6}$ | 0   | 1     | $\frac{7}{12}$   | $\frac{3}{4}$  | $-\frac{7}{12}$ | $-\frac{3}{4}$ | $\frac{147}{4}$ |
| $x$            | 1   | $\frac{2}{3}$ | 0   | 0     | $-\frac{1}{3}$   | 0              | $\frac{1}{3}$   | 0              | 3               |
| $z$            | 0   | $\frac{1}{6}$ | 1   | 0     | $-\frac{1}{12}$  | $-\frac{1}{4}$ | $\frac{1}{12}$  | $\frac{1}{4}$  | $\frac{7}{4}$   |
| $P$            | 0   | $\frac{5}{6}$ | 0   | 0     | $-\frac{11}{12}$ | $-\frac{3}{4}$ | $\frac{11}{12}$ | $\frac{3}{4}$  | $\frac{45}{4}$  |
| $A$            | 0   | 0             | 0   | 0     | 0                | 0              | 1               | 1              | 0               |

- (c) (i) Explain how the above tableau shows that a basic feasible solution has been found for the original linear programming problem.

- (ii) Write down the basic feasible solution for the second stage.

(3)

- (d) Taking the most negative number in the profit row to indicate the pivot column, perform one complete iteration of the second stage of the two-stage simplex method, to obtain a new tableau,  $T$ . Make your method clear by stating the row operations you use.

(5)

- (e) (i) Explain, using  $T$ , whether or not an optimal solution to the original linear programming problem has been found.
- (ii) Write down the value of the objective function.
- (iii) State the values of the basic variables.

(3)

---

(Total for Question 7 is 17 marks)

---

**TOTAL FOR PAPER IS 75 MARKS**

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Other names

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**Wednesday 17 June 2020**

Afternoon (Time: 1 hour 30 minutes)

Paper Reference **9FM0/3D**

**Further Mathematics**

**Advanced**

**Paper 3D: Decision Mathematics 1**

**Answer book**

Do not return the question paper with the answer book.

Total Marks

Turn over ►

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1.

|   | A  | B  | C  | D  | E  | F  | G  |
|---|----|----|----|----|----|----|----|
| A | –  | 24 | –  | 22 | 35 | –  | –  |
| B | 24 | –  | 25 | 27 | –  | –  | –  |
| C | –  | 25 | –  | 33 | 31 | 36 | 26 |
| D | 22 | 27 | 33 | –  | –  | 42 | –  |
| E | 35 | –  | 31 | –  | –  | 37 | 29 |
| F | –  | –  | 36 | 42 | 37 | –  | 40 |
| G | –  | –  | 26 | –  | 29 | 40 | –  |

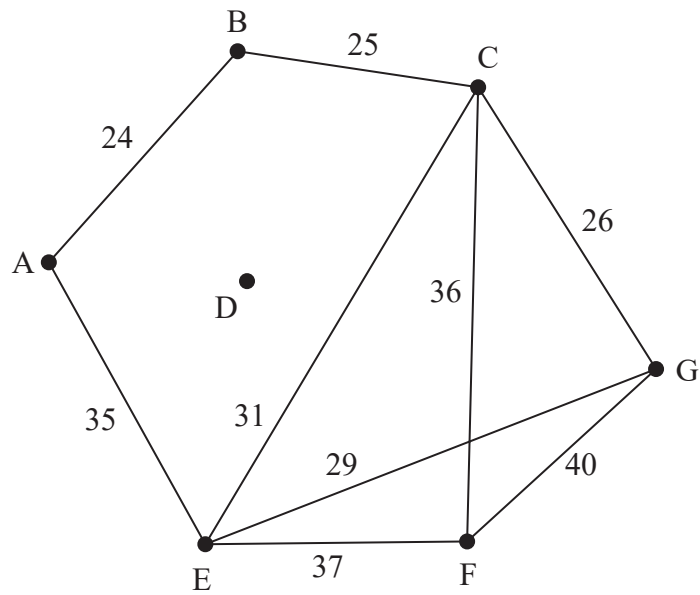


Diagram 1



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. (a)

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(b)

| Activity | Immediately preceding activities |
|----------|----------------------------------|
| A        |                                  |
| B        |                                  |
| C        |                                  |
| D        |                                  |

| Activity | Immediately preceding activities |
|----------|----------------------------------|
| E        |                                  |
| F        |                                  |
| G        |                                  |
| H        |                                  |

| Activity | Immediately preceding activities |
|----------|----------------------------------|
| I        |                                  |
| J        |                                  |
| K        |                                  |
| L        |                                  |

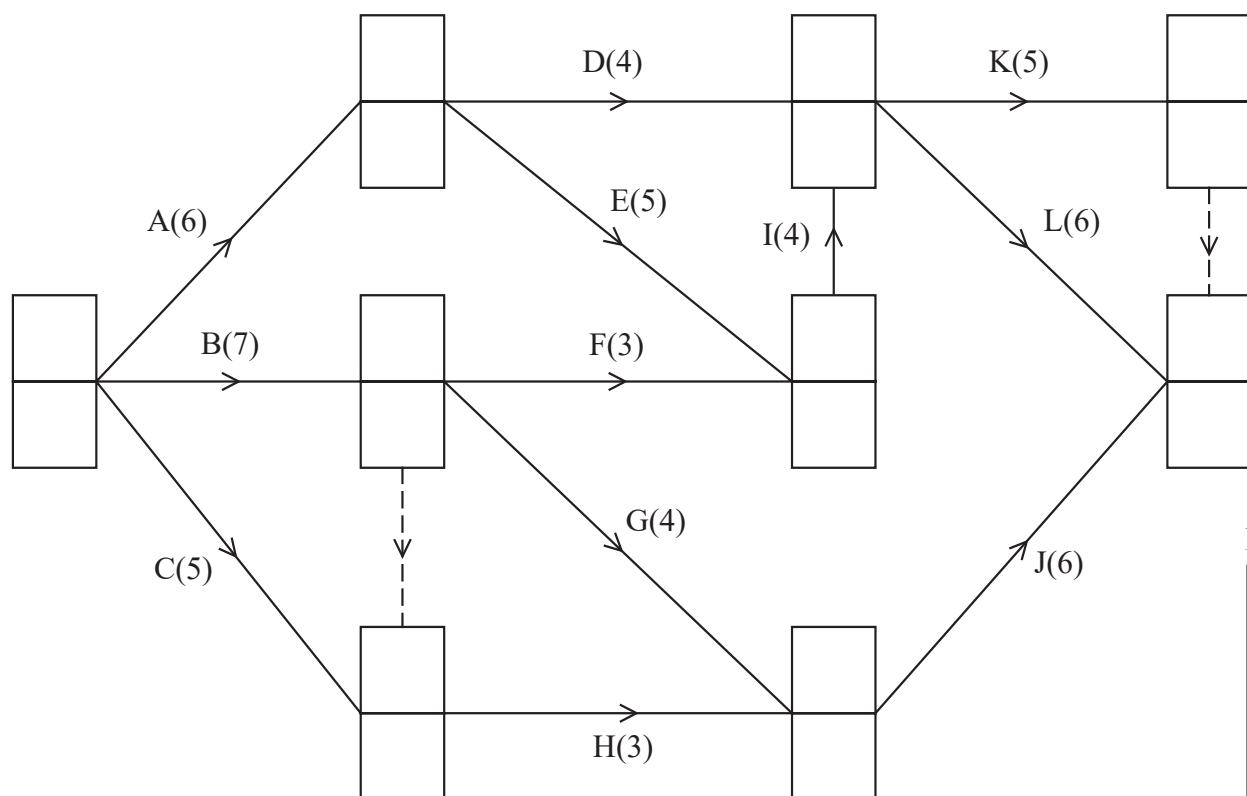


Diagram 1

Key:

|                  |
|------------------|
| Early event time |
| Late event time  |



[illegible]

## Grid 1

3. (a)

Initial time matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Initial route matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

(b)

Time matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Time matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

If you make an error there are spare copies of these matrices on Page 9.



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



**Question 3 continued**

Only use these matrices if you have made an error in your working.

(a)

Spare copy of initial time matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Spare copy of initial route matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

(b)

Spare copy of time matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Spare copy of route matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Spare copy of time matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Spare copy of route matrix

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

(Total for Question 3 is 9 marks)



4.

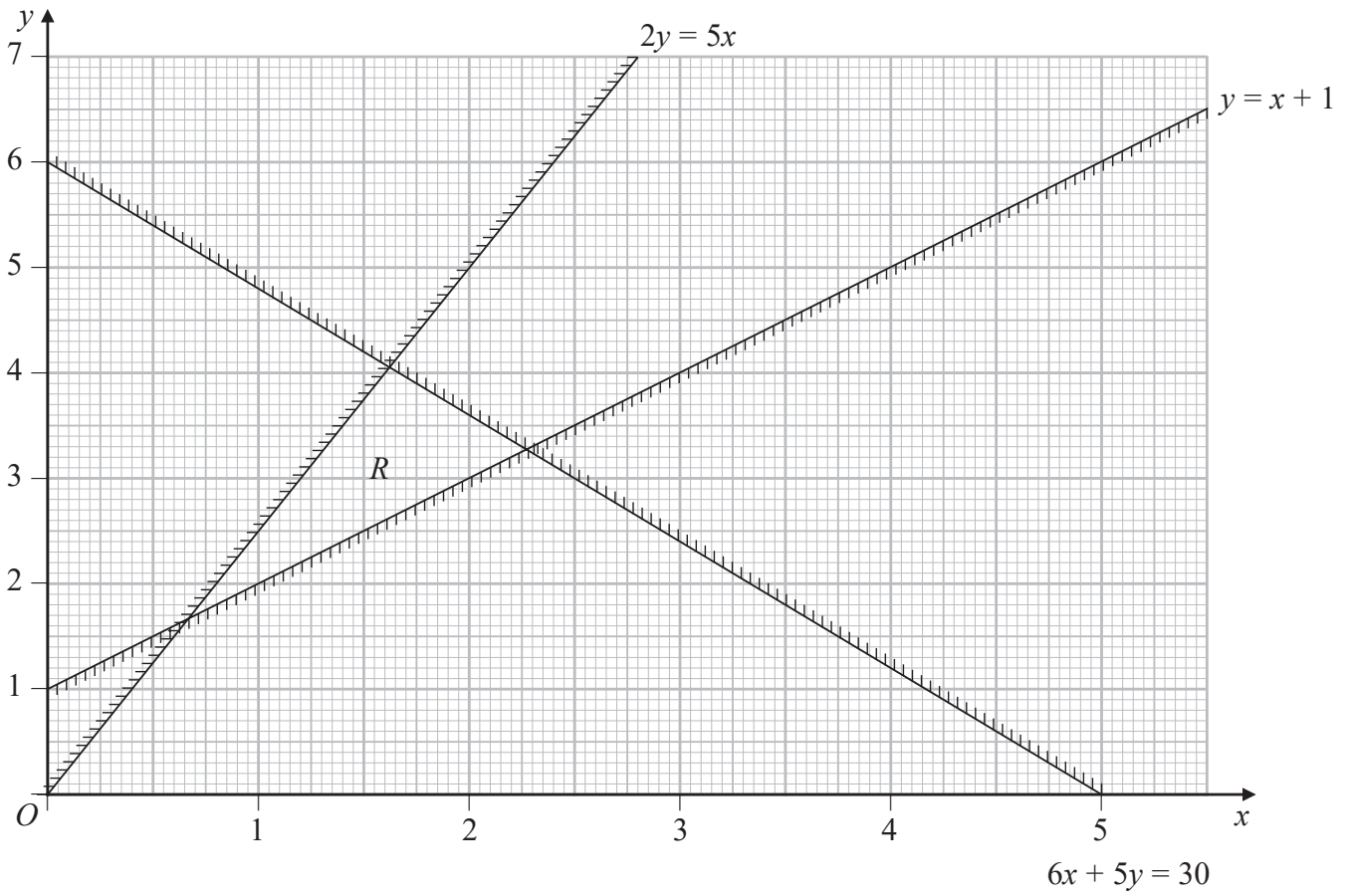


Figure 3



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 10 marks)



5.                      23      17      19       $x$       24      8      18      10      21

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Question 5 continued

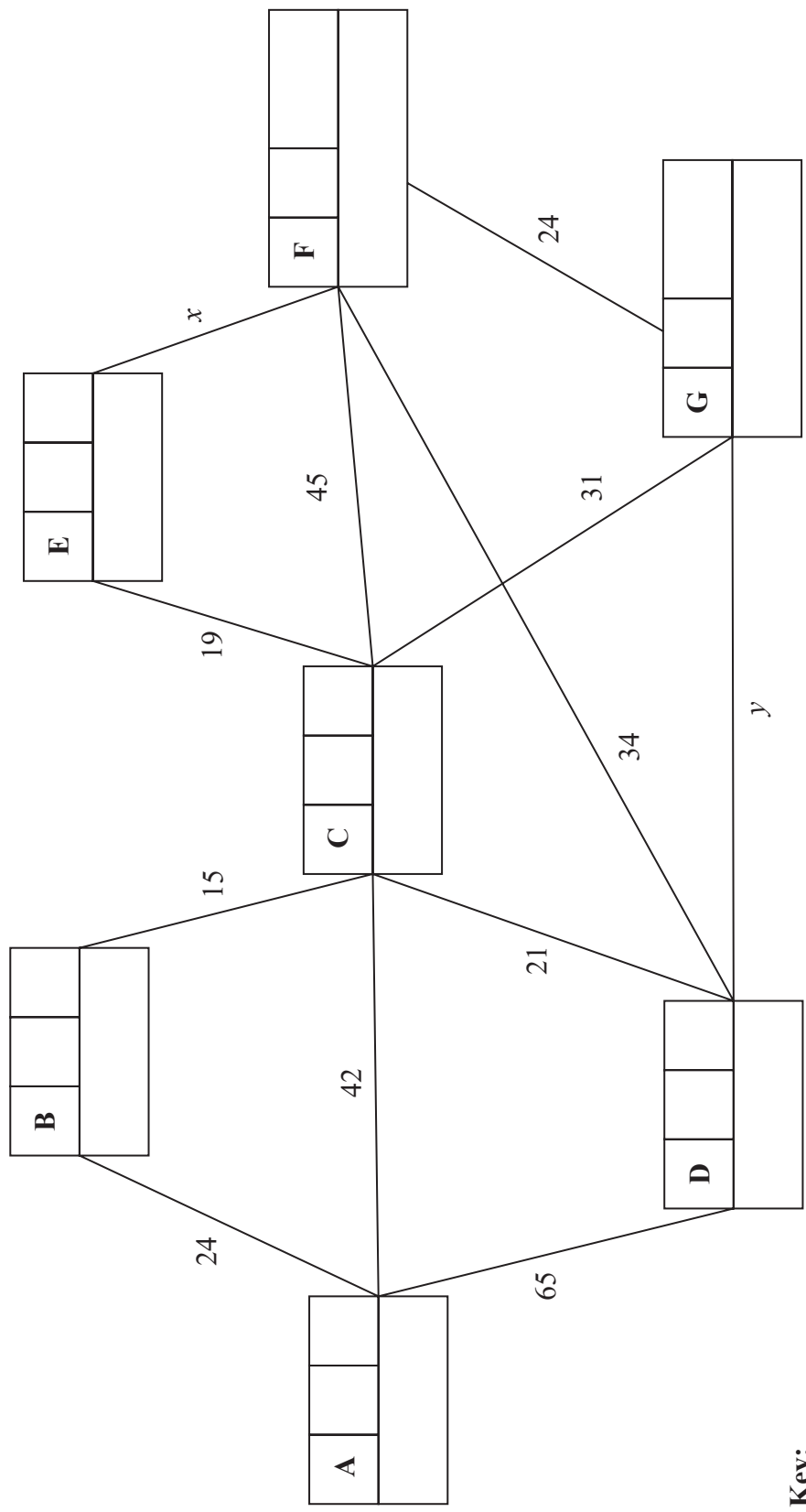
Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)





6. (a)



Key:

| Vertex         | Order of labelling | Final value |
|----------------|--------------------|-------------|
| Working values |                    |             |

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 11 marks)



7.

| Basic variable | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $a_1$ | $a_2$ | Value |
|----------------|-----|-----|-----|-------|-------|-------|-------|-------|-------|
| $s_1$          | 1   | 2   | 3   | 1     | 0     | 0     | 0     | 0     | 45    |
| $a_1$          | 3   | 2   | 0   | 0     | -1    | 0     | 1     | 0     | 9     |
| $a_2$          | -1  | 0   | 4   | 0     | 0     | -1    | 0     | 1     | 4     |
| $P$            | -2  | -1  | -3  | 0     | 0     | 0     | 0     | 0     | 0     |
| $A$            |     |     |     |       |       |       |       |       |       |

Table 1



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Question 7 continued

| b.v.  | $x$ | $y$           | $z$ | $s_1$ | $s_2$            | $s_3$          | Value           |
|-------|-----|---------------|-----|-------|------------------|----------------|-----------------|
| $s_1$ | 0   | $\frac{5}{6}$ | 0   | 1     | $\frac{7}{12}$   | $\frac{3}{4}$  | $\frac{147}{4}$ |
| $x$   | 1   | $\frac{2}{3}$ | 0   | 0     | $-\frac{1}{3}$   | 0              | 3               |
| $z$   | 0   | $\frac{1}{6}$ | 1   | 0     | $-\frac{1}{12}$  | $-\frac{1}{4}$ | $\frac{7}{4}$   |
| $P$   | 0   | $\frac{5}{6}$ | 0   | 0     | $-\frac{11}{12}$ | $-\frac{3}{4}$ | $\frac{45}{4}$  |

| b.v. | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | Value | Row Ops |
|------|-----|-----|-----|-------|-------|-------|-------|---------|
|      |     |     |     |       |       |       |       |         |
|      |     |     |     |       |       |       |       |         |
|      |     |     |     |       |       |       |       |         |
| $P$  |     |     |     |       |       |       |       |         |

Spare copy

| b.v. | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | Value | Row Ops |
|------|-----|-----|-----|-------|-------|-------|-------|---------|
|      |     |     |     |       |       |       |       |         |
|      |     |     |     |       |       |       |       |         |
|      |     |     |     |       |       |       |       |         |
| $P$  |     |     |     |       |       |       |       |         |



**Question 7 continued**

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**(Total for Question 7 is 17 marks)**

**TOTAL FOR PAPER IS 75 MARKS**



**Pearson Edexcel Level 3 GCE**

**Monday 24 June 2019**

Morning (Time: 1 hour 30 minutes)

Paper Reference **9FM0/3D**

## **Further Mathematics**

**Advanced**

**Paper 3D: Decision Mathematics 1**

### **You must have:**

Mathematical Formulae and Statistical Tables (Green), calculator,  
Decision Mathematics Answer Book (enclosed)

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### **Instructions**

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.  
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

### **Information**

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### **Advice**

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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**Answer ALL questions. Write your answers in the answer book provided.**

**1.**                2.1      1.7      3.0      1.9      3.2      1.2      3.3      1.4      1.5      0.2

- (a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 5

**(2)**

The list of numbers is now to be sorted into descending order.

- (b) Perform a quick sort on the original list to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

**(4)**

For a list of  $n$  numbers, the quick sort algorithm has, on average, order  $n \log n$ .

Given that it takes 2.32 seconds to run the algorithm when  $n = 450$

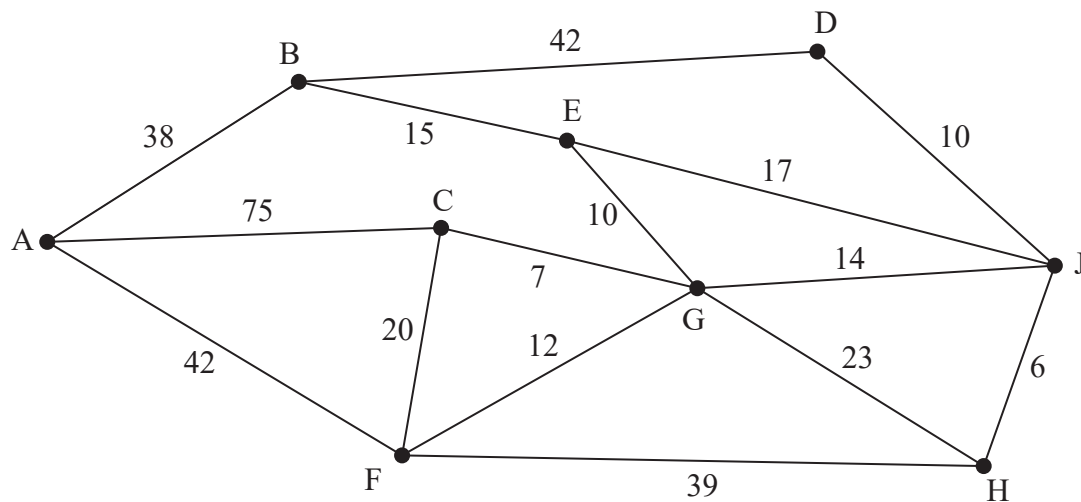
- (c) calculate approximately how long it will take, to the nearest tenth of a second, to run the algorithm when  $n = 11\,250$ . You should make your method and working clear.

**(2)**

---

**(Total for Question 1 is 8 marks)**

2.



**Figure 1**

[The total weight of the network is 370]

Figure 1 represents a network of corridors in a building. The number on each arc represents the length, in metres, of the corresponding corridor.

- (a) Use Dijkstra's algorithm to find the shortest path from A to D, stating the path and its length.

(6)

On a particular day, Naasir needs to check the paintwork along each corridor. Naasir must find a route of minimum length. It must traverse each corridor at least once, starting at B and finishing at G.

- (b) Use an appropriate algorithm to find the arcs that will need to be traversed twice. You must make your method and working clear.

(4)

- (c) Find the length of Naasir's route.

(1)

On a different day, all the corridors that start or finish at B are closed for redecorating. Naasir needs to check all the remaining corridors and may now start at any vertex and finish at any vertex. A route is required that excludes all those corridors that start or finish at B.

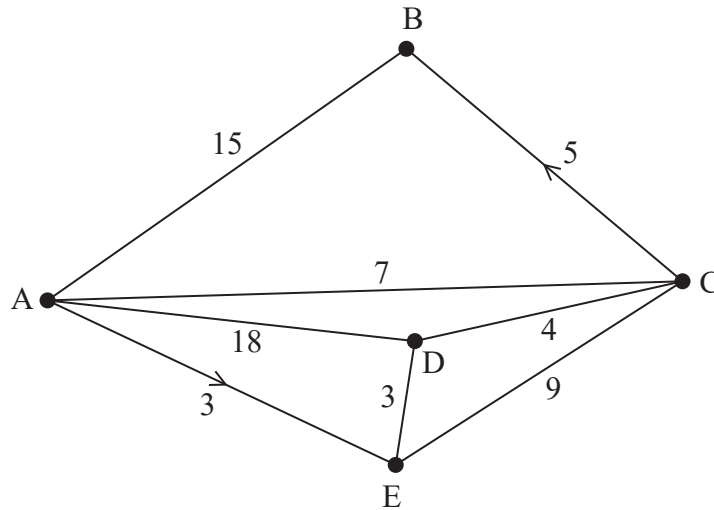
- (d) (i) Determine the possible starting and finishing points so that the length of Naasir's route is minimised. You must give reasons for your answer.

- (ii) Find the length of Naasir's new route.

(3)

(Total for Question 2 is 14 marks)

3.



**Figure 2**

The network in Figure 2 shows the direct roads linking five villages, A, B, C, D and E. The number on each arc represents the length, in miles, of the corresponding road. The roads from A to E and from C to B are one-way, as indicated by the arrows.

(a) Complete the initial distance and route tables for the network provided in the answer book. (2)

(b) Perform the first three iterations of Floyd's algorithm. You should show the distance table and the route table after each of the three iterations. (5)

After five iterations of Floyd's algorithm the final distance table and partially completed final route table are shown below.

Distance table

|   | A  | B  | C  | D  | E  |
|---|----|----|----|----|----|
| A | –  | 12 | 7  | 6  | 3  |
| B | 15 | –  | 22 | 21 | 18 |
| C | 7  | 5  | –  | 4  | 7  |
| D | 11 | 9  | 4  | –  | 3  |
| E | 14 | 12 | 7  | 3  | –  |

Route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A | A |   |   |   |   |
| B | A | B |   |   |   |
| C | A | B | C |   |   |
| D | C | C | C | D |   |
| E | D | D | D | D | E |

(c) (i) Explain how the partially completed final route table can be used to find the shortest route from E to A.

(ii) State this route.

(3)

Mabintou decides to use the distance table to try to find the shortest cycle that passes through each vertex. Starting at D, she applies the nearest neighbour algorithm to the final distance table.

(d) (i) State the cycle obtained using the nearest neighbour algorithm.

(ii) State the length of this cycle.

(iii) Interpret the cycle in terms of the actual villages visited.

(iv) Prove that Mabintou's cycle is not optimal.

(4)

(Total for Question 3 is 14 marks)

4.

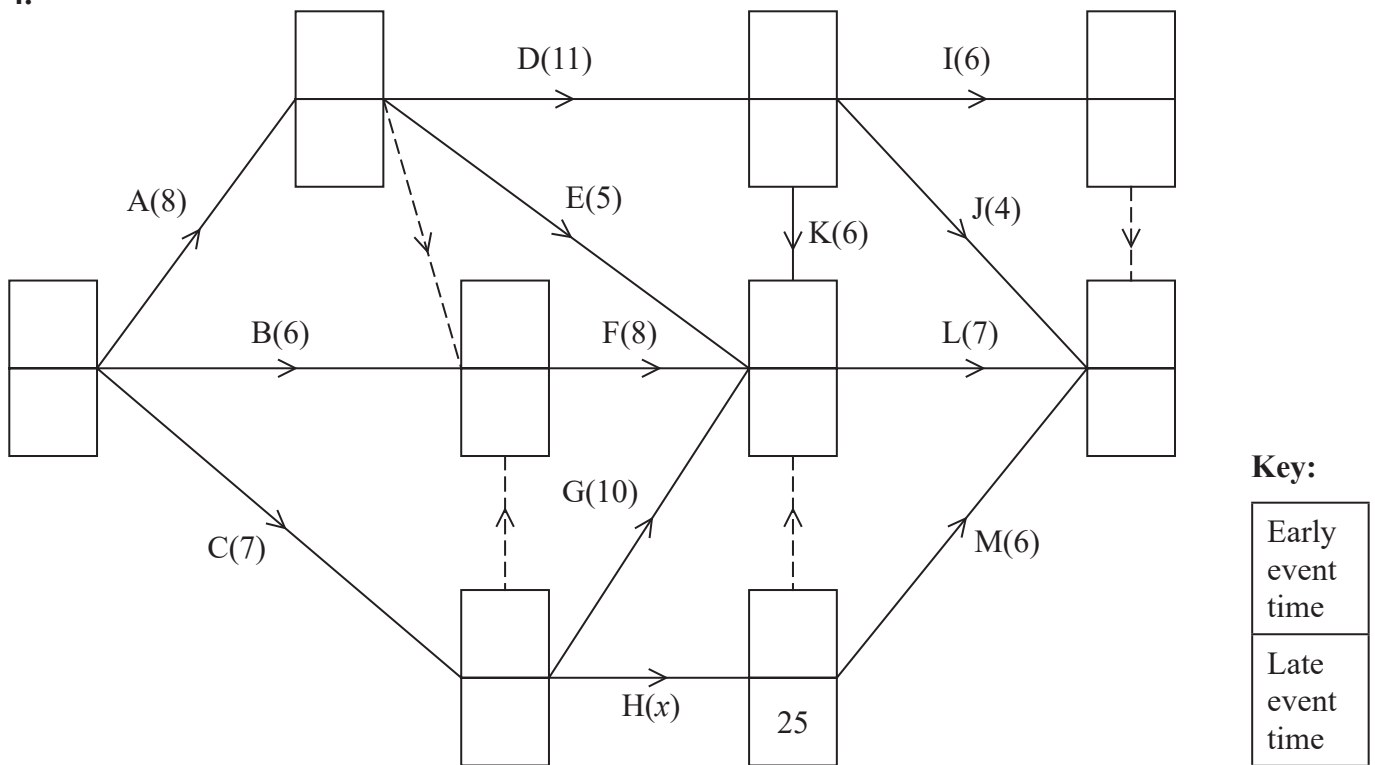


Figure 3

The network in Figure 3 shows the activities that need to be undertaken to complete a project. Each activity is represented by an arc and the duration of the activity, in days, is shown in brackets. The early event times and late event times are to be shown at each vertex and one late event time has been completed for you.

The total float of activity H is 7 days.

(a) Explain, with detailed reasoning, why  $x = 11$  (2)

(b) Determine the missing early event times and late event times, and hence complete Diagram 1 in your answer book. (3)

Each activity requires one worker and the project must be completed in the shortest possible time using as few workers as possible.

(c) Calculate a lower bound for the number of workers needed to complete the project in the shortest possible time. (1)

(d) Schedule the activities using Grid 1 in the answer book. (3)

(Total for Question 4 is 9 marks)

5.

| Activity | Immediately preceding activities |
|----------|----------------------------------|
| A        | –                                |
| B        | –                                |
| C        | –                                |
| D        | A                                |
| E        | C                                |
| F        | B, C, D                          |
| G        | A                                |
| H        | B, C, D                          |
| I        | B, C, D, G                       |
| J        | B, C, D, G                       |
| K        | E, H                             |

- (a) Draw the activity network described in the precedence table above, using activity on arc. Your activity network must contain only the minimum number of dummies.

(5)

Given that all the activities shown in the precedence table have the same duration,

- (b) state the critical path for the network.

(1)

(Total for Question 5 is 6 marks)

6. A linear programming problem in  $x$ ,  $y$  and  $z$  is described as follows.

$$\begin{aligned} \text{Maximise} \quad & P = 2x + 2y - z \\ \text{subject to} \quad & 3x + y + 2z \leq 30 \\ & x - y + z \geq 8 \\ & 4y + 2z \geq 15 \\ & x, y, z \geq 0 \end{aligned}$$

(a) Explain why the Simplex algorithm cannot be used to solve this linear programming problem.

(1)

(b) Set up the initial tableau for solving this linear programming problem using the big-M method.

(7)

After a first iteration of the big-M method, the tableau is

| b.v.  | $x$        | $y$ | $z$        | $s_1$ | $s_2$ | $s_3$          | $a_1$ | $a_2$         | Value          |
|-------|------------|-----|------------|-------|-------|----------------|-------|---------------|----------------|
| $s_1$ | 3          | 0   | 1.5        | 1     | 0     | 0.25           | 0     | -0.25         | 26.25          |
| $a_1$ | 1          | 0   | 1.5        | 0     | -1    | -0.25          | 1     | 0.25          | 11.75          |
| $y$   | 0          | 1   | 0.5        | 0     | 0     | -0.25          | 0     | 0.25          | 3.75           |
| $P$   | $-(2 + M)$ | 0   | $2 - 1.5M$ | 0     | $M$   | $-0.5 + 0.25M$ | 0     | $0.5 + 0.75M$ | $7.5 - 11.75M$ |

(c) State the value of each variable after the first iteration.

(1)

(d) Explain why the solution given by the first iteration is not feasible.

(1)

Taking the most negative entry in the profit row to indicate the pivot column,

(e) obtain the most efficient pivot for a second iteration. You must give reasons for your answer.

(2)

(Total for Question 6 is 12 marks)

7. A shop sells two types of watch, analogue watches and digital watches.

The shop manager knows that, each month, she should order at least 60 watches in total. In addition, at most 80% of the watches she orders must be digital.

Let  $x$  be the number of analogue watches ordered and let  $y$  be the number of digital watches ordered.

- (a) Write down inequalities, in terms of  $x$  and  $y$ , to model these constraints.

(2)

Two further constraints are

$$y + 3x \geq 140$$

$$4y + x \geq 80$$

- (b) Represent all these constraints on Diagram 1 in the answer book. Hence determine, and label, the feasible region,  $R$ .

(4)

The cost to the shop of ordering an analogue watch is five times the cost of ordering a digital watch. The shop manager wishes to minimise the total cost.

- (c) Determine the number of each type of watch the shop manager should order. You must make your method clear.

(3)

Given that the minimum total cost of ordering the watches is £4455

- (d) determine the cost of ordering one analogue watch and the cost of ordering one digital watch. You must make your method clear.

(3)

---

(Total for Question 7 is 12 marks)

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**TOTAL FOR PAPER IS 75 MARKS**

Please check the examination details below before entering your candidate information

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Other names

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Level 3 GCE**

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**Monday 24 June 2019**

Morning (Time: 1 hour 30 minutes)

Paper Reference **9FM0/3D**

**Further Mathematics**

**Advanced**

**Paper 3D: Decision Mathematics 1**

**Answer Book**

Do not return the question paper with the answer book.

Total Marks

Turn over ►

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1. 2.1 1.7 3.0 1.9 3.2 1.2 3.3 1.4 1.5 0.2

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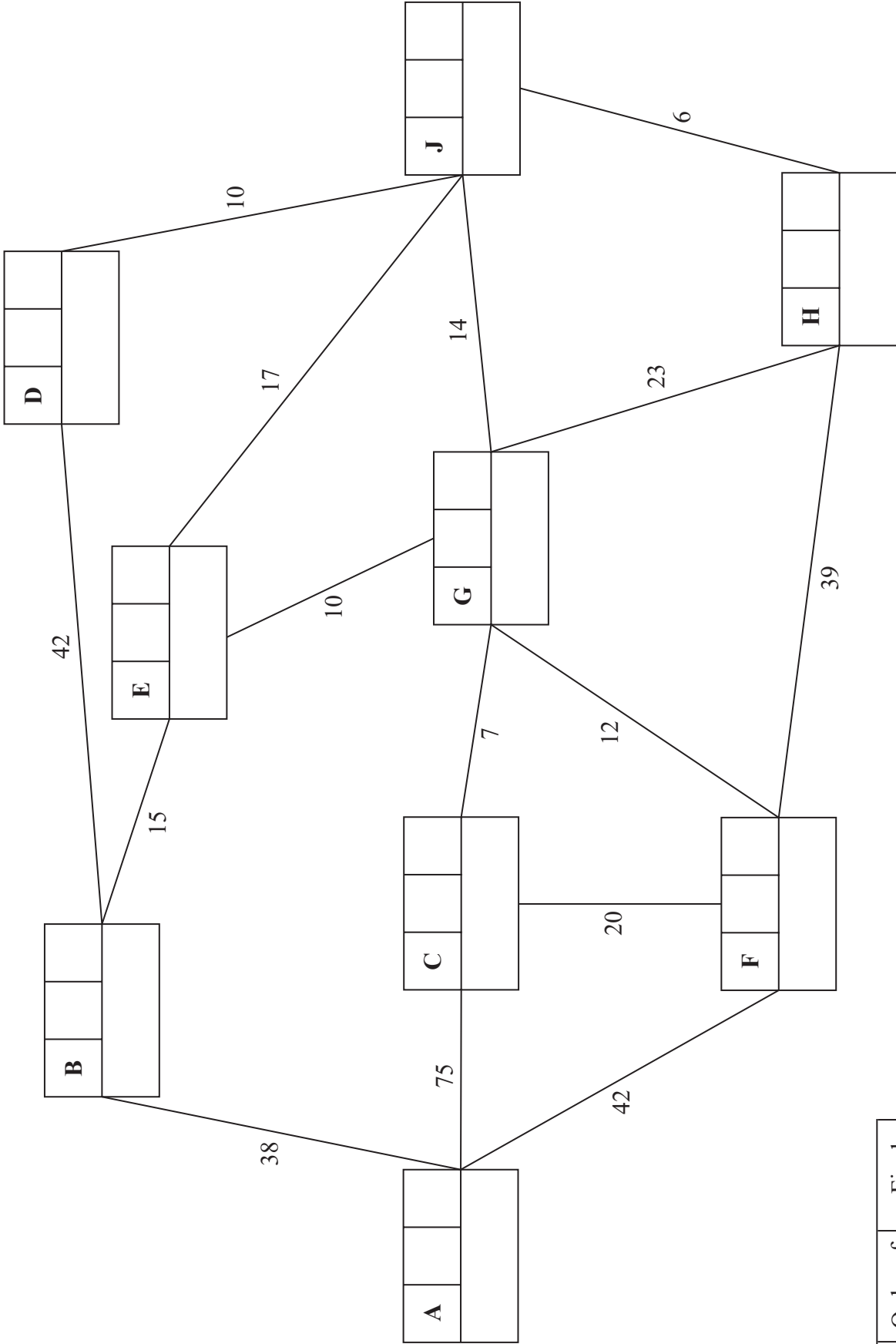


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2.1      1.7      3.0      1.9      3.2      1.2      3.3      1.4      1.5      0.2



2.



Key:

| Vertex         | Order of labelling | Final value |
|----------------|--------------------|-------------|
| Working values |                    |             |

Shortest path from A to D:

Length of shortest path from A to D:

## Question 2 continued

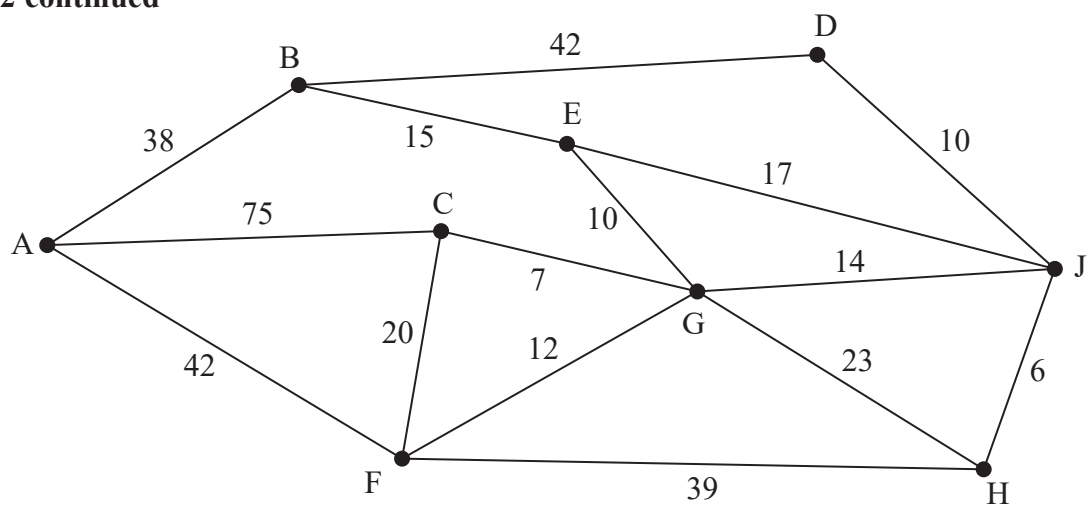


Figure 1

Question 2 continued

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 14 marks)



3. (a)

Initial distance table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Initial route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

(b) 1st iteration

Distance table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

2nd iteration

Distance table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

3rd iteration

Distance table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Turn to page 11 for spare copies of these tables if you need to correct your work.



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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**Question 3 continued****Only use these tables if you need to correct your work.**

(a)

Initial distance table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Initial route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

(b) 1st iteration

Distance table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

2nd iteration

Distance table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

3rd iteration

Distance table

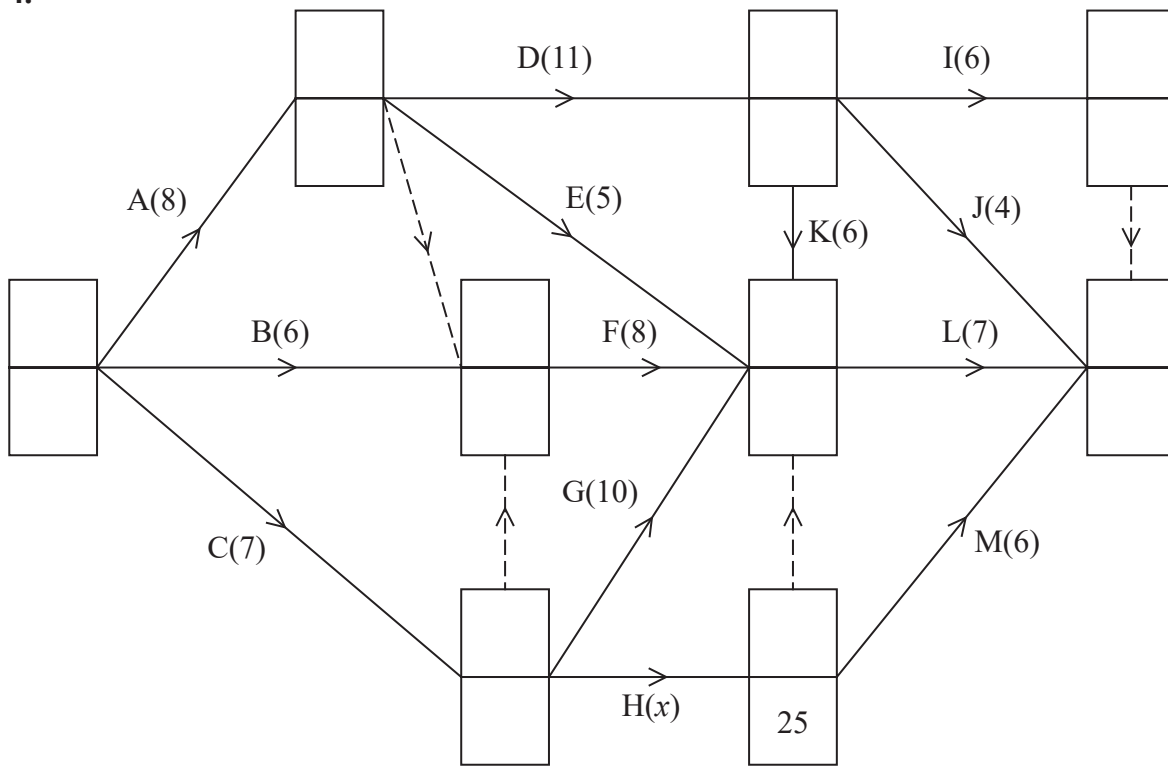
|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

Route table

|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A |   |   |   |   |   |
| B |   |   |   |   |   |
| C |   |   |   |   |   |
| D |   |   |   |   |   |
| E |   |   |   |   |   |

**(Total for Question 3 is 14 marks)**

4.

**Key:**Early  
event  
timeLate  
event  
time**Diagram 1**

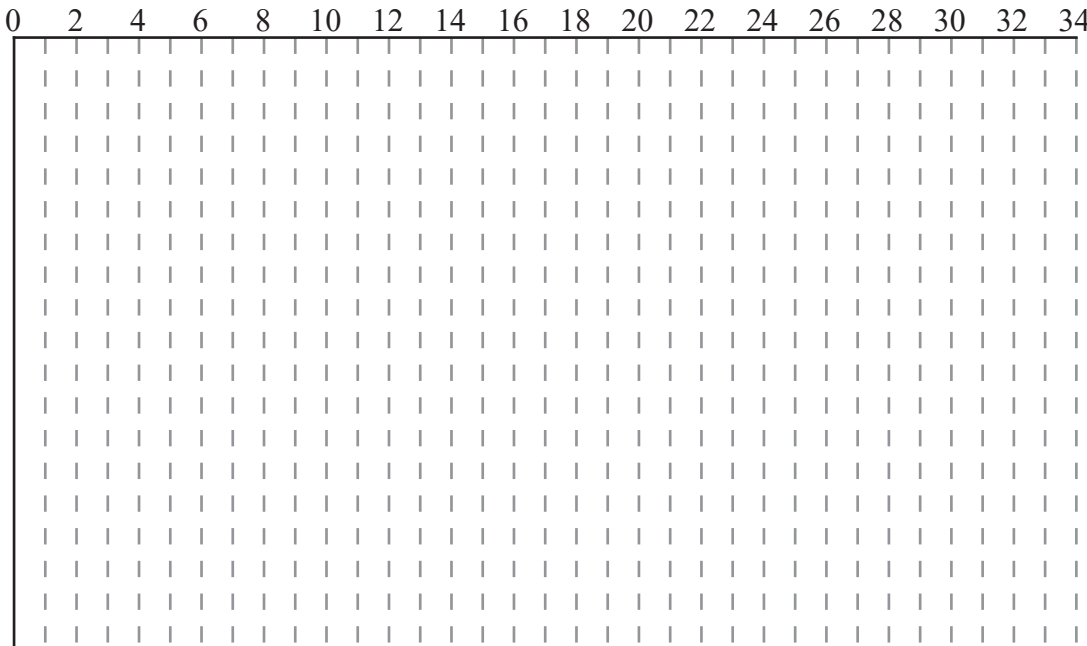
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Question 4 continued

Handwriting practice area with 20 horizontal lines.



Grid 1

(Total for Question 4 is 9 marks)



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5.



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



6.

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**Question 6 continued**

(b)

[illegible]

**Only use the tableau below if you need to correct the above tableau.**

[illegible]

Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7.

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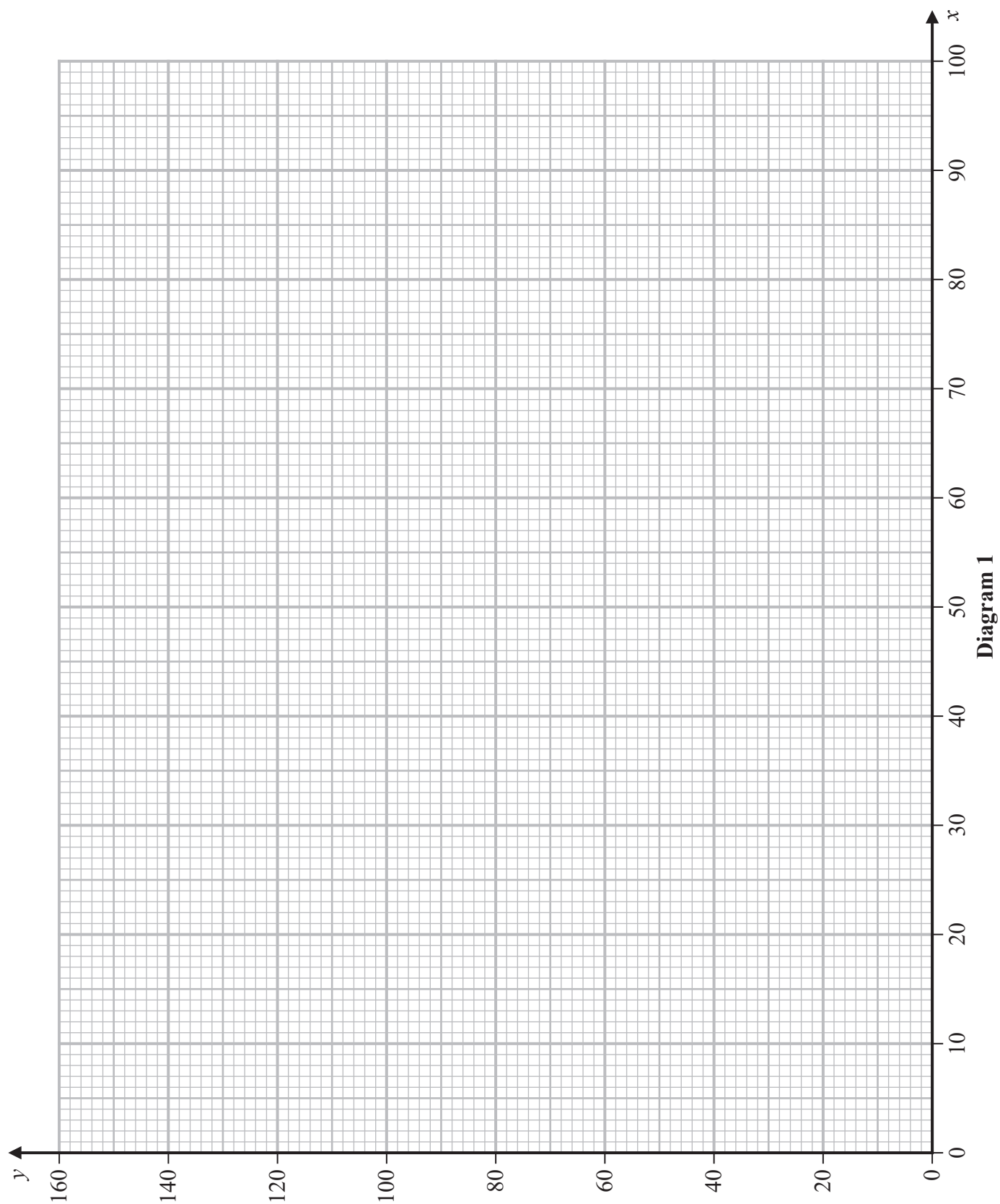
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Question 7 continued

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**Question 7 continued**

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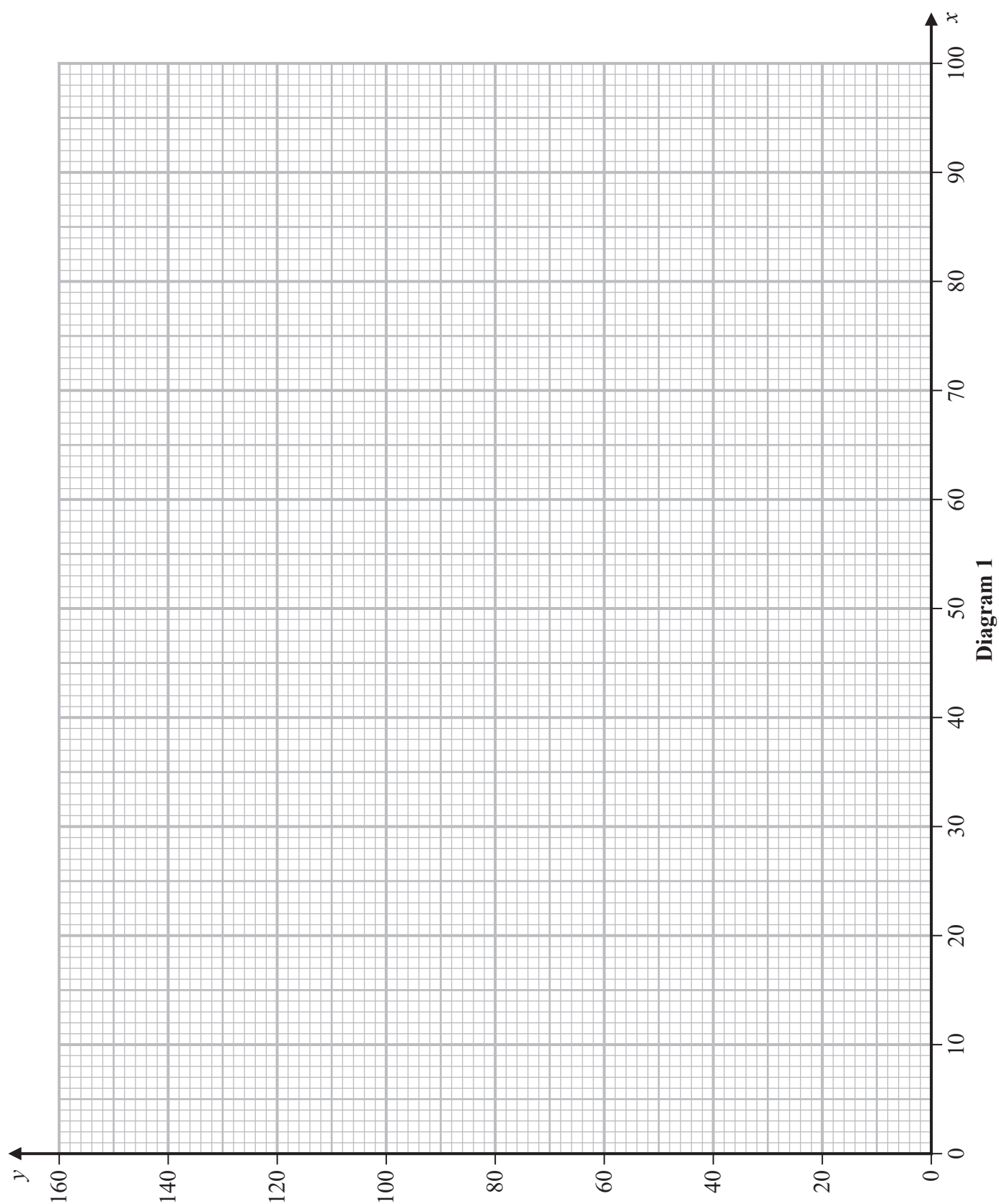
Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

Only use these axes if you need to correct your work.



(Total for Question 7 is 12 marks)

**TOTAL FOR PAPER IS 75 MARKS**

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# Pearson Edexcel Level 3 GCE

## Mock Paper Set 1

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/3D**

## Further Mathematics

### Advanced

### Paper 3D: Decision Mathematics 1

#### You must have:

Decision Mathematics Answer Book (enclosed), calculator

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

#### Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.  
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

#### Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

#### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the answer book provided.

1.

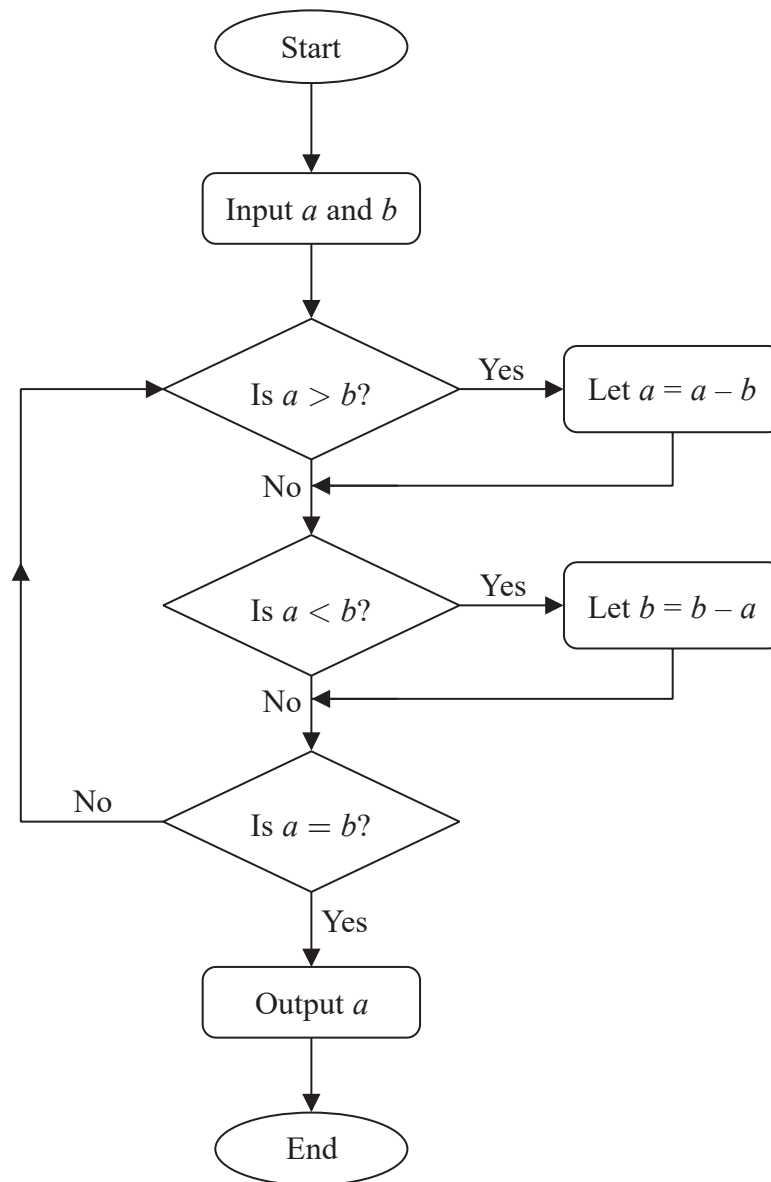


Figure 1

An algorithm is described in Figure 1. The algorithm is run with  $a = 70$  and  $b = 168$

- (a) Complete the table in the answer book to show the result obtained at each step when the algorithm is applied.

(4)

When the algorithm is applied using new values for  $a$  and  $b$ , where  $a = x$  and  $b = y$  the output is  $y$ .

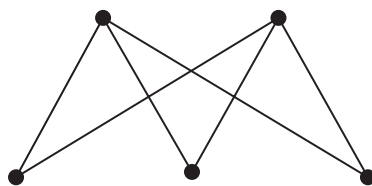
- (b) Write down a conclusion about the relationship between the values of  $x$  and  $y$  in this case, giving a reason for your answer. You do not need to rerun the algorithm.

(2)

(Total for Question 1 is 6 marks)

2. (a) Define the term “Hamiltonian cycle”.

(2)



**Figure 2**

- (b) Redraw the graph in Figure 2 to show that it is planar.

(1)

- (c) Explain why the planarity algorithm cannot be used to show that the graph shown in Figure 2 is a planar graph.

(1)

---

**(Total for Question 2 is 4 marks)**

3.

|   | A       | B       | C       | D       | E       | F       |
|---|---------|---------|---------|---------|---------|---------|
| A | –       | 19      | $x + 4$ | $x - 2$ | $x - 2$ | $x - 5$ |
| B | 19      | –       | $x + 3$ | $x$     | $x - 3$ | $x - 2$ |
| C | $x + 4$ | $x + 3$ | –       | $x - 3$ | $x + 6$ | $x - 7$ |
| D | $x - 2$ | $x$     | $x - 3$ | –       | $x - 5$ | $x + 1$ |
| E | $x - 2$ | $x - 3$ | $x + 6$ | $x - 5$ | –       | $x$     |
| F | $x - 5$ | $x - 2$ | $x - 7$ | $x + 1$ | $x$     | –       |

Claudia must check 6 ponds, A, B, C, D, E and F, to see which plant species are growing in them. The times, in minutes, that she will need to take the direct routes between the ponds are shown in the table.

Initially, some of the times are only known relative to the time it will take to travel between B and D, which is shown as  $x$  minutes in the table. It is known that  $7 < x < 20$

The length of the route obtained when the Nearest Neighbour Algorithm is applied from A takes 2 minutes less than the length of the route obtained when the Nearest Neighbour Algorithm is applied from D.

(a) Determine the value of  $x$ .

(5)

A student claims that the quickest time required for Claudia to check all 6 ponds, starting and finishing at A, can be found by substituting the value of  $x$  found in (a) into the expression obtained from the Nearest Neighbour Algorithm starting from A.

(b) Without any further working, state, giving a reason, whether or not the student's claim is correct.

(2)

(Total for Question 3 is 7 marks)

4.

|   | A | B | C | D |
|---|---|---|---|---|
| A | – | 1 | 2 | 4 |
| B | 1 | – | 4 | 2 |
| C | 2 | 4 | – | 3 |
| D | 6 | 8 | 5 | – |

The distance table shown above represents the paths between four checkpoints, A, B, C and D, on an orienteering course. The numbers represent the time, in minutes, needed to run along the corresponding path. Most of the paths have been designated as one-way routes, to ease possible congestion.

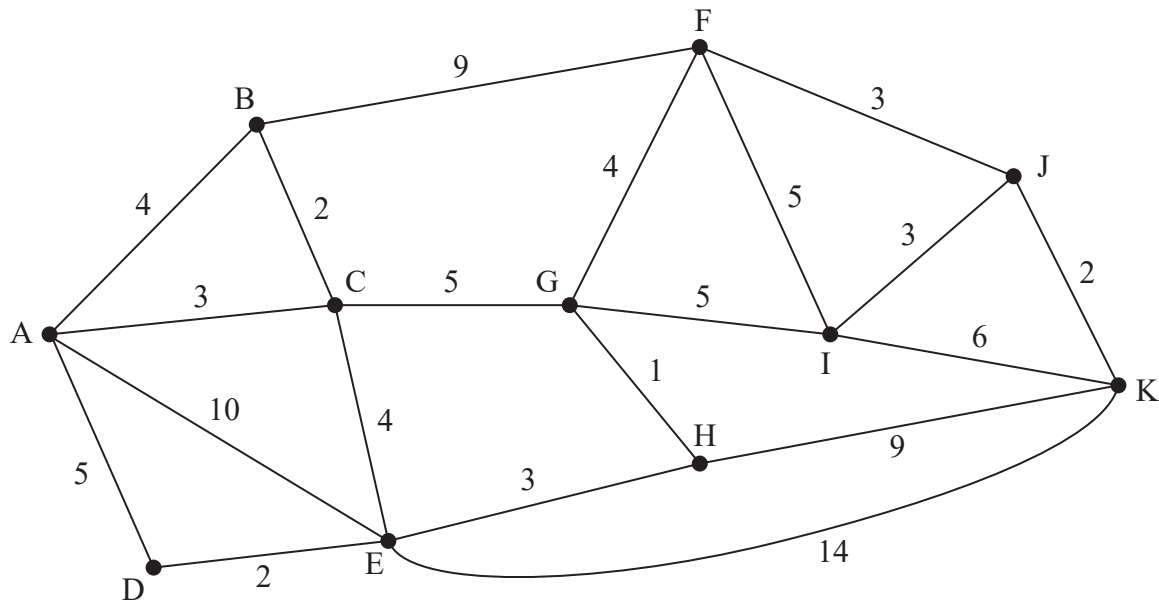
The initial distance and route tables for the network are given in the answer book.

(a) Draw the network using the nodes given in Diagram 1 in the answer book. (1)

(b) Use Floyd's algorithm to find the complete network of shortest distances. Show the distance and route tables after each iteration. (7)

**(Total for Question 4 is 8 marks)**

5.



**Figure 3**

Figure 3 represents a network of paths in a park. The number on each arc shows the time it takes, in minutes, to walk along the corresponding path. Lola wants to walk from A to K in the shortest possible time.

(a) Using Dijkstra's algorithm,

(i) determine the minimum time for Lola's route,

(ii) determine the route she should take.

(5)

The following day, Lola wants to walk from A to K again. A market at G means that there will be a 6-minute delay for anyone passing through G.

(b) Explain how the network may be adapted to allow for this change.

(2)

Dijkstra's algorithm has approximate order

$$x \ln(y)$$

where  $x$  is the number of arcs in the network and  $y$  is the number of nodes.

(c) Explain why the order given is only approximate.

(1)

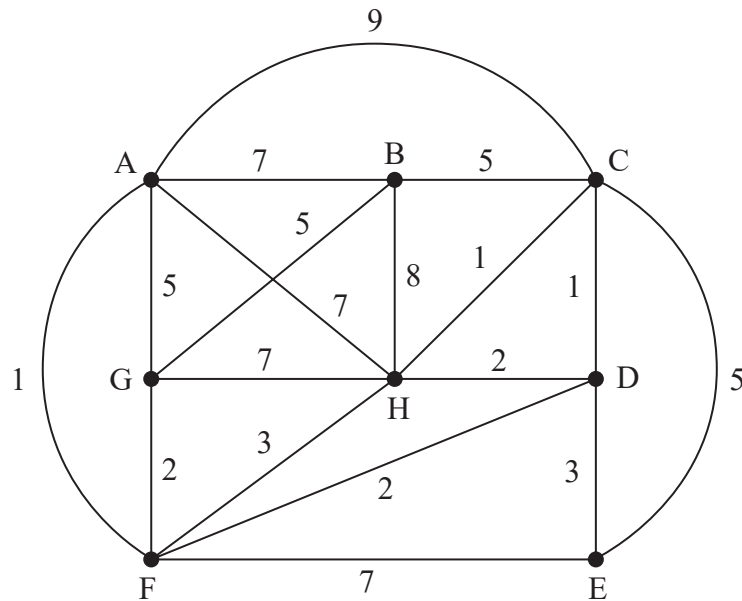
A computer takes exactly 0.07 seconds to apply Dijkstra's algorithm to a network that has 720 arcs and 250 nodes.

(d) Find an estimate for the time that the computer will take to apply Dijkstra's algorithm to a network of 1100 arcs and 625 nodes.

(2)

**(Total for Question 5 is 10 marks)**

6.



**Figure 4**

[The total weight of the network is 80]

Figure 4 models a network of roads that have to be inspected. The number on each arc is the length, in km, of the corresponding section of road.

Each road must be traversed at least once and the length of the inspection route must be minimised.

The inspection route must start and finish at the same vertex.

- (a) Use the route inspection algorithm to find the length of a shortest route. You must make your method and working clear, stating any arcs that must be repeated. (5)

The road that runs directly from C to D is closed.

George decides to drive along all the remaining roads. He will start at D and wants a route that minimises the distance he drives.

- (b) Determine where George should finish his route and the distance he will drive. You should make your reasoning clear. (3)
- (c) Write down a possible route for George. (1)

(Total for Question 6 is 9 marks)

7. Darren makes three types of muesli, Traditional, Fruity and Nutty, to sell in his café. Each type of muesli includes dried fruit and nuts.

Let  $x$ ,  $y$  and  $z$  be the weight in kg of the amount of Traditional muesli, Fruity muesli and Nutty muesli he makes respectively.

Each kg of Traditional, Fruity and Nutty muesli contains 100 g, 75 g and 200 g of nuts respectively.

Darren has 2 kg of nuts available.

He wants to ensure that the amount of Traditional muesli he makes is at least 40% of the total.

He has enough dried fruit to make 3 kg of Traditional muesli or 2 kg of Fruity muesli or 6 kg of Nutty muesli.

Darren wants to make as much muesli as possible.

Express this information as a linear programming problem. You should simplify the constraints, where possible, leaving your answers with integer coefficients. (You may assume the non-negativity constraints.)

---

**(Total for Question 7 is 6 marks)**

8. Robert is planning a conservation project. The activities he must complete, along with the immediately preceding activities, are shown in the table below.

| Activity                         | A | B | C | D | E | F | G | H | I       | J    | K    |
|----------------------------------|---|---|---|---|---|---|---|---|---------|------|------|
| Immediately preceding activities | – | – | – | A | B | C | C | D | D, E, F | G, I | G, I |

- (a) (i) Draw an activity network for this project using activity on arc and exactly two dummies.  
(ii) Explain why each of the dummies is necessary.

(7)

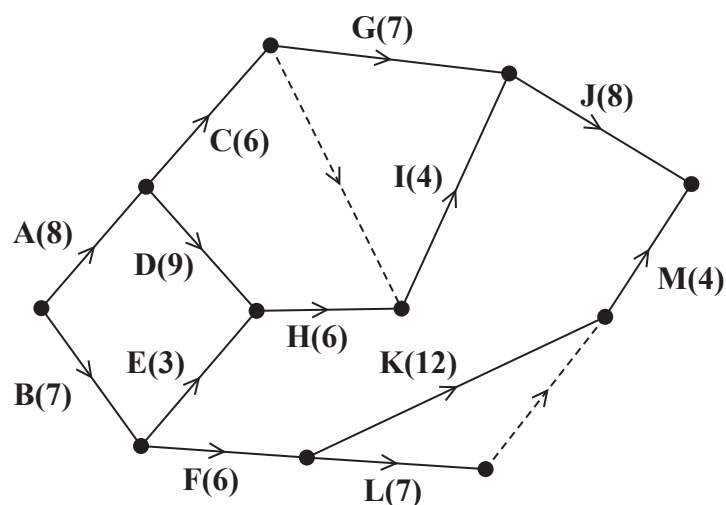


Figure 5

Another conservation project is modelled by the activity network shown in Figure 5. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete the corresponding activity. The project is to be completed in the shortest possible time.

- (b) Add the early event times and the late event times to Diagram 1 in the answer book.  
(c) Write down the critical activities.

(1)

Just as the project is about to start, the project organiser discovers that activity G will take an additional 10 days to complete.

- (d) Find the effect of this change on the earliest completion time of the project, giving a reason for your answer.

(2)

(Total for Question 8 is 13 marks)

9. The tableau shown below is the initial tableau for a linear programming problem in  $x$ ,  $y$  and  $z$ . The objective is to maximise the profit,  $P$ .

| Basic variable | $x$ | $y$ | $z$ | $r$ | $s$ | $t$ | Value |
|----------------|-----|-----|-----|-----|-----|-----|-------|
| $r$            | 1   | 2   | -6  | 1   | 0   | 0   | 12    |
| $s$            | 2   | 1   | -7  | 0   | 1   | 0   | 5     |
| $t$            | 4   | 5   | -8  | 0   | 0   | 1   | 7     |
| $P$            | 6   | -12 | 1   | 0   | 0   | 0   | 0     |

- (a) Perform **one** iteration of the Simplex algorithm to obtain a new tableau, T. Make your method clear by explaining how you select the pivot and stating the row operations you use. (5)
- (b) Write down the profit equation given by T and state the current values of all the variables. (2)
- (c) Use T to explain why the Simplex algorithm cannot be used to find the optimal value of  $P$  for this problem. (1)

The linear programming problem given above is changed to give the tableau shown below.

| Basic variable | $x$ | $y$ | $z$ | $r$ | $s$ | $t$ | Value |
|----------------|-----|-----|-----|-----|-----|-----|-------|
| $r$            | 1   | 2   | 6   | 1   | 0   | 0   | 12    |
| $s$            | 2   | 1   | 7   | 0   | 1   | 0   | 5     |
| $t$            | 4   | 5   | 8   | 0   | 0   | 1   | 7     |
| $P$            | 6   | -12 | 1   | 0   | 0   | 0   | 0     |

The additional constraint  $x \geq 1$  is added to the problem.

- (d) Without performing any iterations, complete Tableau S in the answer book so that the two-stage Simplex method could be applied to solve the problem. (4)

(Total for Question 9 is 12 marks)

**TOTAL FOR PAPER IS 75 MARKS**

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# Pearson Edexcel Level 3 GCE

## Mock Paper Set 2

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/3D**

## Further Mathematics

**Advanced**

**Paper 3D: Decision Mathematics 1**

### You must have:

Decision Mathematics Answer Book (enclosed), calculator

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.  
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

### Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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**Answer ALL questions. Write your answers in the answer book provided.**

1. (a) Explain what it means for two graphs to be **isomorphic**.

(1)

- (b) On Diagram 1 in the answer booklet, draw a graph corresponding to the adjacency matrix given below

|   | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| A | 2 | 1 | 1 | 1 | 0 | 1 | 1 |
| B | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| C | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
| D | 1 | 1 | 1 | 0 | 1 | 1 | 0 |
| E | 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| F | 1 | 0 | 1 | 1 | 1 | 0 | 0 |
| G | 1 | 1 | 0 | 0 | 1 | 0 | 0 |

(2)

- (c) Determine whether the graph in (b) is Eulerian, semi-Eulerian or neither, giving a reason for your answer.

(2)

- (d) Use the planarity algorithm to determine whether the graph is planar. You must make your working clear.

(4)

**(Total for Question 1 is 9 marks)**

2. Esme is constructing a treehouse. The planks of wood she will buy are 13 m long and cost £32 each. She needs to cut the following lengths of wood from these planks in order to carry out her construction.

7      3      12      9       $p$       4      5

where  $p$  is a constant,  $p \leq 13$ ,  $p \in \mathbb{R}$

Esme knows that she will require at least 5 planks of wood.

- (a) Find the range of possible values for  $p$ . (2)
- (b) Carry out a bubble sort to produce a list of the required lengths in descending order. Give the state of the list after each pass. (4)
- (c) Apply the first-fit decreasing bin packing algorithm to your ordered list. (2)

Esme buys the minimum number of planks required to construct the treehouse. She is given all other items required at no cost to her. She sells all the excess wood from the planks to a woodchip manufacturer for 80p per metre. The total cost of the treehouse to Esme is £150.16

- (d) Determine the value of  $p$ . (3)

---

(Total for Question 2 is 11 marks)

3. (a) Explain what is meant by a **Hamiltonian cycle**.

(1)

At a funfair, there are six rides, A, B, C, D, E and F. Barnaby needs to visit each ride at least once, starting and finishing at A.

The table gives the times, in minutes, that it takes Barnaby to walk between each pair of rides, where  $d > 3$

|   | A   | B   | C | D   | E   | F |
|---|-----|-----|---|-----|-----|---|
| A | –   | 7.5 | 3 | $d$ | 5   | 8 |
| B | 7.5 | –   | 5 | 4   | 9   | 6 |
| C | 3   | 5   | – | 6   | 5   | 7 |
| D | $d$ | 4   | 6 | –   | 8.5 | 6 |
| E | 5   | 9   | 5 | 8.5 | –   | 3 |
| F | 8   | 6   | 7 | 6   | 3   | – |

Barnaby wants to minimise the total time he spends walking. He finds the following nearest neighbour route starting from A

A      C      E      F      D      B      A

- (b) Use this nearest neighbour route to calculate an upper bound for the least total walking time.

(1)

- (c) (i) Find two additional nearest neighbour routes starting from A.

- (ii) Calculate the corresponding upper bounds for the least total walking time, giving your answer in terms of  $d$  where necessary.

(2)

By deleting vertex F and all of its arcs, a residual minimum spanning tree is found. Given that this tree contains AD,

- (d) calculate a corresponding lower bound, in terms of  $d$ , for the total walking time. You must make your method clear.

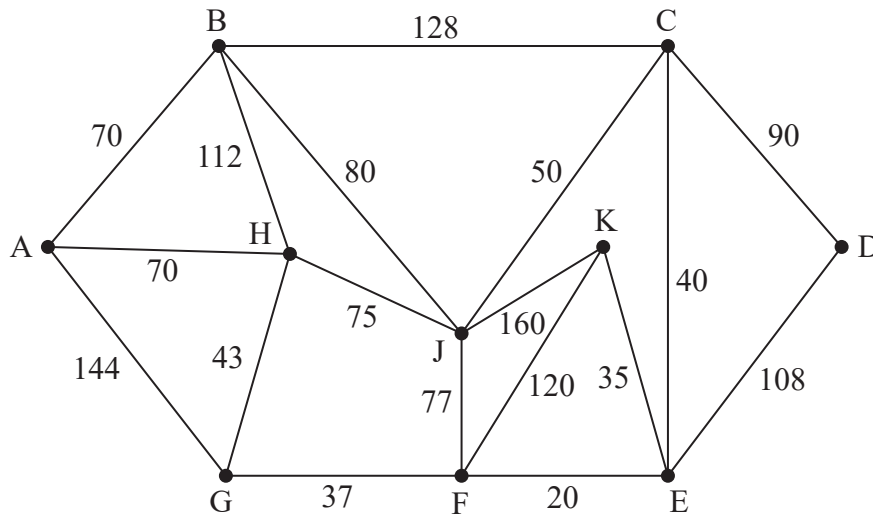
(3)

- (e) By considering the upper and lower bounds obtained, write down the shortest interval that must contain  $d$ , justifying your answer.

(3)

(Total for Question 3 is 10 marks)

4.



**Figure 1**

*[The total weight of the network is 1459]*

The network in Figure 1 represents the passageways through a farmers' market. The number on each arc represents the length, in metres, of the corresponding passageway. A customer wants to walk along each passageway, at least once, to see all the stalls that are set up in the market. He must start and finish at the same vertex.

- (a) Use an appropriate algorithm to find the length of a shortest route. You must make your method and working clear, stating any arcs that must be repeated.

(5)

The customer discovers that if he finishes his walk at K then he can take a tractor ride from K back to any starting vertex.

- (b) Find the maximum reduction in walking distance that the customer can achieve by taking the tractor ride from K back to his starting vertex. You must make your reasoning clear.

(3)

**(Total for Question 4 is 8 marks)**

5. A linear programming problem in  $x$  and  $y$  is described as follows.

$$\text{Maximise } P = 2x - y$$

$$\begin{aligned} \text{Subject to: } 3x - 2y &\leq 100 \\ x + 3y &\geq 60 \\ x, y &\geq 0 \end{aligned}$$

- (a) Explain why the Simplex algorithm cannot be used to solve this linear programming problem.

(1)

The big-M method is to be used to solve this linear programming problem.

- (b) Rewrite the first two constraints using slack, surplus and artificial variables as appropriate.

(2)

- (c) Set up the initial tableau for this linear programming problem so it can be solved using the big-M method.

(4)

- (d) (i) Perform **one** iteration of the big-M method, making your row operations clear.

- (ii) Determine whether the resulting tableau represents an optimal solution, justifying your answer.

(4)

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(Total for Question 5 is 11 marks)

6.

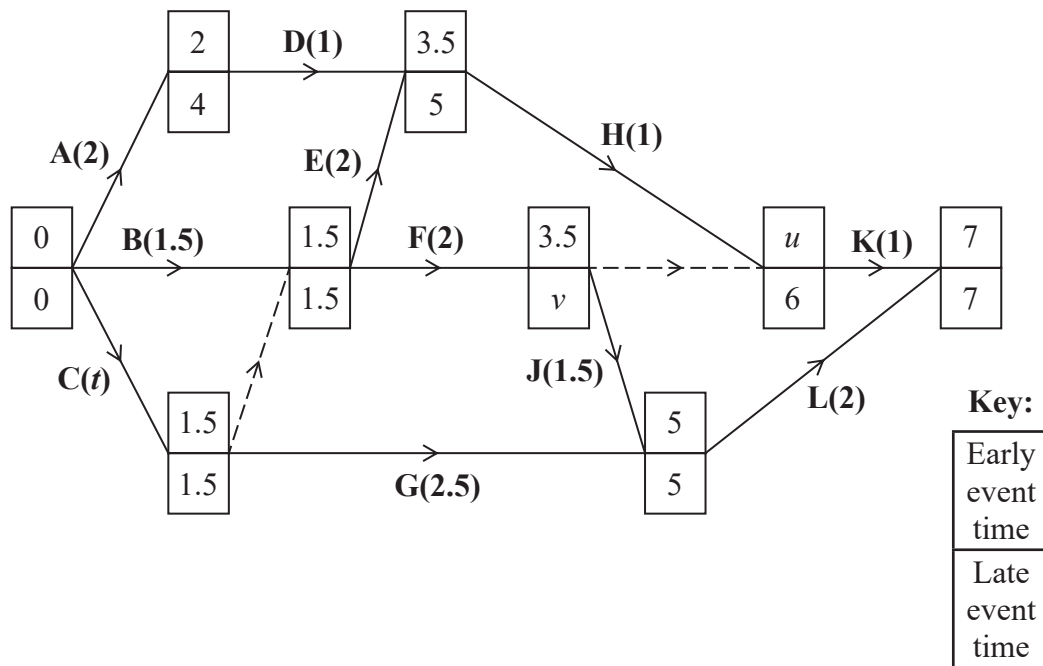


Figure 2

Some volunteers are preparing for a fundraising event. The activity network in Figure 2 represents the activities required to prepare for the fundraising event. The number in brackets on each arc represents the time, in hours, to complete the corresponding activity. Each activity must be completed by exactly one volunteer.

Given that C is a critical activity,

- write down the values of  $t$ ,  $u$  and  $v$ ,
  - state the minimum length of time, in hours, needed to prepare for the fundraising event.

(3)
- Explain how critical activities may be identified.
  - Write down two critical paths for the network given in Figure 2.

(2)
- Draw a Gantt chart on Grid 1 in the answer book to represent the activity network.
 

(3)

The preparation for the fundraising event must be completed in the minimum possible time. Assuming that preparation for the fundraising event begins at 11 am,

- identify the earliest time from which only two volunteers are needed to finish the preparation. Justify your answer by referring to times and activities.
 

(3)

(Total for Question 6 is 11 marks)

7. (i) Ishi raises money for charity by selling tickets for a school talent show.

*Golden* tickets will be sold for £6 each and *Platinum* tickets will be sold for £10 each. Let  $g$  be the number of *Golden* tickets sold and let  $p$  be the number of *Platinum* tickets sold.

Ishi will sell at most 250 tickets.

Based on past experience, she knows that at most 30% of the tickets sold will be *Platinum* tickets.

The number of *Platinum* tickets sold must be at least one sixth of the number of *Golden* tickets sold.

Ishi wants to maximise the amount raised from ticket sales.

By modelling the constraints as simplified inequalities with integer coefficients, formulate this situation as a linear programming problem. You must not solve this problem.

(5)

- (ii) The following month Ishi will sell tickets for a school concert.

She will sell  $x$  tickets of type *A* and  $y$  tickets of type *B*. She will sell type *A* tickets for £2 each and type *B* tickets for £5 each. Ishi formulates the following linear programme to model ticket sales.

Maximise  $P = 2x + 5y$

Subject to:  $x + y \leq 250$

$$7x \geq 13y$$

$$x \leq 5y$$

- (a) Using Grid 2 in the answer booklet and a graphical approach, solve this linear programming problem and hence determine the number of each type of ticket that should be sold to maximise the amount raised. You must make your method and working clear.

- (b) Find the maximum possible amount raised.

(10)

---

(Total for Question 7 is 15 marks)

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**TOTAL FOR PAPER IS 75 MARKS**

# Pearson Edexcel Level 3 GCE

## Further Mathematics

Advanced

Further Mathematics Option 1

Paper 3: Decision Mathematics 1

Further Mathematics Option 2

Paper 4: Decision Mathematics 1

Sample Assessment Material for first teaching September 2017

**Time: 1 hour 30 minutes**

Paper Reference

**9FM0/3D**

**9FM0/4D**

**You must have:**

Decision Mathematics Answer Book (enclosed), calculator

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

### Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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**Answer ALL questions. Write your answers in the answer book provided.**

- 1.** A list of  $n$  numbers needs to be sorted into descending order starting at the left-hand end of the list.

(a) Describe how to carry out the first pass of a bubble sort on the numbers in the list.

(2)

Bubble sort is a quadratic order algorithm.

A computer takes approximately 0.021 seconds to apply a bubble sort to a list of 2000 numbers.

(b) Estimate the time it would take the computer to apply a bubble sort to a list of 50 000 numbers. Make your method clear.

(2)

---

**(Total for Question 1 is 4 marks)**

2.

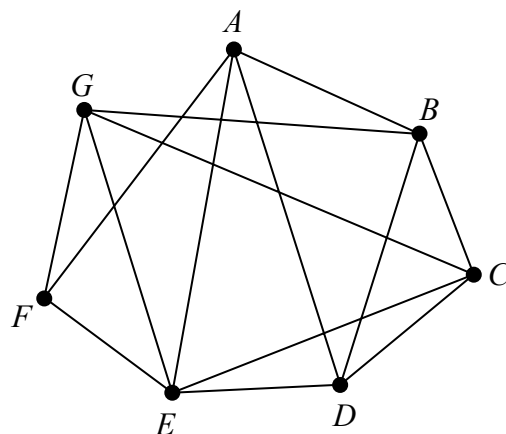


Figure 1

(a) Define what is meant by a **planar** graph.

(2)

(b) Starting at A, find a Hamiltonian cycle for the graph in Figure 1.

(1)

Arc AG is added to Figure 1 to create the graph shown in Figure 2.

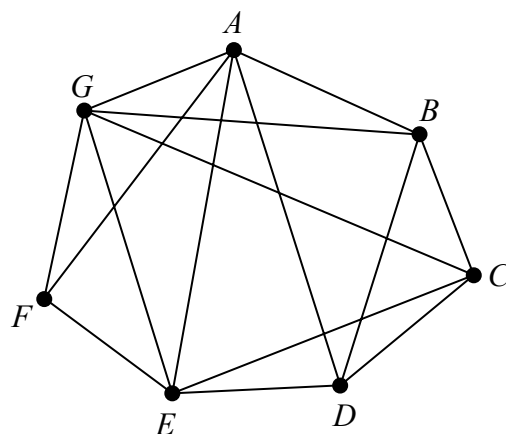


Figure 2

Taking ABCDEFGA as the Hamiltonian cycle,

(c) use the planarity algorithm to determine whether the graph shown in Figure 2 is planar. You must make your working clear and justify your answer.

(4)

(Total for Question 2 is 7 marks)

3. (a) Explain clearly the difference between the classical travelling salesperson problem and the practical travelling salesperson problem.

(2)

|   | A  | B   | C  | D  | E  | F  | G   |
|---|----|-----|----|----|----|----|-----|
| A | –  | 17  | 24 | 16 | 21 | 18 | 41  |
| B | 17 | –   | 35 | 25 | 30 | 31 | $x$ |
| C | 24 | 35  | –  | 28 | 20 | 35 | 32  |
| D | 16 | 25  | 28 | –  | 29 | 19 | 45  |
| E | 21 | 30  | 20 | 29 | –  | 22 | 35  |
| F | 18 | 31  | 35 | 19 | 22 | –  | 37  |
| G | 41 | $x$ | 32 | 45 | 35 | 37 | –   |

The table shows the least distances, in km, by road between seven towns, A, B, C, D, E, F and G. The least distance between B and G is  $x$  km, where  $x > 25$

Preety needs to visit each town at least once, starting and finishing at A. She wishes to minimise the total distance she travels.

- (b) Starting by deleting B and all of its arcs, find a lower bound for Preety's route.

(3)

Preety found the nearest neighbour routes from each of A and C. Given that the sum of the lengths of these routes is 331 km,

- (c) find  $x$ , making your method clear.

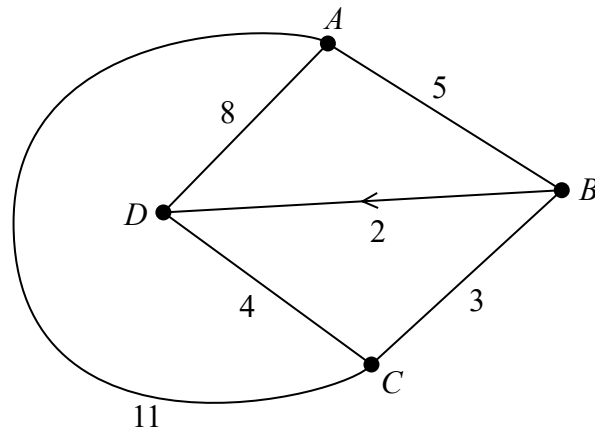
(4)

- (d) Write down the smallest interval that you can be confident contains the optimal length of Preety's route. Give your answer as an inequality.

(2)

**(Total for Question 3 is 11 marks)**

4.



**Figure 3**

The network in Figure 3 shows the roads linking a depot, D, and three collection points A, B and C. The number on each arc represents the length, in miles, of the corresponding road. The road from B to D is a one-way road, as indicated by the arrow.

- (a) Explain clearly if Dijkstra's algorithm can be used to find a route from D to A. (1)

The initial distance and route tables for the network are given in the answer book.

- (b) Use Floyd's algorithm to find a table of least distances. You should show both the distance table and the route table after each iteration. (7)

- (c) Explain how the final route table can be used to find the shortest route from D to B. State this route. (2)

There are items to collect at A, B and C. A van will leave D to make these collections in any order and then return to D. A minimum route is required.

Using the final distance table and the Nearest Neighbour algorithm starting at D,

- (d) find a minimum route and state its length. (2)

Floyd's algorithm and Dijkstra's algorithm are applied to a network. Each will find the shortest distance between vertices of the network.

- (e) Describe how the results of these algorithms differ. (2)

**(Total for Question 4 is 14 marks)**

5. A garden centre makes hanging baskets to sell to its customers. Three types of hanging basket are made, *Sunshine*, *Drama* and *Peaceful*. The plants used are categorised as *Impact*, *Flowering* or *Trailing*.

Each *Sunshine* basket contains 2 *Impact* plants, 4 *Flowering* plants and 3 *Trailing* plants. Each *Drama* basket contains 3 *Impact* plants, 2 *Flowering* plants and 4 *Trailing* plants. Each *Peaceful* basket contains 1 *Impact* plant, 3 *Flowering* plants and 2 *Trailing* plants.

The garden centre can use at most 80 *Impact* plants, at most 140 *Flowering* plants and at most 96 *Trailing* plants each day.

The profit on *Sunshine*, *Drama* and *Peaceful* baskets are £12, £20 and £16 respectively. The garden centre wishes to maximise its profit.

Let  $x$ ,  $y$  and  $z$  be the number of *Sunshine*, *Drama* and *Peaceful* baskets respectively, produced each day.

- (a) Formulate this situation as a linear programming problem, giving your constraints as inequalities.

(5)

- (b) State the further restriction that applies to the values of  $x$ ,  $y$  and  $z$  in this context.

(1)

The Simplex algorithm is used to solve this problem. After one iteration, the tableau is

| b.v. | $x$            | $y$ | $z$            | $r$ | $s$ | $t$            | Value |
|------|----------------|-----|----------------|-----|-----|----------------|-------|
| $r$  | $-\frac{1}{4}$ | 0   | $-\frac{1}{2}$ | 1   | 0   | $-\frac{3}{4}$ | 8     |
| $s$  | $\frac{5}{2}$  | 0   | 2              | 0   | 1   | $-\frac{1}{2}$ | 92    |
| $y$  | $\frac{3}{4}$  | 1   | $\frac{1}{2}$  | 0   | 0   | $\frac{1}{4}$  | 24    |
| $P$  | 3              | 0   | -6             | 0   | 0   | 5              | 480   |

- (c) State the variable that was increased in the first iteration. Justify your answer.

(2)

- (d) Determine how many plants in total are being used after only one iteration of the Simplex algorithm.

(1)

- (e) Explain why for a second iteration of the Simplex algorithm the 2 in the  $z$  column is the pivot value.

(2)

After a second iteration, the tableau is

| b.v. | $x$            | $y$ | $z$ | $r$ | $s$            | $t$            | Value |
|------|----------------|-----|-----|-----|----------------|----------------|-------|
| $r$  | $\frac{3}{8}$  | 0   | 0   | 1   | $\frac{1}{4}$  | $-\frac{7}{8}$ | 31    |
| $s$  | $\frac{5}{4}$  | 0   | 1   | 0   | $\frac{1}{2}$  | $-\frac{1}{4}$ | 46    |
| $y$  | $\frac{1}{8}$  | 1   | 0   | 0   | $-\frac{1}{4}$ | $\frac{3}{8}$  | 1     |
| $P$  | $\frac{21}{2}$ | 0   | 0   | 0   | 3              | $\frac{7}{2}$  | 756   |

(f) Use algebra to explain why this tableau is optimal.

(1)

(g) State the optimal number of each type of basket that should be made.

(1)

The manager of the garden centre is able to increase the number of *Impact* plants available each day from 80 to 100. She wants to know if this would increase her profit.

(h) Use your final tableau to determine the effect of this increase. (You should not carry out any further calculations.)

(2)

**(Total for Question 5 is 15 marks)**

6.

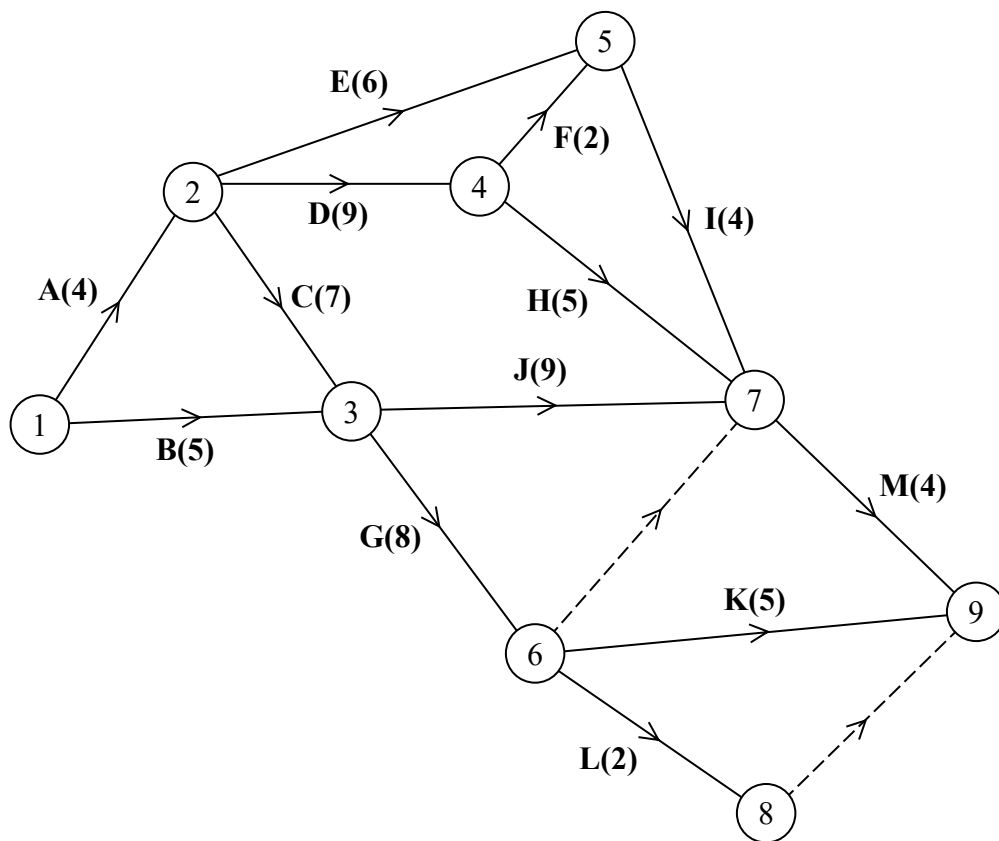


Figure 4

A project is modelled by the activity network shown in Figure 4. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete that activity. Each activity requires one worker. The project is to be completed in the shortest possible time.

- Calculate the early time and the late time for each event, using Diagram 1 in the answer book. (3)
- On Grid 1 in the answer book, complete the cascade (Gantt) chart for this project. (3)
- On Grid 2 in the answer book, draw a resource histogram to show the number of workers required each day when each activity begins at its earliest time. (3)

The supervisor of the project states that only three workers are required to complete the project in the minimum time.

- Use Grid 2 to determine if the project can be completed in the minimum time by only three workers. Give reasons for your answer. (3)

(Total for Question 6 is 12 marks)

7. A linear programming problem in  $x$ ,  $y$  and  $z$  is described as follows.

Maximise  $P = 3x + 2y + 2z$

subject to  $2x + 2y + z \leq 25$

$x + 4y \leq 15$

$x \geq 3$

(a) Explain why the Simplex algorithm cannot be used to solve this linear programming problem.

(1)

The big-M method is to be used to solve this linear programming problem.

(b) Define, in this context, what  $M$  represents. You must use correct mathematical language in your answer.

(1)

The initial tableau for a big-M solution to the problem is shown below.

| b.v.  | $x$        | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $t_1$ | Value |
|-------|------------|-----|-----|-------|-------|-------|-------|-------|
| $s_1$ | 2          | 2   | 1   | 1     | 0     | 0     | 0     | 25    |
| $s_2$ | 1          | 4   | 0   | 0     | 1     | 0     | 0     | 15    |
| $t_1$ | 1          | 0   | 0   | 0     | 0     | -1    | 1     | 3     |
| $P$   | $-(3 + M)$ | -2  | -2  | 0     | 0     | $M$   | 0     | $-3M$ |

(c) Explain clearly how the equation represented in the b.v.  $t_1$  row was derived.

(1)

(d) Show how the equation represented in the b.v.  $P$  row was derived.

(2)

The tableau obtained from the first iteration of the big-M method is shown below.

| b.v.  | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $t_1$   | Value |
|-------|-----|-----|-----|-------|-------|-------|---------|-------|
| $s_1$ | 0   | 2   | 1   | 1     | 0     | 2     | -2      | 19    |
| $s_2$ | 0   | 4   | 0   | 0     | 1     | 1     | -1      | 12    |
| $x$   | 1   | 0   | 0   | 0     | 0     | -1    | 1       | 3     |
| $P$   | 0   | -2  | -2  | 0     | 0     | -3    | $3 + M$ | 9     |

(e) Solve the linear programming problem, starting from this second tableau. You must

- give a detailed explanation of your method by clearly stating the row operations you use and
- state the solution by deducing the final values of  $P$ ,  $x$ ,  $y$  and  $z$ .

(7)

**(Total for Question 7 is 12 marks)**

**TOTAL FOR PAPER IS 75 MARKS**

Write your name here

Surname

Other names

**Pearson Edexcel**  
**Level 3 GCE**

Centre Number

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Candidate Number

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|  |  |  |  |
|--|--|--|--|

# Further Mathematics

**Advanced**

**Further Mathematics Option 1**

**Paper 3: Decision Mathematics 1**

**Further Mathematics Option 2**

**Paper 4: Decision Mathematics 1**

Sample Assessment Material for first teaching September 2017

Paper Reference

**9FM0/3D**

**9FM0/4D**

## Answer Book

Do not return the question paper with the answer book.

Total Marks

Turn over ►

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Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

2.

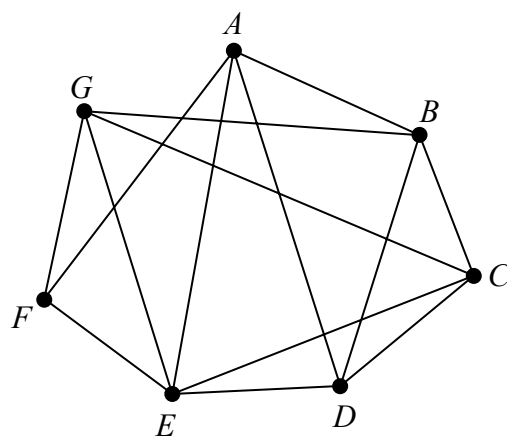
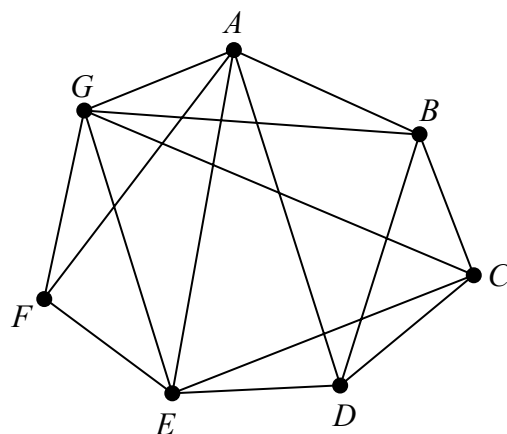


Figure 1

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**Question 2 continued****Figure 2****(Total for Question 2 is 7 marks)**

3.

|   | A  | B   | C  | D  | E  | F  | G   |
|---|----|-----|----|----|----|----|-----|
| A | –  | 17  | 24 | 16 | 21 | 18 | 41  |
| B | 17 | –   | 35 | 25 | 30 | 31 | $x$ |
| C | 24 | 35  | –  | 28 | 20 | 35 | 32  |
| D | 16 | 25  | 28 | –  | 29 | 19 | 45  |
| E | 21 | 30  | 20 | 29 | –  | 22 | 35  |
| F | 18 | 31  | 35 | 19 | 22 | –  | 37  |
| G | 41 | $x$ | 32 | 45 | 35 | 37 | –   |

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## Question 3 continued

|   | A  | B   | C  | D  | E  | F  | G   |
|---|----|-----|----|----|----|----|-----|
| A | –  | 17  | 24 | 16 | 21 | 18 | 41  |
| B | 17 | –   | 35 | 25 | 30 | 31 | $x$ |
| C | 24 | 35  | –  | 28 | 20 | 35 | 32  |
| D | 16 | 25  | 28 | –  | 29 | 19 | 45  |
| E | 21 | 30  | 20 | 29 | –  | 22 | 35  |
| F | 18 | 31  | 35 | 19 | 22 | –  | 37  |
| G | 41 | $x$ | 32 | 45 | 35 | 37 | –   |

(Total for Question 3 is 11 marks)

4.

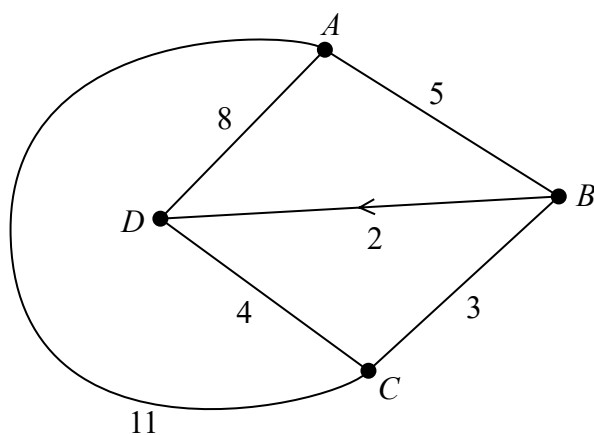


Figure 3

(b)

|   | A  | B        | C  | D |
|---|----|----------|----|---|
| A | –  | 5        | 11 | 8 |
| B | 5  | –        | 3  | 2 |
| C | 11 | 3        | –  | 4 |
| D | 8  | $\infty$ | 4  | – |

|   | A | B | C | D |
|---|---|---|---|---|
| A | A | B | C | D |
| B | A | B | C | D |
| C | A | B | C | D |
| D | A | B | C | D |

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

## Question 4 continued

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

|   | A | B | C | D |
|---|---|---|---|---|
| A |   |   |   |   |
| B |   |   |   |   |
| C |   |   |   |   |
| D |   |   |   |   |

(Total for Question 4 is 14 marks)

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6.

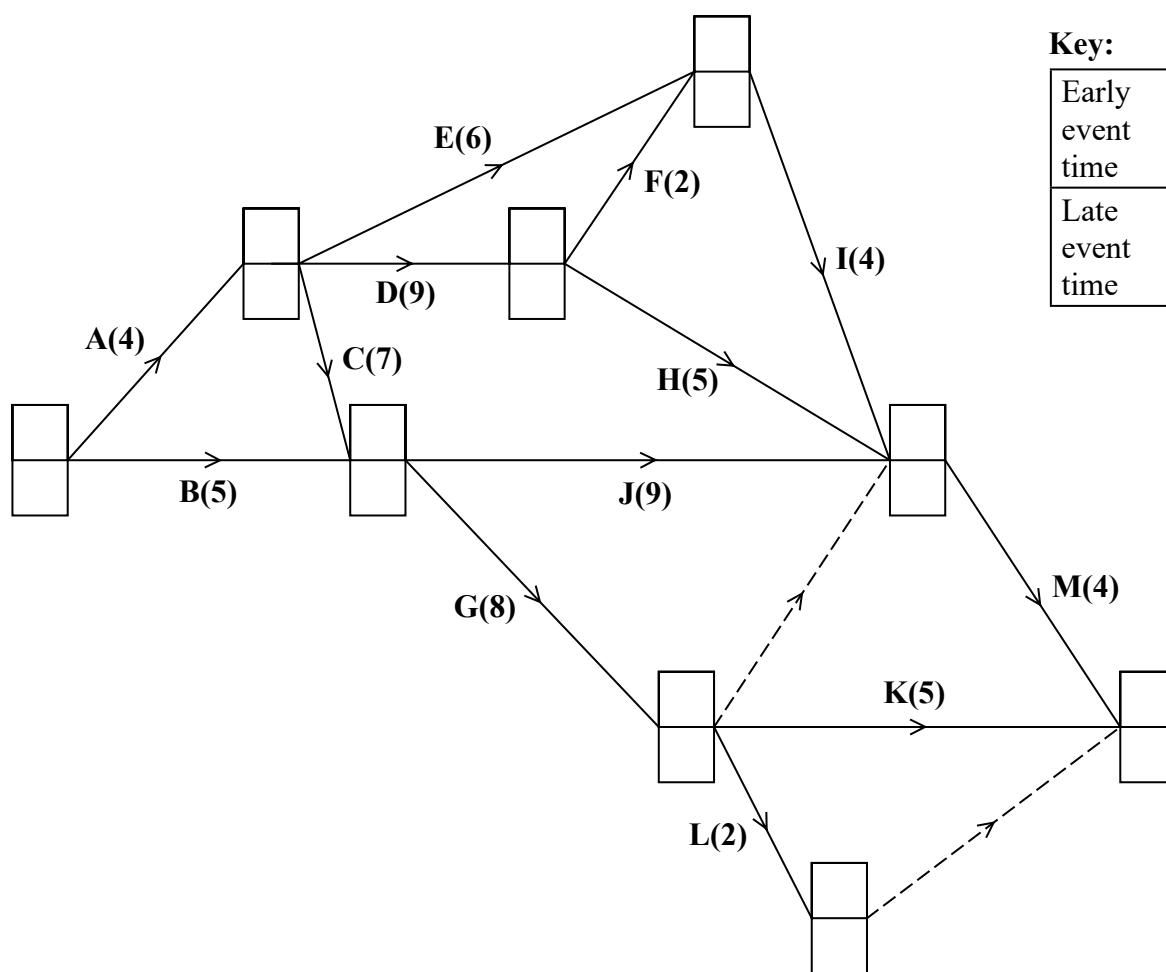


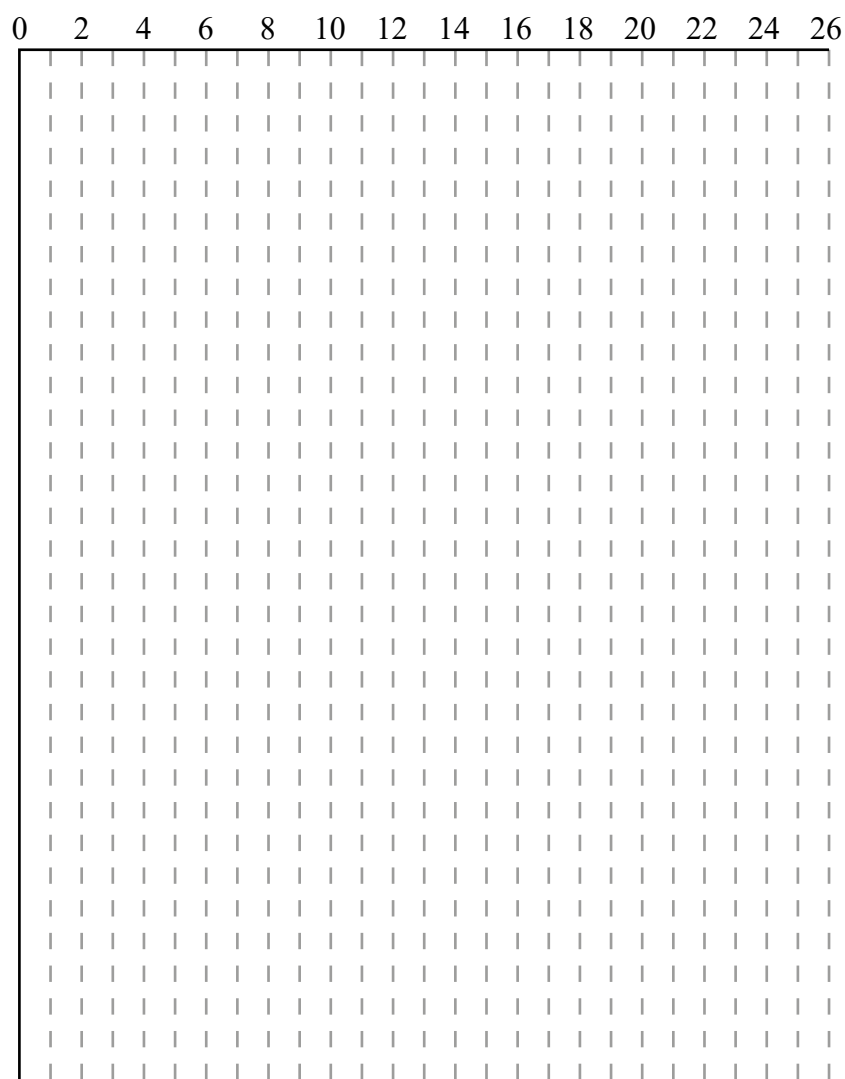
Diagram 1

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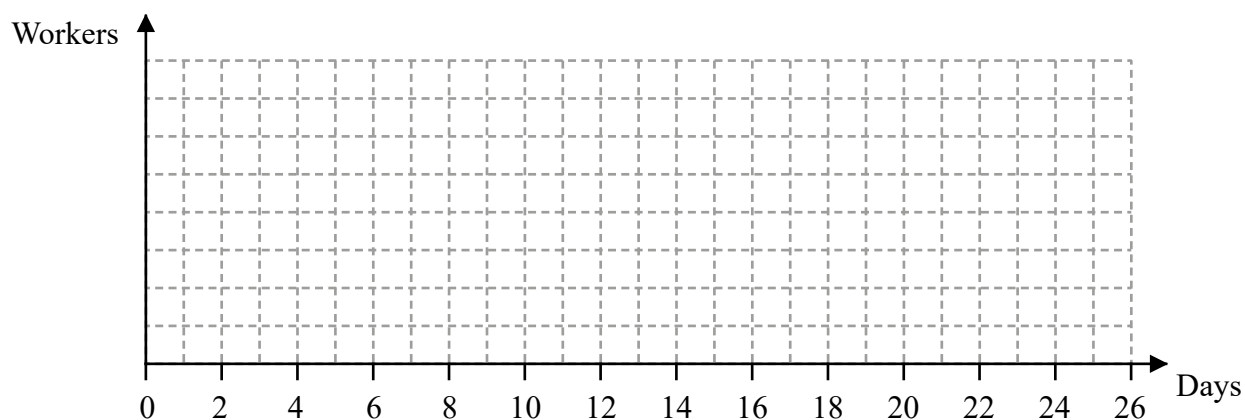
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**Question 6 continued**



**Grid 1**

(There is a spare grid on the next page)



**Grid 2**

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316

7.

| b.v.  | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $t_1$   | Value |
|-------|-----|-----|-----|-------|-------|-------|---------|-------|
| $s_1$ | 0   | 2   | 1   | 1     | 0     | 2     | -2      | 19    |
| $s_2$ | 0   | 4   | 0   | 0     | 1     | 1     | -1      | 12    |
| $x$   | 1   | 0   | 0   | 0     | 0     | -1    | 1       | 3     |
| $P$   | 0   | -2  | -2  | 0     | 0     | -3    | $3 + M$ | 9     |

| b.v. | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $t_1$ | Value | Row Ops |
|------|-----|-----|-----|-------|-------|-------|-------|-------|---------|
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
| $P$  |     |     |     |       |       |       |       |       |         |

| b.v. | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $t_1$ | Value | Row Ops |
|------|-----|-----|-----|-------|-------|-------|-------|-------|---------|
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
| $P$  |     |     |     |       |       |       |       |       |         |

**Question 7 continued**

| b.v. | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $t_1$ | Value | Row Ops |
|------|-----|-----|-----|-------|-------|-------|-------|-------|---------|
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
| $P$  |     |     |     |       |       |       |       |       |         |

| b.v. | $x$ | $y$ | $z$ | $s_1$ | $s_2$ | $s_3$ | $t_1$ | Value | Row Ops |
|------|-----|-----|-----|-------|-------|-------|-------|-------|---------|
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
|      |     |     |     |       |       |       |       |       |         |
| $P$  |     |     |     |       |       |       |       |       |         |

(Total for Question 7 is 12 marks)

**TOTAL FOR PAPER IS 75 MARKS**

# Pearson Edexcel Level 3 GCE

Specimen Paper

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/3D**

## Further Mathematics Advanced Paper 3D: Decision Mathematics 1

### You must have:

Decision Mathematics Answer Book (enclosed), calculator

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.  
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

### Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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**Answer ALL questions. Write your answers in the answer book provided.**

**1.**                    6     1     9     14     18     7     10     4     17     13

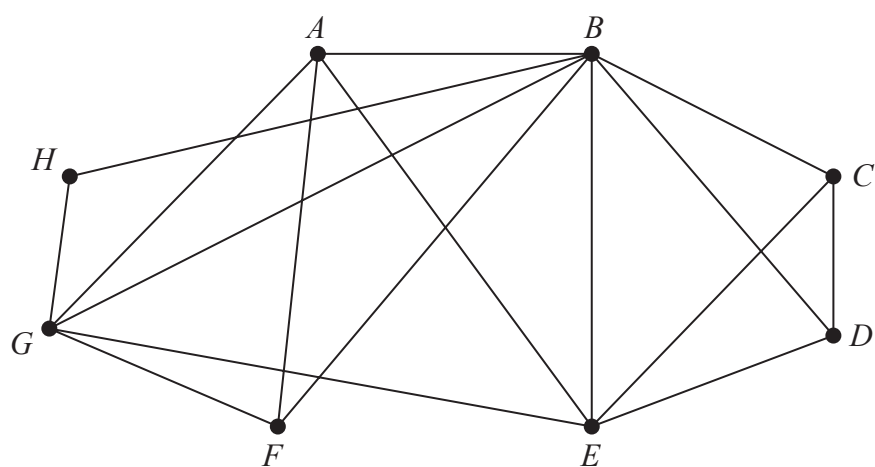
(a) The list of numbers shown above is to be sorted into descending order. Apply the quick sort algorithm to obtain the sorted list. You must make your pivots clear. (3)

(b) Apply the first-fit decreasing bin packing algorithm to your ordered list to pack the numbers into bins of size 30 (2)

**(Total for Question 1 is 5 marks)**

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2.

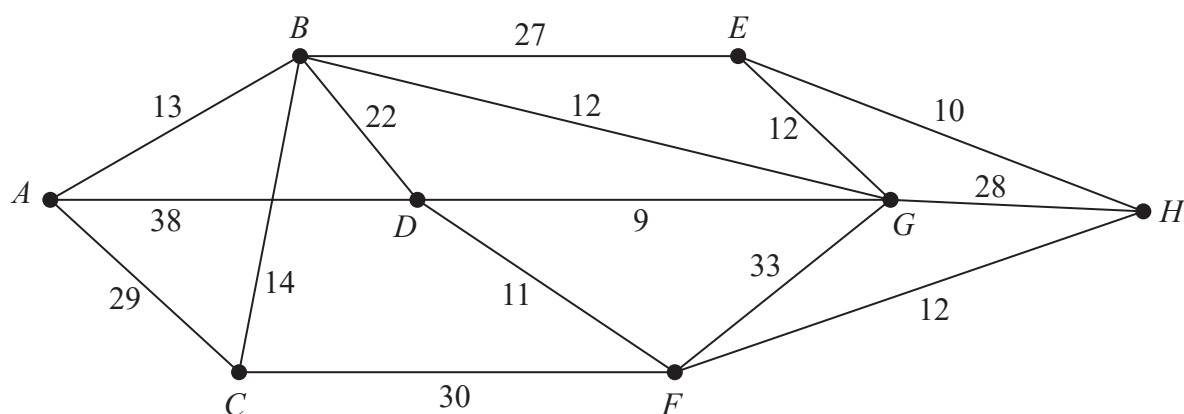


**Figure 1**

- (a) Write down the number of arcs in a minimum spanning tree for the graph shown in Figure 1. (1)
- (b) Determine the minimum number of arcs that would need to be added to make the graph semi-Eulerian. You must justify your answer. (2)
- Taking AFGHBCDEA as the Hamiltonian cycle,
- (c) use the planarity algorithm to determine whether or not the graph shown in Figure 1 is planar. You must make your working clear and justify your answer. (3)

**(Total for Question 2 is 6 marks)**

3.



**Figure 2**

[The total weight of the network is 300]

Figure 2 represents a network of train tracks. The vertices A, B, C, D, E, F, G and H represent train stations.

- (i) The number on each edge represents the time, in minutes, to travel on the corresponding track.

- (a) Use Dijkstra's algorithm to find the time required to travel the quickest route from A to H.

(4)

A train is to travel from A to F via H without stopping.

- (b) Find the quickest route from A to F via H. State the route and the time taken.

(2)

A train is to travel from A to H, stopping for 1 minute at each train station it passes through on its route.

- (c) Explain how you would adapt the network so that Dijkstra's algorithm could be used to find the quickest route for this train. You do not need to find this route.

(2)

- (ii) The number on each edge **now** represent the length, in km, of the corresponding track. A route that starts at C, finishes at G and traverses each track at least once needs to be found.

- (a) Use an appropriate algorithm to find the length of the shortest route. You must make your method clear.

(4)

- (b) State the tracks that will need to be traversed twice.

(1)

(Total for Question 3 is 13 marks)

4. The table below lists the durations (in minutes), the immediate preceding activities and the number of workers required for each activity in a project.

| Activity | Duration (minutes) | Immediate preceding activities | Number of workers |
|----------|--------------------|--------------------------------|-------------------|
| A        | 20                 | -                              | 1                 |
| B        | 15                 | -                              | 1                 |
| C        | 35                 | -                              | 2                 |
| D        | 10                 | C                              | 1                 |
| E        | 35                 | A                              | 1                 |
| F        | 20                 | A                              | 2                 |
| G        | 10                 | A, B, D                        | 1                 |
| H        | 25                 | C                              | 1                 |
| I        | 30                 | E                              | 1                 |
| J        | 20                 | E, F, G, H                     | 2                 |

The project is modelled by the activity network shown in Figure 3. The activities are represented by the arcs. The project is to be completed in the shortest possible time.

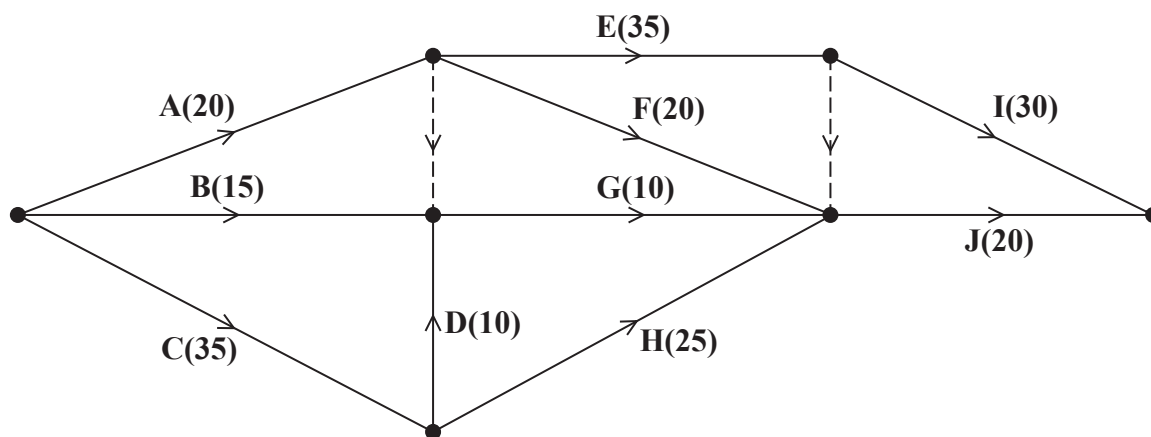


Figure 3

- (a) (i) Calculate the early time and the late time for each event, using Diagram 1 in the answer book.
- (ii) Determine the minimum project completion time.
- (iii) Identify the critical activities. (5)
- (b) On Grid 1 in the answer book, draw a resource histogram to show the number of workers required at each time when each activity begins at its earliest possible start time. (3)
- (c) Determine whether or not the project can be completed in the minimum possible time by at most four workers. You must justify your answer with reference to your resource histogram. (2)

(Total for Question 4 is 10 marks)

5. Dale is planning a production run of three types of desk. The three types are lectern desk, roll top desk and writing desk.

In total, Dale has  $400\text{ m}^2$  of wood available; each lectern desk requires  $3\text{ m}^2$ , each roll top desk requires  $5\text{ m}^2$ , and each writing desk requires  $8\text{ m}^2$

In total, Dale has 350 hours available; each lectern desk requires 3 hours, each roll top desk requires 6 hours, and each writing desk requires 10 hours.

Once complete, the desks need to be stored in a warehouse. The warehouse has  $75\text{ m}^3$  of storage space available; each lectern desk requires  $1\text{ m}^3$ , each roll top desk requires  $1.5\text{ m}^3$  and each writing desk requires  $1.25\text{ m}^3$

The profit on each lectern desk sold is £40, the profit on each roll top desk sold is £50 and the profit on each writing desk sold is £65

Dale wants to maximise his profit.

Let  $x$ ,  $y$  and  $z$  be the number of lectern desks, roll top desks and writing desks made respectively during the production run.

- (a) Formulate this situation as a linear programming problem, giving your constraints as inequalities. (4)
- (b) Complete the initial tableau in the answer book for this linear programming problem. (2)
- (c) Taking the most negative number in the profit row to indicate the pivot column, perform one complete iteration of the Simplex algorithm. Give an explanation of the method by clearly stating the row operations you use. (4)

After a second iteration, the exact values in the tableau are

| b.v.  | $x$ | $y$   | $z$ | $s_1$ | $s_2$ | $s_3$ | Value |
|-------|-----|-------|-----|-------|-------|-------|-------|
| $s_1$ | 0   | -0.52 | 0   | 1     | -0.68 | -0.96 | 90    |
| $z$   | 0   | 0.24  | 1   | 0     | 0.16  | -0.48 | 20    |
| $x$   | 1   | 1.2   | 0   | 0     | -0.2  | 1.6   | 50    |
| $P$   | 0   | 13.6  | 0   | 0     | 2.4   | 32.8  | 3300  |

- (d) Use algebra to explain how you know that this tableau is optimal. (1)
- (e) (i) State the optimal number of each type of desk that should be made.  
(ii) State the maximum total profit. (2)
- (f) Explain, in context, the meaning of the 90 in the value column. (2)
- (g) Give a reason why the profit may be less than the value stated in (e)(ii). (1)

(Total for Question 5 is 16 marks)

6.

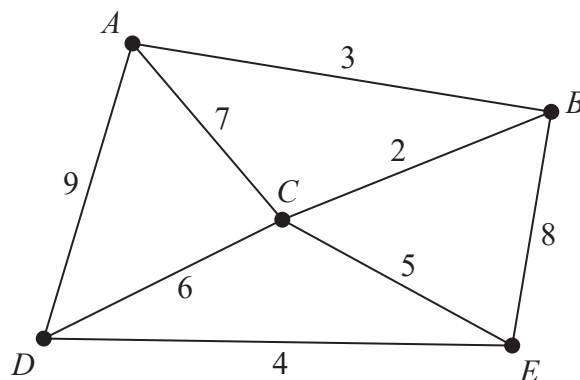


Figure 4

The network in Figure 4 shows the direct routes linking five towns, A, B, C, D and E. The edge weights represent distances, in km, between the towns.

- (a) For this network, complete the initial distance and route tables provided in the answer book. (2)
- (b) Perform the first and second iterations of Floyd's algorithm. You should show the distance table and the route table after each iteration. (4)

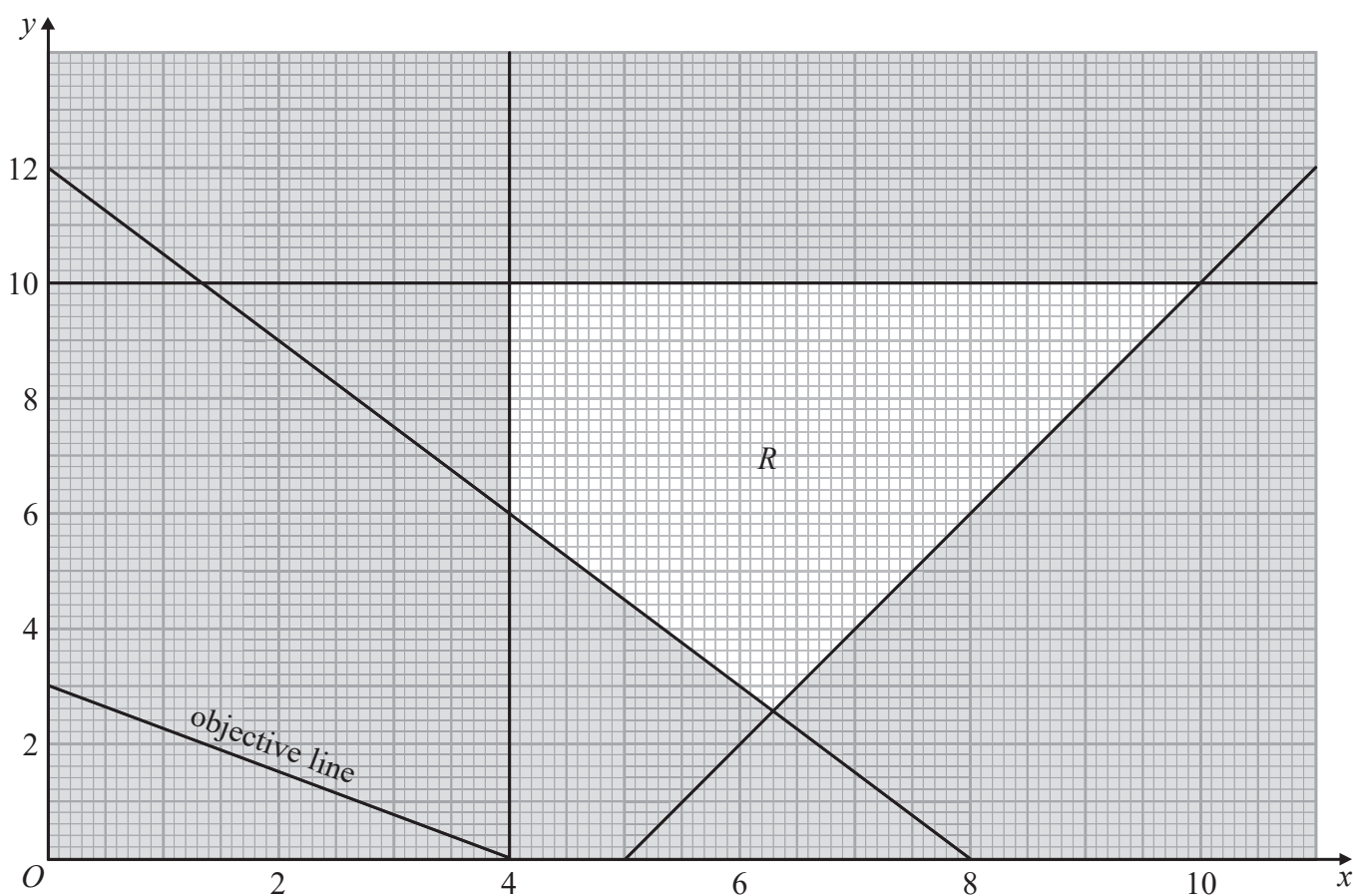
The final tables after five iterations of Floyd's algorithm are shown below.

| Distance table |    |   |   |   |    | Route table |   |   |   |   |   |
|----------------|----|---|---|---|----|-------------|---|---|---|---|---|
|                | A  | B | C | D | E  |             | A | B | C | D | E |
| A              | –  | 3 | 5 | 9 | 10 | A           | A | B | B | D | C |
| B              | 3  | – | 2 | 8 | 7  | B           | A | B | C | C | C |
| C              | 5  | 2 | – | 6 | 5  | C           | B | B | C | D | E |
| D              | 9  | 8 | 6 | – | 4  | D           | A | C | C | D | E |
| E              | 10 | 7 | 5 | 4 | –  | E           | C | C | C | D | E |

- (c) Draw the complete network of shortest distances. (2)
- (d) (i) Use the nearest neighbour algorithm, starting at vertex E, to produce a Hamiltonian cycle for the complete network. (2)
- (ii) Write down the length of the Hamiltonian cycle. (2)
- (e) Interpret the Hamiltonian cycle from (d) in terms of the towns actually visited. (1)
- (f) Starting by deleting A from the complete network of shortest distances, and all of its edges, find a lower bound for the solution to the Travelling Salesman problem. (2)
- (g) Use your results to write down the smallest interval that you can be confident contains the optimal solution to the Travelling Salesman problem. (2)

(Total for Question 6 is 15 marks)

7.



**Figure 5**

Figure 5 shows the constraints of a maximisation linear programming problem in  $x$  and  $y$ , where  $R$  is the feasible region. An objective line is also shown and labelled on Figure 5.

A student decides to find the optimal vertex of  $R$  by using the two-stage Simplex algorithm.

Set up an initial tableau for the two-stage Simplex algorithm. (You should not solve the linear programming problem.)

**(Total for Question 7 is 10 marks)**

**TOTAL FOR PAPER IS 75 MARKS**